DISPENSARY MANAGEMENT SYSTEM

REST API Documentation

Release 1.0 - base path /api/v1 - 49 operations

Document	03-REST-API-Documentation.pdf
Version	1.0.0
Prepared	12 October 2026
Status	Draft for owner review - implementation baseline for release 1.0
Purpose	Human-readable rendering of openapi.yaml with business rules, validation and concurrency notes per endpoint.

Document set

File	Content
01-Software-Requirements-Specification.pdf	What the system must do: scope, roles, FR/NFR, business rules, workflows, use cases, validation, security, audit, open questions, traceability.
02-Database-and-System-Architecture.pdf	How it is built: architecture, components, sequences, data model, data dictionary, DDL, concurrency, deployment.
03-REST-API-Documentation.pdf	Every endpoint with roles, parameters, examples, validation, errors and transactions (generated from openapi.yaml).
openapi.yaml	Machine-readable OpenAPI 3.1 contract - the source of truth for the API.
backend-requirements.md / frontend-requirements.md	Implementation instructions for Claude Code.

All documents in the set are generated from the same sources (database migration, openapi.yaml and requirement catalogues) and are cross-checked automatically; identifiers such as FR-012, BR-Q-03 or ERR codes are stable across documents.

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1. Introduction

This document is the human-readable rendering of `openapi.yaml` (OpenAPI 3.1, contract version 1.0.0). Every endpoint section, example, validation table and error list below is generated from that file, so the two cannot diverge; business rules, affected entities and concurrency notes come from the `x-` extensions of the same file. The contract covers **49 operations** on 41 paths with 79 schemas and 55 error codes. Requirements are in 01-Software-Requirements-Specification.pdf (SRS); tables and locking in 02-Database-and-System-Architecture.pdf (Architecture).

All request and response examples were validated automatically against their JSON schemas (formats, enums, required fields, unknown properties, mutually exclusive fields) when this document was built.

2. Conventions

2.1 Base path, versions and formats

- Base path `/api/v1` for all business endpoints; health checks under `/actuator`. A breaking change would introduce `/api/v2` alongside.
- Browsers call the API on the same origin as the frontend (`https://<host>/api/v1/...`) through the reverse proxy or the Next.js proxy; direct calls to the backend (`http://localhost:8080`) are for development and Swagger UI.
- Request and response bodies are UTF-8 JSON (`application/json`); errors are `application/problem+json`. Maximum request body 256 KB.
- Identifiers are UUID strings. Human identifiers: `patientNumber` (`P-0001`), `prescriptionNumber` (`RX-000123`), `queueNumber` (integer).
- Instants: ISO-8601 UTC with `Z` (`2026-10-12T04:30:00Z`). Dates: `YYYY-MM-DD` in the dispensary time zone.
- Decimal quantities: JSON numbers with at most two decimals (`7.5`, `60`), mapped to Java `BigDecimal` (never `double`).
- Optional fields are always present in responses, with `null` when empty (schemas list them as `type: [..., 'null']`). In requests, optional fields may be omitted or `null`.
- Requests with unknown properties are rejected (`400 MALFORMED_REQUEST`); strings are trimmed by the server; empty optional strings become `null`.
- Enumerations are upper-case strings; unknown enum values are `400 MALFORMED_REQUEST`.

2.2 Authentication

Step	Request	Result
Log in	`POST /api/v1/auth/login` with `{username, password}`	`200 {accessToken, tokenType: "Bearer", expiresIn: 900, user}` and cookie `dms_refresh`
Call the API	`Authorization: Bearer <accessToken>`	normal responses; `401 TOKEN_EXPIRED` after 15 min
Refresh	`POST /api/v1/auth/refresh` with the cookie and `X-Requested-With: XMLHttpRequest`	new access token and rotated cookie
Log out	`POST /api/v1/auth/logout` (cookie + header)	`204`, cookie cleared, token family revoked

Cookie attributes: `HttpOnly; SameSite=Strict; Path=/api/v1/auth; Max-Age=<remaining family lifetime>` and `Secure` when `APP_REFRESH_COOKIE_SECURE=true`. The access token must be kept in memory only. JWT claims: `sub` (user id), `roles`, `pwd_change`, `iss`, `iat`, `exp`, `jti`.

2.3 Authorisation

Each operation lists its roles (`x-roles`): `PUBLIC` (no token), `ANY_STAFF` (any authenticated user) or explicit roles. A user's roles are the union of their grants. Wrong role → `403 FORBIDDEN`; ownership rule failed → `403 NOT_RESOURCE_OWNER`; while `mustChangePassword` is true every endpoint except `GET /auth/me`, `POST /auth/change-password` and `POST /auth/logout` returns `403 PASSWORD_CHANGE_REQUIRED`. The complete matrix is in section 4.2.

2.4 Standard request headers

Header	When	Notes
`Authorization: Bearer <token>`	all non-public endpoints	
`Content-Type: application/json`	requests with a body	
`Idempotency-Key`	operations marked *Idempotency: Required*	see 2.6
`X-Request-Id`	optional on every request	echoed in the response header and in `Problem.requestId`
`X-Requested-With: XMLHttpRequest`	`auth/refresh`, `auth/logout`	CSRF protection for cookie-authenticated calls
`If-None-Match`	`GET` queue status endpoints	conditional polling, `304 Not Modified`

2.5 Pagination, sorting and filtering

- Query parameters `page` (0-based, default 0) and `size` (1-100, default 20; medical history 1-50, default 10).
- Response envelope: `{ "items": [...], "page": 0, "size": 20, "totalItems": 57, "totalPages": 3 }`.
- `sort` accepts only the values listed per endpoint (whitelist); other values are `400 VALIDATION_FAILED`.
- Repeatable filters use the form style: `?status=PENDING&status=IN_PROGRESS`.
- Time-range filters are `from` inclusive, `to` exclusive (`submittedFrom`, `submittedTo`).

2.6 Idempotency

Operations `createPatient`, `createConsultation`, `addPrescriptionToConsultation`, `createBooking` and `dispensePrescription` require an `Idempotency-Key` header (`^[A-Za-z0-9_-]{8,100}$`, UUID v4 recommended):

- The client generates one key per user intent (one form submission) and reuses it for every retry of that submission; a new submission gets a new key.
- Same user + same key + same operation and body within 24 h → the original status code and the **current** representation of the created resource (replay). No duplicate is created, even when the requests arrive concurrently.
- Same key with a different body or operation → `422 IDEMPOTENCY_KEY_REUSED`. Missing or malformed key -> `400 IDEMPOTENCY_KEY_REQUIRED`.
- Implementation: Architecture section 7.5.

2.7 Concurrency control

Mechanism	Endpoints	Client behaviour on conflict
Optimistic `version` in the body	`PUT /patients/{id}`, `PUT /users/{id}`, `PUT /medicines/{id}`	`409 CONCURRENT_MODIFICATION`: reload, show the newer values, let the user re-apply
Compare-and-set `expectedLastCalledNumber`	`POST /queue-sessions/{id}/call-next`	`409 QUEUE_STATE_CHANGED`: refetch the session; never retry automatically

Mechanism	Endpoints	Client behaviour on conflict
Server-side row locks + state checks	prescription start/dispense/cancel, queue entry transitions, bookings	`409 INVALID_STATUS_TRANSITION` with `currentStatus`: refresh and show the current state
ETag / If-None-Match	`GET /public/queue-sessions/{id}`, `GET /queue-sessions/{id}`	`304`: keep the cached data
Lock timeout (5 s)	any write	`503 SERVICE_UNAVAILABLE`: retry later (same idempotency key)

2.8 HTTP status codes

Status	Used for
200 OK	Successful read, update, state transition (including idempotent no-ops) and replayed `dispense`
201 Created	Resource created (`Location` header set); also the status of a replayed create
204 No Content	Logout, password change, password reset
304 Not Modified	Conditional GET with a matching ETag
400 Bad Request	Malformed JSON, unknown properties or enum values, field validation, missing Idempotency-Key
401 Unauthorized	Missing/invalid/expired token, wrong credentials, locked/disabled account, invalid refresh token
403 Forbidden	Role or ownership rule not satisfied, password change pending
404 Not Found	Resource in the path does not exist (or a public session is in the past)
409 Conflict	Duplicates, stale versions, invalid status transitions, queue state changed, session not open/closed/full
422 Unprocessable Content	Well-formed request violating a business rule that involves other data (inactive medicine, quantity above prescribed, referenced id not found, date window, idempotency key reuse)
429 Too Many Requests	Rate limit; `Retry-After` header
500 Internal Server Error	Unexpected error; body contains only the generic message and `requestId`
503 Service Unavailable	Database unavailable, lock timeout, shutting down; readiness probe

2.9 Rate limits

`POST /auth/login`: `APP_RATE_LIMIT_LOGIN_PER_MINUTE` (default 10) per client IP. `/api/v1/public/**`: `APP_RATE_LIMIT_PUBLIC_PER_MINUTE` (default 120) per client IP - enough for one display polling every 5 s plus manual reloads.

2.10 Canonical statuses and transitions

Prescription status (`prescriptions.status`, enum `PrescriptionStatus`)

From	Action	To / result	Who
(new)	create consultation / add prescription	PENDING	DOCTOR
PENDING	start	IN_PROGRESS	PHARMACIST
PENDING	dispense	DISPENSED	PHARMACIST

From	Action	To / result	Who
IN_PROGRESS	dispense	DISPENSED	PHARMACIST
PENDING, IN_PROGRESS	cancel (reason)	CANCELLED	prescribing DOCTOR, any PHARMACIST
IN_PROGRESS	start (repeat)	IN_PROGRESS (no-op, 200)	PHARMACIST
CANCELLED	cancel (repeat)	CANCELLED (no-op, 200)	as cancel
DISPENSED	start / cancel / dispense	409 INVALID_STATUS_TRANSITION	-
CANCELLED	start / dispense	409 INVALID_STATUS_TRANSITION	-

Dispensing outcome (`dispensings.outcome`, enum `DispensingOutcome`): `FULL` when every item is fully dispensed, otherwise `PARTIAL`. **Dispensed item status** (`dispensing_items.status`, enum `DispensedItemStatus`): `FULL` (= prescribed), `PARTIAL` (0 < qty < prescribed), `NOT_DISPENSED` (0). Both are derived by the server, never sent by the client.

Queue entry status (`queue_entries.status`, enum `QueueEntryStatus` - a separate lifecycle from prescriptions)

From	Action	To / result	Who
(new)	create booking	WAITING	DOCTOR, PHARMACIST
WAITING	call-next / call	SERVING	DOCTOR (session OPEN)
WAITING	cancel (reason)	CANCELLED	DOCTOR, PHARMACIST
SERVING	complete / call-next completeCurrentEntryId	COMPLETED	DOCTOR
SERVING	skip	SKIPPED	DOCTOR
SKIPPED	requeue	WAITING (same number)	DOCTOR, PHARMACIST
WAITING	session close force=true	CANCELLED ("Session closed")	system
SERVING	session close force=true	COMPLETED	system
same as target	repeat call / complete / skip / cancel / requeue	unchanged (no-op, 200)	-
any other	any	409 INVALID_STATUS_TRANSITION	-

Queue session status (`queue_sessions.status`, enum `QueueSessionStatus`)

From	Action	To / result	Who
(new)	create	SCHEDULED	DOCTOR, PHARMACIST
SCHEDULED	open (only on sessionDate)	OPEN	DOCTOR, PHARMACIST
SCHEDULED, OPEN	close (force if active entries)	CLOSED	DOCTOR, PHARMACIST, nightly job
OPEN	open (repeat)	OPEN (no-op)	-
CLOSED	any mutation or booking	409 QUEUE_SESSION_CLOSED	-

2.11 How the required special cases are handled

Case	Behaviour
Validation errors	`400 VALIDATION_FAILED`; `errors[]` lists every failing field with JSON path, constraint code and message.
Duplicate patient identifier	`409 DUPLICATE_PATIENT_NUMBER` with `existingPatientId` so the UI can open the existing record; enforced by `uq_patients_patient_number` even under concurrent registration.
Missing patient	`GET /patients/by-number/{n}` → `404 PATIENT_NOT_FOUND` (UI offers registration); unknown `patientId` in a body → `422 REFERENCED_RESOURCE_NOT_FOUND`.
Duplicate requests (double click, retry)	Creates: Idempotency-Key replay. Transitions: same-state no-op (200) or `409 INVALID_STATUS_TRANSITION`. Call-next: compare-and-set (`409 QUEUE_STATE_CHANGED`). Bookings by phone: `409 DUPLICATE_BOOKING` unless confirmed.
Unavailable medicines	Catalogue medicine deactivated → cannot be prescribed (`422 MEDICINE_INACTIVE`); at dispensing time the pharmacist records `quantityDispensed` 0 or partial with a note (outcome `PARTIAL`); if nothing can be supplied the prescription is cancelled. There is no stock check (no inventory).
Invalid queue transitions	`409 INVALID_STATUS_TRANSITION` with `currentStatus`; session not open `409 QUEUE_SESSION_NOT_OPEN`; closed `409 QUEUE_SESSION_CLOSED`; full `409 QUEUE_SESSION_FULL`.
Queue-only consumers	Public and queue endpoints return queue DTOs only; they never include medical history, patient ids or phone numbers (public) - BR-Q-10.

3. Error format and error codes

3.1 Problem Details

Every error body follows RFC 9457 with DMS extensions:

```
{
  "type": "https://dms.example/problems/validation-failed",
  "title": "Validation failed",
  "status": 400,
  "detail": "One or more fields are invalid.",
  "instance": "/api/v1/patients",
  "code": "VALIDATION_FAILED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {"field": "fullName", "code": "NotBlank", "message": "must not be blank"},
    {"field": "dateOfBirth", "code": "PastOrPresent", "message": "must not be in the future"}
  ]
}
```

`code` is stable and is what clients switch on. Context fields appear only for specific codes: `existingPatientId` (DUPLICATE_PATIENT_NUMBER), `existingQueueNumber` (DUPLICATE_BOOKING), `currentStatus` (INVALID_STATUS_TRANSITION), `currentServingNumber` (QUEUE_STATE_CHANGED), `waitingCount` / `servingCount` (QUEUE_SESSION_HAS_ACTIVE_ENTRIES), `lockedUntil` (ACCOUNT_LOCKED), `retryAfterSeconds` (RATE_LIMITED). `rejectedValue` is never included for passwords or clinical free text.

3.2 Error code catalogue

Code	HTTP	Title	When it is returned
`MALFORMED_REQUEST`	400	Malformed request	Body is not valid JSON, has wrong JSON types, or an unknown enum value.
`VALIDATION_FAILED`	400	Validation failed	Bean Validation (field or cross-field) failed for the body, path or query parameters. errors[] lists every failing field.
`IDEMPOTENCY_KEY_REQUIRED`	400	Idempotency key required	A POST that requires the Idempotency-Key header was sent without it, or the key does not match ^[A-Za-z0-9_-]{8,100}$.
`UNAUTHENTICATED`	401	Authentication required	No bearer token, or the token is malformed or has an invalid signature.
`TOKEN_EXPIRED`	401	Access token expired	The bearer token is well-formed but expired. Client should call POST /auth/refresh once and retry.
`INVALID_CREDENTIALS`	401	Invalid credentials	Unknown username or wrong password (same response for both).
`ACCOUNT_LOCKED`	401	Account temporarily locked	Correct password but the account is locked after 5 consecutive failures. Returned only when the password is correct, so it cannot be used to enumerate users.
`ACCOUNT_DISABLED`	401	Account disabled	Correct password but the account is disabled.
`REFRESH_TOKEN_INVALID`	401	Session expired	Refresh cookie missing, unknown, expired or revoked. Client must show the login screen.
`REFRESH_TOKEN_ROTATED`	401	Refresh token already rotated	Refresh token was rotated less than 30 s ago (parallel refresh from another tab). The browser already holds the new cookie; client retries refresh once after 500 ms.
`SESSION_REVOKED`	401	Session revoked	A rotated refresh token was reused after the 30 s grace window (possible theft). The whole token family is revoked.
`FORBIDDEN`	403	Access denied	Authenticated user lacks the role required by the endpoint.
`PASSWORD_CHANGE_REQUIRED`	403	Password change required	users.must_change_password is TRUE; only GET /auth/me, POST /auth/change-password and POST /auth/logout are allowed.
`NOT_RESOURCE_OWNER`	403	Not the owner of this resource	Role is sufficient but an ownership rule fails, e.g. a doctor cancels another doctor's prescription.
`RESOURCE_NOT_FOUND`	404	Resource not found	Unknown route or generic missing resource.
`PATIENT_NOT_FOUND`	404	Patient not found	No patient with the given id or patient number. For lookup by number the UI offers registration.
`CONSULTATION_NOT_FOUND`	404	Consultation not found	No consultation with the given id.
`PRESCRIPTION_NOT_FOUND`	404	Prescription not found	No prescription with the given id.
`QUEUE_SESSION_NOT_FOUND`	404	Queue session not found	No queue session with the given id.
`QUEUE_ENTRY_NOT_FOUND`	404	Queue entry not found	No queue entry with the given id.
`USER_NOT_FOUND`	404	User not found	No user with the given id.

Code	HTTP	Title	When it is returned
`MEDICINE_NOT_FOUND`	404	Medicine not found	No medicine with the given id (path parameter).
`DUPLICATE_PATIENT_NUMBER`	409	Patient number already exists	uq_patients_patient_number violated. errors[0].field = patientNumber; existingPatientId is returned so the UI can open that record.
`DUPLICATE_USERNAME`	409	Username already exists	uq_users_username violated.
`DUPLICATE_MEDICINE`	409	Medicine already exists	uq_medicines_name_strength_form violated.
`DUPLICATE_QUEUE_SESSION`	409	Queue session already exists	uq_queue_sessions_date_name violated.
`DUPLICATE_BOOKING`	409	Possible duplicate booking	An active (WAITING/SERVING) entry in the same session has the same contact phone and allowDuplicate was not true. existingQueueNumber is returned.
`CONCURRENT_MODIFICATION`	409	Record changed by someone else	Optimistic-lock version in the request does not match (JPA OptimisticLockException). Client reloads and re-applies.
`INVALID_STATUS_TRANSITION`	409	Invalid status transition	Requested transition is not allowed from the current status (prescription or queue entry). currentStatus is returned.
`QUEUE_STATE_CHANGED`	409	Queue changed	call-next expectedLastCalledNumber differs from the session's last_called_number (someone else already called). currentServingNumber is returned; UI refreshes.
`QUEUE_SESSION_NOT_OPEN`	409	Queue session not open	Calling, completing or skipping requires status OPEN.
`QUEUE_SESSION_CLOSED`	409	Queue session closed	Booking or any mutation on a CLOSED session.
`QUEUE_SESSION_FULL`	409	Queue session full	last_issued_number = 999.
`QUEUE_SESSION_HAS_ACTIVE_ENTRIES`	409	Queue session has active entries	close without force=true while WAITING or SERVING entries exist. waitingCount and servingCount are returned.
`QUEUE_ENTRY_ALREADY_LINKED`	409	Queue entry already linked	uq_consultations_queue_entry violated: another consultation already references the queue entry.
`ACTIVE_PRESCRIPTION_EXISTS`	409	Active prescription exists	uq_prescriptions_active_per_consultation violated: the consultation already has a non-cancelled prescription.
`REFERENCED_RESOURCE_NOT_FOUND`	422	Referenced resource not found	An id inside the request body (patientId, queueEntryId, medicineId, assignedDoctorId, prescriptionItemId, supersedesPrescriptionId) does not exist. errors[0].field names it.
`MEDICINE_INACTIVE`	422	Medicine is not available	A prescription item references a catalogue medicine with active = false.
`CONSULTATION_LOCKED`	422	Consultation locked	A prescription is added to a consultation more than 24 h after consultation_at. (A different doctor gets 403 NOT_RESOURCE_OWNER.)
`INVALID_SUPERSEDED_PRESCRIPTION`	422	Invalid superseded prescription	supersedesPrescriptionId is not a CANCELLED prescription of the same consultation, or is already superseded.

Code	HTTP	Title	When it is returned
`QUEUE_ENTRY_NOT_IN_SESSION`	422	Queue entry not in session	completeCurrentEntryId (call-next) or queueEntryId (consultation) refers to an entry of a different session or a non-SERVING entry where SERVING is required.
`DISPENSE_ITEMS_MISMATCH`	422	Dispense items do not match	items[] must contain every prescription item of the prescription exactly once.
`DISPENSE_QUANTITY_EXCEEDS_PRESCRIBED`	422	Quantity exceeds prescribed	quantityDispensed > prescribed quantity for an item.
`NOTHING_DISPENSED`	422	Nothing dispensed	Every item has quantityDispensed = 0. Cancel the prescription instead.
`DISPENSE_NOTE_REQUIRED`	422	Note required	An item with quantityDispensed below the prescribed quantity has no note.
`INVALID_SESSION_DATE`	422	Invalid session date	sessionDate is before today or more than 30 days ahead (APP_TIMEZONE).
`ASSIGNED_USER_NOT_DOCTOR`	422	Assigned user is not a doctor	assignedDoctorId refers to a user without the DOCTOR role or a disabled user.
`SELF_MODIFICATION_NOT_ALLOWED`	422	Cannot change own account this way	An admin tries to disable themselves or remove their own ADMIN role.
`LAST_ADMIN_REQUIRED`	422	At least one admin required	The change would leave no enabled user with the ADMIN role.
`CURRENT_PASSWORD_INCORRECT`	422	Current password incorrect	change-password currentPassword is wrong (422, not 401, so the client does not trigger token refresh).
`PASSWORD_POLICY_VIOLATION`	422	Password does not meet policy	New password equals the current password, contains the username, or is on the deny-list.
`IDEMPOTENCY_KEY_REUSED`	422	Idempotency key reused	The same Idempotency-Key was used by the same user for a different operation or a different request body within 24 h.
`RATE_LIMITED`	429	Too many requests	Rate limit exceeded (login 10/min/IP, public queue 120/min/IP). Retry-After header is set.
`INTERNAL_ERROR`	500	Internal server error	Unexpected exception. Details are logged server-side with requestId; nothing internal is returned.
`SERVICE_UNAVAILABLE`	503	Service unavailable	Database unreachable, a row lock could not be obtained within lock_timeout (5 s; client may retry), or the application is shutting down.

3.3 Standard error examples

These bodies apply to every endpoint whose response table lists the code.

400 VALIDATION_FAILED

```
{
  "type": "https://dms.example/problems/validation-failed",
  "title": "Validation failed",
  "status": 400,
  "detail": "One or more fields are invalid.",
```

```
  "instance": "/api/v1/auth/login",
  "code": "VALIDATION_FAILED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "fullName",
      "code": "NotBlank",
      "message": "must not be blank"
    }
  ]
}
```

400 MALFORMED_REQUEST

```
{
  "type": "https://dms.example/problems/malformed-request",
  "title": "Malformed request",
  "status": 400,
  "detail": "Request body could not be parsed.",
  "instance": "/api/v1/auth/login",
  "code": "MALFORMED_REQUEST",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

401 UNAUTHENTICATED

```
{
  "type": "https://dms.example/problems/unauthenticated",
  "title": "Authentication required",
  "status": 401,
  "detail": "A valid access token is required.",
  "instance": "/api/v1/auth/me",
  "code": "UNAUTHENTICATED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

401 TOKEN_EXPIRED

```
{
  "type": "https://dms.example/problems/token-expired",
  "title": "Access token expired",
  "status": 401,
  "detail": "The access token has expired.",
  "instance": "/api/v1/auth/me",
  "code": "TOKEN_EXPIRED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

403 FORBIDDEN

```
{
  "type": "https://dms.example/problems/forbidden",
  "title": "Access denied",
  "status": 403,
  "detail": "You do not have permission to perform this action.",
  "instance": "/api/v1/users",
  "code": "FORBIDDEN",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

403 PASSWORD_CHANGE_REQUIRED

```
{
  "type": "https://dms.example/problems/password-change-required",
  "title": "Password change required",
```

```
  "status": 403,
  "detail": "You must change your password before continuing.",
  "instance": "/api/v1/users",
  "code": "PASSWORD_CHANGE_REQUIRED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


400 IDEMPOTENCY_KEY_REQUIRED


```
{
  "type": "https://dms.example/problems/idempotency-key-required",
  "title": "Idempotency key required",
  "status": 400,
  "detail": "Header Idempotency-Key is required for this operation.",
  "instance": "/api/v1/patients",
  "code": "IDEMPOTENCY_KEY_REQUIRED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


422 IDEMPOTENCY_KEY_REUSED


```
{
  "type": "https://dms.example/problems/idempotency-key-reused",
  "title": "Idempotency key reused",
  "status": 422,
  "detail": "Idempotency-Key was already used with a different request.",
  "instance": "/api/v1/patients",
  "code": "IDEMPOTENCY_KEY_REUSED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


500 INTERNAL_ERROR


```
{
  "type": "https://dms.example/problems/internal-error",
  "title": "Internal server error",
  "status": 500,
  "detail": "An unexpected error occurred. Quote request id 7f3c... when reporting.",
  "instance": "/api/v1/auth/login",
  "code": "INTERNAL_ERROR",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

4. Endpoint index and permissions

4.1 Endpoint index

Resource	Method	Path	Purpose	Roles	operationId
Authentication	POST	`/api/v1/auth/login`	Log in with username and password	PUBLIC	`login`
Authentication	POST	`/api/v1/auth/refresh`	Exchange the refresh cookie for a new access token	PUBLIC	`refreshToken`
Authentication	POST	`/api/v1/auth/logout`	Log out and revoke the refresh-token family	PUBLIC	`logout`
Authentication	GET	`/api/v1/auth/me`	Get the authenticated user	ANY_STAFF	`getCurrentUser`
Authentication	POST	`/api/v1/auth/change-password`	Change own password	ANY_STAFF	`changePassword`
Users and staff	GET	`/api/v1/users`	List staff accounts	ADMIN	`listUsers`
Users and staff	POST	`/api/v1/users`	Create a staff account	ADMIN	`createUser`
Users and staff	GET	`/api/v1/users/{userId}`	Get a staff account	ADMIN	`getUser`
Users and staff	PUT	`/api/v1/users/{userId}`	Update a staff account (name, roles, enabled, registration number)	ADMIN	`updateUser`
Users and staff	POST	`/api/v1/users/{userId}/reset-password`	Set a temporary password for a user	ADMIN	`resetUserPassword`
Users and staff	GET	`/api/v1/staff`	Minimal directory of enabled staff (e.g. doctors for queue-session assignment)	ADMIN, DOCTOR, PHARMACIST	`listStaff`
Patients	GET	`/api/v1/patients`	Search patients by name fragment, phone number or patient-number prefix	DOCTOR	`searchPatients`
Patients	POST	`/api/v1/patients`	Register a new patient	DOCTOR	`createPatient`
Patients	GET	`/api/v1/patients/by-number/{patientNumber}`	Look up a patient by patient number (doctor's patient-ID search)	DOCTOR	`getPatientByNumber`
Patients	GET	`/api/v1/patients/{patientId}`	Get patient demographics	DOCTOR	`getPatient`
Patients	PUT	`/api/v1/patients/{patientId}`	Update patient demographics and allergies	DOCTOR	`updatePatient`
Medical history	GET	`/api/v1/patients/{patientId}/medical-history`	Get a patient's medical history (conditions summary + paginated consultations)	DOCTOR	`getMedicalHistory`
Consultations	POST	`/api/v1/consultations`	Create a consultation with diagnoses and an optional prescription (one transaction)	DOCTOR	`createConsultation`

Resource	Method	Path	Purpose	Roles	operationId
Consultations	GET	`/api/v1/consultations/{consultationId}`	Get a consultation with diagnoses, prescriptions and dispensing status	DOCTOR	`getConsultation`
Prescriptions	POST	`/api/v1/consultations/{consultationId}/prescriptions`	Add (or re-issue) a prescription for an existing consultation	DOCTOR	`addPrescriptionToConsultation`
Prescriptions	GET	`/api/v1/prescriptions`	List prescriptions (pharmacist queue / doctor's own prescriptions)	PHARMACIST, DOCTOR	`listPrescriptions`
Prescriptions	GET	`/api/v1/prescriptions/{prescriptionId}`	Get a prescription with items, minimal patient data and dispensing result	PHARMACIST, DOCTOR	`getPrescription`
Pharmacy and dispensing	POST	`/api/v1/prescriptions/{prescriptionId}/start`	Mark a prescription IN_PROGRESS (pharmacist has started preparing it)	PHARMACIST	`startPrescription`
Pharmacy and dispensing	POST	`/api/v1/prescriptions/{prescriptionId}/dispense`	Confirm dispensing ("Mark as Dispensed") with per-item quantities	PHARMACIST	`dispensePrescription`
Prescriptions	POST	`/api/v1/prescriptions/{prescriptionId}/cancel`	Cancel a prescription that has not been dispensed	DOCTOR, PHARMACIST	`cancelPrescription`
Medicines	GET	`/api/v1/medicines`	Search the medicine catalogue (prescription autocomplete and catalogue management)	DOCTOR, PHARMACIST, ADMIN	`listMedicines`
Medicines	POST	`/api/v1/medicines`	Add a medicine to the catalogue	PHARMACIST, ADMIN	`createMedicine`
Medicines	GET	`/api/v1/medicines/{medicineId}`	Get a catalogue medicine	DOCTOR, PHARMACIST, ADMIN	`getMedicine`
Medicines	PUT	`/api/v1/medicines/{medicineId}`	Update or (de)activate a catalogue medicine	PHARMACIST, ADMIN	`updateMedicine`
Queue management	GET	`/api/v1/queue-sessions`	List queue sessions for a date	DOCTOR, PHARMACIST	`listQueueSessions`
Queue management	POST	`/api/v1/queue-sessions`	Create a queue session (e.g. today's clinic for a room/doctor)	DOCTOR, PHARMACIST	`createQueueSession`
Queue management	GET	`/api/v1/queue-sessions/{queueSessionId}`	Get queue session status for staff (counts, now serving, next number)	DOCTOR, PHARMACIST	`getQueueSession`
Queue management	POST	`/api/v1/queue-sessions/{queueSessionId}/open`	Open a scheduled session so patients can be called	DOCTOR, PHARMACIST	`openQueueSession`
Queue management	POST	`/api/v1/queue-sessions/{queueSessionId}/close`	Close a session (end of clinic)	DOCTOR, PHARMACIST	`closeQueueSession`
Bookings	GET	`/api/v1/queue-sessions/{queueSessionId}/entries`	List bookings / queue entries of a session (staff only)	DOCTOR, PHARMACIST	`listQueueEntries`

Resource	Method	Path	Purpose	Roles	operationId
Bookings	POST	`/api/v1/queue-sessions/{queueSessionId}/entries`	Register a telephone or walk-in booking and assign the next queue number	DOCTOR, PHARMACIST	`createBooking`
Queue management	POST	`/api/v1/queue-sessions/{queueSessionId}/call-next`	Call the next waiting patient (the doctor's "Next patient" button)	DOCTOR	`callNextPatient`
Bookings	GET	`/api/v1/queue-entries/{queueEntryId}`	Get one booking / queue entry	DOCTOR, PHARMACIST	`getQueueEntry`
Queue management	POST	`/api/v1/queue-entries/{queueEntryId}/call`	Call a specific waiting entry out of order (e.g. emergency or priority patient)	DOCTOR	`callQueueEntry`
Queue management	POST	`/api/v1/queue-entries/{queueEntryId}/complete`	Mark the served entry as completed without calling the next one	DOCTOR	`completeQueueEntry`
Queue management	POST	`/api/v1/queue-entries/{queueEntryId}/skip`	Mark a called patient as not present (no-show)	DOCTOR	`skipQueueEntry`
Queue management	POST	`/api/v1/queue-entries/{queueEntryId}/requeue`	Return a skipped patient to the waiting list (arrived late)	DOCTOR, PHARMACIST	`requeueQueueEntry`
Bookings	POST	`/api/v1/queue-entries/{queueEntryId}/cancel`	Cancel a waiting booking (patient called to cancel)	DOCTOR, PHARMACIST	`cancelBooking`
Public queue display	GET	`/api/v1/public/queue-sessions`	Today's queue sessions (public, no authentication)	PUBLIC	`listPublicQueueSessions`
Public queue display	GET	`/api/v1/public/queue-sessions/{queueSessionId}`	Privacy-safe live status of one queue (waiting-room display and patients' phones)	PUBLIC	`getPublicQueueStatus`
Audit	GET	`/api/v1/audit-events`	Search the audit trail	ADMIN	`listAuditEvents`
Health checks	GET	`/actuator/health`	Aggregate health (UP/DOWN); no component details are exposed	PUBLIC	`health`
Health checks	GET	`/actuator/health/liveness`	Liveness probe (process is running; does not check the database)	PUBLIC	`healthLiveness`
Health checks	GET	`/actuator/health/readiness`	Readiness probe (includes database connectivity)	PUBLIC	`healthReadiness`

4.2 Permission matrix

Endpoint (under /api/v1 unless /actuator)	operationId	ADMIN	DOCTOR	PHARMACIST
POST `/auth/login`	`login`	public	public	public

Endpoint (under /api/v1 unless /actuator)	operationId	ADMIN	DOCTOR	PHARMACIST
POST `/auth/refresh`	`refreshToken`	public	public	public
POST `/auth/logout`	`logout`	public	public	public
GET `/auth/me`	`getCurrentUser`	yes	yes	yes
POST `/auth/change-password`	`changePassword`	yes	yes	yes
GET `/users`	`listUsers`	yes	-	-
POST `/users`	`createUser`	yes	-	-
GET `/users/{userId}`	`getUser`	yes	-	-
PUT `/users/{userId}`	`updateUser`	yes	-	-
POST `/users/{userId}/reset-password`	`resetUserPassword`	yes	-	-
GET `/staff`	`listStaff`	yes	yes	yes
GET `/patients`	`searchPatients`	-	yes	-
POST `/patients`	`createPatient`	-	yes	-
GET `/patients/by-number/{patientNumber}`	`getPatientByNumber`	-	yes	-
GET `/patients/{patientId}`	`getPatient`	-	yes	-
PUT `/patients/{patientId}`	`updatePatient`	-	yes	-
GET `/patients/{patientId}/medical-history`	`getMedicalHistory`	-	yes	-
POST `/consultations`	`createConsultation`	-	yes	-
GET `/consultations/{consultationId}`	`getConsultation`	-	yes	-
POST `/consultations/{consultationId}/prescriptions`	`addPrescriptionToConsultation`	-	yes	-
GET `/prescriptions`	`listPrescriptions`	-	yes	yes
GET `/prescriptions/{prescriptionId}`	`getPrescription`	-	yes	yes
POST `/prescriptions/{prescriptionId}/start`	`startPrescription`	-	-	yes
POST `/prescriptions/{prescriptionId}/dispense`	`dispensePrescription`	-	-	yes
POST `/prescriptions/{prescriptionId}/cancel`	`cancelPrescription`	-	yes	yes
GET `/medicines`	`listMedicines`	yes	yes	yes
POST `/medicines`	`createMedicine`	yes	-	yes
GET `/medicines/{medicineId}`	`getMedicine`	yes	yes	yes
PUT `/medicines/{medicineId}`	`updateMedicine`	yes	-	yes
GET `/queue-sessions`	`listQueueSessions`	-	yes	yes
POST `/queue-sessions`	`createQueueSession`	-	yes	yes
GET `/queue-sessions/{queueSessionId}`	`getQueueSession`	-	yes	yes
POST `/queue-sessions/{queueSessionId}/open`	`openQueueSession`	-	yes	yes

Endpoint (under /api/v1 unless /actuator)	operationId	ADMIN	DOCTOR	PHARMACIST
POST `/queue-sessions/{queueSessionId}/close`	`closeQueueSession`	-	yes	yes
GET `/queue-sessions/{queueSessionId}/entries`	`listQueueEntries`	-	yes	yes
POST `/queue-sessions/{queueSessionId}/entries`	`createBooking`	-	yes	yes
POST `/queue-sessions/{queueSessionId}/call-next`	`callNextPatient`	-	yes	-
GET `/queue-entries/{queueEntryId}`	`getQueueEntry`	-	yes	yes
POST `/queue-entries/{queueEntryId}/call`	`callQueueEntry`	-	yes	-
POST `/queue-entries/{queueEntryId}/complete`	`completeQueueEntry`	-	yes	-
POST `/queue-entries/{queueEntryId}/skip`	`skipQueueEntry`	-	yes	-
POST `/queue-entries/{queueEntryId}/requeue`	`requeueQueueEntry`	-	yes	yes
POST `/queue-entries/{queueEntryId}/cancel`	`cancelBooking`	-	yes	yes
GET `/public/queue-sessions`	`listPublicQueueSessions`	public	public	public
GET `/public/queue-sessions/{queueSessionId}`	`getPublicQueueStatus`	public	public	public
GET `/audit-events`	`listAuditEvents`	yes	-	-
GET `/actuator/health`	`health`	public	public	public
GET `/actuator/health/liveness`	`healthLiveness`	public	public	public
GET `/actuator/health/readiness`	`healthReadiness`	public	public	public

5. Authentication

Login, token refresh, logout, current user, password change.

5.1 POST /api/v1/auth/login

Item	Value
Method / path	`POST /api/v1/auth/login`
operationId	`login`
Purpose	Log in with username and password. Verifies credentials, issues a 15-minute access token in the body and a refresh token in the `dms_refresh` HttpOnly cookie.
Required role(s)	Public - no role
Authentication	None (public)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters: none.

Request body - schema `LoginRequest` (required)

```
{
  "username": "dr.perera",
  "password": "DevPassword#2026"
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`username`	string	yes	min length 1; max length 50	
`password`	string (password)	yes	min length 1; max length 128	

Business rules

- Username is trimmed and lower-cased before lookup.
- Unknown user and wrong password both return 401 INVALID_CREDENTIALS after a comparable BCrypt delay (dummy hash check) to prevent user enumeration.
- ACCOUNT_LOCKED / ACCOUNT_DISABLED are returned only when the password is correct.
- 5 consecutive failures set locked_until = now + 15 min and reset the counter; success resets failed_login_count to 0 and sets last_login_at.
- A new refresh-token family is created per login; the family expires after APP_JWT_REFRESH_TOKEN_TTL (default 12 h).
- When mustChangePassword is true the client must route to the change-password screen; other endpoints return 403 PASSWORD_CHANGE_REQUIRED.

Responses

Status	Meaning / error codes	Body
200	Authenticated.	TokenResponse
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`	Problem
401	`INVALID_CREDENTIALS`, `ACCOUNT_LOCKED`, `ACCOUNT_DISABLED`	Problem
429	`RATE_LIMITED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)

```
{
  "accessToken":
"eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiIwMDAwMDAwMC0wMDAwLTQwMDAtODAwMC0wMDAwMDAwMDAwMDIifQ.sig",
  "tokenType": "Bearer",
  "expiresIn": 900,
  "user": {
    "id": "00000000-0000-4000-8000-000000000002",
    "username": "dr.perera",
```

```
    "fullName": "Dr. Nimal Perera",
    "roles": [
      "DOCTOR"
    ],
    "mustChangePassword": false,
    "professionalRegistrationNumber": "MC-12345"
  }
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

401 INVALID_CREDENTIALS


```
{
  "type": "https://dms.example/problems/invalid-credentials",
  "title": "Invalid credentials",
  "status": 401,
  "detail": "Username or password is incorrect.",
  "instance": "/api/v1/auth/login",
  "code": "INVALID_CREDENTIALS",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


401 ACCOUNT_LOCKED


```
{
  "type": "https://dms.example/problems/account-locked",
  "title": "Account temporarily locked",
  "status": 401,
  "detail": "Account is locked until 2026-10-12T09:15:00Z.",
  "instance": "/api/v1/auth/login",
  "code": "ACCOUNT_LOCKED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "lockedUntil": "2026-10-12T09:15:00Z"
}
```


401 ACCOUNT_DISABLED


```
{
  "type": "https://dms.example/problems/account-disabled",
  "title": "Account disabled",
  "status": 401,
  "detail": "This account has been disabled. Contact an administrator.",
  "instance": "/api/v1/auth/login",
  "code": "ACCOUNT_DISABLED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


429 RATE_LIMITED


```
{
  "type": "https://dms.example/problems/rate-limited",
  "title": "Too many requests",
  "status": 429,
  "detail": "Too many requests. Retry after 30 seconds.",
  "instance": "/api/v1/auth/login",
  "code": "RATE_LIMITED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "retryAfterSeconds": 30
}
```

Aspect	Details
Database entities affected	users (R/W), user_roles (R), refresh_tokens (W), audit_events (W)
Transaction / concurrency	Failed-login counter is updated with a single UPDATE ... SET failed_login_count = failed_login_count + 1 statement to avoid lost updates.
Audit events	`AUTH_LOGIN_SUCCEEDED`, `AUTH_LOGIN_FAILED`, `AUTH_ACCOUNT_LOCKED`

5.2 POST /api/v1/auth/refresh

Item	Value
Method / path	`POST /api/v1/auth/refresh`
operationId	`refreshToken`
Purpose	Exchange the refresh cookie for a new access token. Rotates the refresh token (old one is marked rotated, a new cookie is set) and returns a new access token. Called on page load and whenever an API call returns 401 TOKEN_EXPIRED.
Required role(s)	Public - no role
Authentication	Refresh cookie `dms_refresh` + header `X-Requested-With: XMLHttpRequest` (no bearer token)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Accept	no	application/json (errors: application/problem+json)
X-Requested-With	yes	Must be `XMLHttpRequest`. Cross-site forms cannot set it, which (with SameSite=Strict) protects the cookie-authenticated endpoints against CSRF.
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.
Cookie	yes	dms_refresh=<opaque refresh token>

Path and query parameters: none.

Request body: none.

Business rules

- Token is looked up by SHA-256 hash; revoked or expired tokens → 401 REFRESH_TOKEN_INVALID and the cookie is cleared.
- A token already rotated less than 30 s ago → 401 REFRESH_TOKEN_ROTATED (no revocation; parallel tab race). Client retries once after 500 ms.
- A token rotated more than 30 s ago → the whole family is revoked (REUSE_DETECTED) → 401 SESSION_REVOKED.
- Disabled users → family revoked (USER_DISABLED), 401 ACCOUNT_DISABLED.
- The new token keeps the family's absolute expiry; the cookie Max-Age is the remaining lifetime.
- Roles are re-read from the database, so role changes take effect at the next refresh (<= 15 min).

Responses

Status	Meaning / error codes	Body
200	New access token issued.	TokenResponse

Status	Meaning / error codes	Body
401	`REFRESH_TOKEN_INVALID`, `REFRESH_TOKEN_ROTATED`, `SESSION_REVOKED`, `ACCOUNT_DISABLED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "accessToken":
"eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiIwMDAwMDAwMC0wMDAwLTQwMDAtODAwMC0wMDAwMDAwMDAwMDQifQ.sig",
  "tokenType": "Bearer",
  "expiresIn": 900,
  "user": {
    "id": "00000000-0000-4000-8000-000000000004",
    "username": "ph.silva",
    "fullName": "Kamal Silva",
    "roles": [
      "PHARMACIST"
    ],
    "mustChangePassword": false,
    "professionalRegistrationNumber": "PH-3456"
  }
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

401 REFRESH_TOKEN_INVALID


```
{
  "type": "https://dms.example/problems/refresh-token-invalid",
  "title": "Session expired",
  "status": 401,
  "detail": "Your session has expired. Please sign in again.",
  "instance": "/api/v1/auth/refresh",
  "code": "REFRESH_TOKEN_INVALID",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


401 REFRESH_TOKEN_ROTATED


```
{
  "type": "https://dms.example/problems/refresh-token-rotated",
  "title": "Refresh token already rotated",
  "status": 401,
  "detail": "Refresh token was already used; retry.",
  "instance": "/api/v1/auth/refresh",
  "code": "REFRESH_TOKEN_ROTATED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


401 SESSION_REVOKED


```
{
  "type": "https://dms.example/problems/session-revoked",
  "title": "Session revoked",
  "status": 401,
  "detail": "This session was revoked for security reasons. Please sign in again.",
  "instance": "/api/v1/auth/refresh",
  "code": "SESSION_REVOKED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


401 ACCOUNT_DISABLED

```
{
  "type": "https://dms.example/problems/account-disabled",
  "title": "Account disabled",
  "status": 401,
  "detail": "This account has been disabled. Contact an administrator.",
  "instance": "/api/v1/auth/refresh",
  "code": "ACCOUNT_DISABLED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

Aspect	Details
Database entities affected	refresh_tokens (R/W), users (R), user_roles (R), audit_events (W, reuse only)
Transaction / concurrency	The presented token row is locked with SELECT … FOR UPDATE so two simultaneous refreshes with the same token cannot both rotate it.
Audit events	`AUTH_REFRESH_REUSE_DETECTED`

5.3 POST /api/v1/auth/logout

Item	Value
Method / path	`POST /api/v1/auth/logout`
operationId	`logout`
Purpose	Log out and revoke the refresh-token family. Revokes the refresh-token family of the presented cookie and clears the cookie. Always returns 204 (also when the cookie is missing or already invalid), so logout is idempotent.
Required role(s)	Public - no role
Authentication	Refresh cookie `dms_refresh` + header `X-Requested-With: XMLHttpRequest` (no bearer token)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Accept	no	application/json (errors: application/problem+json)
X-Requested-With	yes	Must be `XMLHttpRequest`. Cross-site forms cannot set it, which (with SameSite=Strict) protects the cookie-authenticated endpoints against CSRF.
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.
Cookie	yes	dms_refresh=<opaque refresh token>

Path and query parameters: none.

Request body: none.

Business rules

- The access token is not required; it simply expires (max 15 min). The frontend discards it from memory immediately.

Responses

Status	Meaning / error codes	Body
204	Logged out; `Set-Cookie` clears dms_refresh (Max-Age=0).	(no body)

Status	Meaning / error codes	Body
500	`INTERNAL_ERROR`	Problem

Error response example (500 INTERNAL_ERROR; all codes for this endpoint are listed under Responses)


```
{
  "type": "https://dms.example/problems/internal-error",
  "title": "Internal server error",
  "status": 500,
  "detail": "An unexpected error occurred. Quote request id 7f3c... when reporting.",
  "instance": "/api/v1/auth/logout",
  "code": "INTERNAL_ERROR",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


Aspect	Details
Database entities affected	refresh_tokens (W), audit_events (W)
Transaction / concurrency	Single @Transactional service call.
Audit events	`AUTH_LOGOUT`

5.4 GET /api/v1/auth/me

Item	Value
Method / path	`GET /api/v1/auth/me`
operationId	`getCurrentUser`
Purpose	Get the authenticated user
Required role(s)	Any authenticated staff user (also while a password change is pending)
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters: none.

Request body: none.

Responses

Status	Meaning / error codes	Body
200	Current user.	CurrentUser
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "00000000-0000-4000-8000-000000000005",
  "username": "dr.jayasinghe",
  "fullName": "Dr. Ruwan Jayasinghe",
  "roles": [
    "DOCTOR",
    "PHARMACIST"
  ],
  "mustChangePassword": false,
  "professionalRegistrationNumber": "MC-34567"
}
```


Error response example (401 UNAUTHENTICATED; all codes for this endpoint are listed under Responses)


```
{
  "type": "https://dms.example/problems/unauthenticated",
  "title": "Authentication required",
  "status": 401,
  "detail": "A valid access token is required.",
  "instance": "/api/v1/auth/me",
  "code": "UNAUTHENTICATED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


Aspect	Details
Database entities affected	users (R), user_roles (R)
Transaction / concurrency	Read-only, single transaction.
Audit events	none

5.5 POST /api/v1/auth/change-password

Item	Value
Method / path	`POST /api/v1/auth/change-password`
operationId	`changePassword`
Purpose	Change own password. Changes the caller's password, clears mustChangePassword and revokes all of the caller's other refresh-token families. The current family stays valid so the user remains logged in.
Required role(s)	Any authenticated staff user (also while a password change is pending)
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters: none.

Request body - schema `ChangePasswordRequest` (required)

```
{
  "currentPassword": "Temp-Password-2026",
  "newPassword": "correct horse battery staple"
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`currentPassword`	string (password)	yes	min length 1; max length 128	
`newPassword`	string (password)	yes	min length 12; max length 128	

Business rules

- newPassword must be 12-128 characters, differ from the current password, not contain the username (case-insensitive) and not be on the deny-list (top 10 000 common passwords bundled as a resource).
- Wrong currentPassword returns 422 (not 401) and counts towards the lockout counter.

Responses

Status	Meaning / error codes	Body
204	Password changed.	(no body)
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
422	`CURRENT_PASSWORD_INCORRECT`, `PASSWORD_POLICY_VIOLATION`	Problem
500	`INTERNAL_ERROR`	Problem

Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

422 CURRENT_PASSWORD_INCORRECT

```
{
  "type": "https://dms.example/problems/current-password-incorrect",
  "title": "Current password incorrect",
  "status": 422,
  "detail": "Current password is incorrect.",
  "instance": "/api/v1/auth/change-password",
  "code": "CURRENT_PASSWORD_INCORRECT",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

422 PASSWORD_POLICY_VIOLATION

```
{
  "type": "https://dms.example/problems/password-policy-violation",
  "title": "Password does not meet policy",
  "status": 422,
  "detail": "Password must not contain the username.",
  "instance": "/api/v1/auth/change-password",
  "code": "PASSWORD_POLICY_VIOLATION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "newPassword",
      "code": "PasswordPolicy",
      "message": "must not contain the username"
```

```
    }
  ]
}
```

Aspect	Details
Database entities affected	users (W), refresh_tokens (W), audit_events (W)
Transaction / concurrency	Single @Transactional service call.
Audit events	`AUTH_PASSWORD_CHANGED`

6. Users and staff

Staff account administration (ADMIN) and a minimal staff directory.

6.1 GET /api/v1/users

Item	Value
Method / path	`GET /api/v1/users`
operationId	`listUsers`
Purpose	List staff accounts
Required role(s)	ADMIN
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`search`	query	string	no	min length 2; max length 50	Case-insensitive match on username or full name (2-50 chars).
`role`	query	Role	no	one of ADMIN, DOCTOR, PHARMACIST	
`enabled`	query	boolean	no		
`page`	query	integer	no	min 0; default 0	Zero-based page index.
`size`	query	integer	no	min 1; max 100; default 20	Page size.
`sort`	query	string	no	default fullName,asc; one of fullName,asc, fullName,desc, username,asc, createdAt,desc	Sort key.

Request body: none.

Responses

Status	Meaning / error codes	Body
200	Page of users.	UserPage
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "items": [
    {
      "id": "00000000-0000-4000-8000-000000000002",
      "username": "dr.perera",
      "fullName": "Dr. Nimal Perera",
      "roles": [
        "DOCTOR"
      ],
      "enabled": true,
      "mustChangePassword": false,
      "professionalRegistrationNumber": "MC-12345",
      "lastLoginAt": "2026-10-12T02:30:00Z",
      "lockedUntil": null,
      "createdAt": "2026-09-01T03:00:00Z",
      "updatedAt": "2026-09-01T03:00:00Z",
      "version": 0
    }
  ],
  "page": 0,
  "size": 20,
  "totalItems": 1,
  "totalPages": 1
}
```


Error response example (400 VALIDATION_FAILED; all codes for this endpoint are listed under Responses)


```
{
  "type": "https://dms.example/problems/validation-failed",
  "title": "Validation failed",
  "status": 400,
  "detail": "One or more fields are invalid.",
  "instance": "/api/v1/users",
  "code": "VALIDATION_FAILED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "fullName",
      "code": "NotBlank",
      "message": "must not be blank"
    }
  ]
}
```


Aspect	Details
Database entities affected	users (R), user_roles (R)
Transaction / concurrency	Read-only, single transaction.
Audit events	none

6.2 POST /api/v1/users

Item	Value
Method / path	`POST /api/v1/users`
operationId	`createUser`
Purpose	Create a staff account
Required role(s)	ADMIN
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters: none.

Request body - schema `CreateUserRequest` (required)

```
{
  "username": "ph.dias",
  "fullName": "Chamari Dias",
  "roles": [
    "PHARMACIST"
  ],
  "professionalRegistrationNumber": "PH-9911",
  "temporaryPassword": "Temp-Password-2026"
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`username`	string	yes	pattern `^[A-Za-z0-9._-]{3,50}$`	Lower-cased by the server.
`fullName`	string	yes	min length 1; max length 150	
`roles`	array of Role	yes	min items 1; max items 3; unique items	
`professionalRegistrationNumber`	string | null	no	max length 50	
`temporaryPassword`	string (password)	yes	min length 12; max length 128	

Business rules

- username is trimmed and lower-cased; must match ^[a-z0-9._-]{3,50}$.
- The account is created with mustChangePassword = true; the admin hands the temporary password to the user out of band.
- roles must contain 1-3 distinct values.

Responses

Status	Meaning / error codes	Body
201	Created.	UserResponse
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
409	`DUPLICATE_USERNAME`	Problem
422	`PASSWORD_POLICY_VIOLATION`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (201)


```
{
  "id": "7a1d1f8e-5b0c-4c1e-9f6e-2c4b8a9d0e11",
  "username": "ph.dias",
  "fullName": "Chamari Dias",
  "roles": [
    "PHARMACIST"
  ],
  "enabled": true,
  "mustChangePassword": true,
  "professionalRegistrationNumber": "PH-9911",
  "lastLoginAt": null,
  "lockedUntil": null,
  "createdAt": "2026-10-12T04:00:00Z",
  "updatedAt": "2026-10-12T04:00:00Z",
  "version": 0
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

409 DUPLICATE_USERNAME


```
{
  "type": "https://dms.example/problems/duplicate-username",
  "title": "Username already exists",
  "status": 409,
  "detail": "Username dr.perera is already taken.",
  "instance": "/api/v1/users",
  "code": "DUPLICATE_USERNAME",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


422 PASSWORD_POLICY_VIOLATION


```
{
  "type": "https://dms.example/problems/password-policy-violation",
  "title": "Password does not meet policy",
  "status": 422,
  "detail": "Password must not contain the username.",
  "instance": "/api/v1/users",
  "code": "PASSWORD_POLICY_VIOLATION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "newPassword",
      "code": "PasswordPolicy",
      "message": "must not contain the username"
    }
```

```
  ]
}
```

Aspect	Details
Database entities affected	users (W), user_roles (W), audit_events (W)
Transaction / concurrency	Uniqueness is enforced by uq_users_username; a violation is mapped to 409 DUPLICATE_USERNAME.
Audit events	`USER_CREATED`

6.3 GET /api/v1/users/{userId}

Item	Value
Method / path	`GET /api/v1/users/{userId}`
operationId	`getUser`
Purpose	Get a staff account
Required role(s)	ADMIN
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`userId`	path	string (uuid)	yes		User UUID.

Request body: none.

Responses

Status	Meaning / error codes	Body
200	User.	UserResponse
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`USER_NOT_FOUND`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)

```
{
  "id": "00000000-0000-4000-8000-000000000004",
  "username": "ph.silva",
  "fullName": "Kamal Silva",
  "roles": [
    "PHARMACIST"
  ],
  "enabled": true,
  "mustChangePassword": false,
  "professionalRegistrationNumber": "PH-3456",
  "lastLoginAt": "2026-10-12T02:00:00Z",
  "lockedUntil": null,
  "createdAt": "2026-09-01T03:00:00Z",
  "updatedAt": "2026-09-01T03:00:00Z",
  "version": 3
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 USER_NOT_FOUND


```
{
  "type": "https://dms.example/problems/user-not-found",
  "title": "User not found",
  "status": 404,
  "detail": "User not found.",
  "instance": "/api/v1/users/00000000-0000-4000-8000-000000000004",
  "code": "USER_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


Aspect	Details
Database entities affected	users (R), user_roles (R)
Transaction / concurrency	Read-only, single transaction.
Audit events	none

6.4 PUT /api/v1/users/{userId}

Item	Value
Method / path	`PUT /api/v1/users/{userId}`
operationId	`updateUser`
Purpose	Update a staff account (name, roles, enabled, registration number)
Required role(s)	ADMIN
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Content-Type	yes	application/json

Header	Required	Value / notes
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`userId`	path	string (uuid)	yes		User UUID.

Request body - schema `UpdateUserRequest` (required)

```
{
  "fullName": "Kamal Silva",
  "roles": [
    "PHARMACIST"
  ],
  "enabled": false,
  "professionalRegistrationNumber": "PH-3456",
  "version": 3
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`fullName`	string	yes	min length 1; max length 150	
`roles`	array of Role	yes	min items 1; max items 3; unique items	
`enabled`	boolean	yes		
`professionalRegistrationNumber`	string | null	no	max length 50	
`version`	integer (int64)	yes	min 0	

Business rules

- username cannot be changed.
- Setting enabled = false revokes all refresh-token families of the user (USER_DISABLED); existing access tokens expire within 15 min.
- An admin cannot disable themselves or remove their own ADMIN role.
- At least one enabled ADMIN must remain.
- Setting enabled = true also clears locked_until and failed_login_count.

Responses

Status	Meaning / error codes	Body
200	Updated.	UserResponse
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`USER_NOT_FOUND`	Problem
409	`CONCURRENT_MODIFICATION`	Problem
422	`SELF_MODIFICATION_NOT_ALLOWED`, `LAST_ADMIN_REQUIRED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "00000000-0000-4000-8000-000000000004",
  "username": "ph.silva",
  "fullName": "Kamal Silva",
  "roles": [
    "PHARMACIST"
  ],
  "enabled": false,
  "mustChangePassword": false,
  "professionalRegistrationNumber": "PH-3456",
  "lastLoginAt": "2026-10-12T02:00:00Z",
  "lockedUntil": null,
  "createdAt": "2026-09-01T03:00:00Z",
  "updatedAt": "2026-10-12T05:00:00Z",
  "version": 4
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 USER_NOT_FOUND


```
{
  "type": "https://dms.example/problems/user-not-found",
  "title": "User not found",
  "status": 404,
  "detail": "User not found.",
  "instance": "/api/v1/users/00000000-0000-4000-8000-000000000004",
  "code": "USER_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 CONCURRENT_MODIFICATION


```
{
  "type": "https://dms.example/problems/concurrent-modification",
  "title": "Record changed by someone else",
  "status": 409,
  "detail": "This record was modified by another user. Reload and try again.",
  "instance": "/api/v1/users/00000000-0000-4000-8000-000000000004",
  "code": "CONCURRENT_MODIFICATION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


422 SELF_MODIFICATION_NOT_ALLOWED


```
{
  "type": "https://dms.example/problems/self-modification-not-allowed",
  "title": "Cannot change own account this way",
  "status": 422,
  "detail": "You cannot disable your own account or remove your own ADMIN role.",
  "instance": "/api/v1/users/00000000-0000-4000-8000-000000000004",
  "code": "SELF_MODIFICATION_NOT_ALLOWED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


422 LAST_ADMIN_REQUIRED


```
{
  "type": "https://dms.example/problems/last-admin-required",
  "title": "At least one admin required",
  "status": 422,
  "detail": "At least one enabled administrator must remain.",
  "instance": "/api/v1/users/00000000-0000-4000-8000-000000000004",
  "code": "LAST_ADMIN_REQUIRED",
```

```
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


Aspect	Details
Database entities affected	users (W), user_roles (W), refresh_tokens (W), audit_events (W)
Transaction / concurrency	Optimistic locking on users.version; stale version → 409 CONCURRENT_MODIFICATION. The last-admin check runs with the ADMIN user rows locked (SELECT ... FOR UPDATE) to prevent two admins demoting each other concurrently.
Audit events	`USER_UPDATED`

6.5 POST /api/v1/users/{userId}/reset-password

Item	Value
Method / path	`POST /api/v1/users/{userId}/reset-password`
operationId	`resetUserPassword`
Purpose	Set a temporary password for a user
Required role(s)	ADMIN
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`userId`	path	string (uuid)	yes		User UUID.

Request body - schema `ResetPasswordRequest` (required)

```
{
  "temporaryPassword": "Reset-Password-2026"
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`temporaryPassword`	string (password)	yes	min length 12; max length 128	

Business rules

- Sets mustChangePassword = true, clears the lockout and revokes all refresh-token families of the user (PASSWORD_RESET).

Responses

Status	Meaning / error codes	Body
204	Password reset.	(no body)
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`USER_NOT_FOUND`	Problem
422	`PASSWORD_POLICY_VIOLATION`	Problem
500	`INTERNAL_ERROR`	Problem

Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 USER_NOT_FOUND


```
{
  "type": "https://dms.example/problems/user-not-found",
  "title": "User not found",
  "status": 404,
  "detail": "User not found.",
  "instance": "/api/v1/users/00000000-0000-4000-8000-000000000004/reset-password",
  "code": "USER_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


422 PASSWORD_POLICY_VIOLATION


```
{
  "type": "https://dms.example/problems/password-policy-violation",
  "title": "Password does not meet policy",
  "status": 422,
  "detail": "Password must not contain the username.",
  "instance": "/api/v1/users/00000000-0000-4000-8000-000000000004/reset-password",
  "code": "PASSWORD_POLICY_VIOLATION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "newPassword",
      "code": "PasswordPolicy",
      "message": "must not contain the username"
    }
  ]
}
```


Aspect	Details
Database entities affected	users (W), refresh_tokens (W), audit_events (W)
Transaction / concurrency	Single @Transactional service call.
Audit events	`USER_PASSWORD_RESET`

6.6 GET /api/v1/staff

Item	Value
Method / path	`GET /api/v1/staff`
operationId	`listStaff`
Purpose	Minimal directory of enabled staff (e.g. doctors for queue-session assignment)
Required role(s)	ADMIN, DOCTOR, PHARMACIST
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`role`	query	Role	no	one of ADMIN, DOCTOR, PHARMACIST	

Request body: none.

Business rules

- Returns only enabled users, sorted by fullName, max 200 entries, no pagination. Exposes id, fullName and roles only.

Responses

Status	Meaning / error codes	Body
200	Staff list.	StaffList
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "items": [
    {
      "id": "00000000-0000-4000-8000-000000000003",
      "fullName": "Dr. Anusha Fernando",
      "roles": [
        "DOCTOR"
      ]
    },
    {
      "id": "00000000-0000-4000-8000-000000000002",
      "fullName": "Dr. Nimal Perera",
      "roles": [
```

```
        "DOCTOR"
      ]
    }
  ]
}
```

Error response example (400 VALIDATION_FAILED; all codes for this endpoint are listed under Responses)

```
{
  "type": "https://dms.example/problems/validation-failed",
  "title": "Validation failed",
  "status": 400,
  "detail": "One or more fields are invalid.",
  "instance": "/api/v1/staff",
  "code": "VALIDATION_FAILED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "fullName",
      "code": "NotBlank",
      "message": "must not be blank"
    }
  ]
}
```

Aspect	Details
Database entities affected	users (R), user_roles (R)
Transaction / concurrency	Read-only, single transaction.
Audit events	none

7. Patients

Patient lookup, search, registration and demographic updates.

7.1 GET /api/v1/patients

Item	Value
Method / path	`GET /api/v1/patients`
operationId	`searchPatients`
Purpose	Search patients by name fragment, phone number or patient-number prefix. Used before registration to avoid duplicates and when a patient has no card. Exact lookup by number uses GET /patients/by-number/{patientNumber}.
Required role(s)	DOCTOR
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)

Header	Required	Value / notes
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`search`	query	string	yes	min length 2; max length 100	Trimmed, 2-100 chars. Matches full_name ILIKE %search%, patient_number prefix (upper-cased) or exact phone (spaces, hyphens and parentheses removed).
`page`	query	integer	no	min 0; default 0	Zero-based page index.
`size`	query	integer	no	min 1; max 100; default 20	Page size.

Request body: none.

Business rules

- Results sorted by fullName then patientNumber. Returns summaries only (no allergies, no address).

Responses

Status	Meaning / error codes	Body
200	Page of matching patients (may be empty).	PatientSummaryPage
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "items": [
    {
      "id": "20000000-0000-4000-8000-000000000001",
      "patientNumber": "P-0001",
      "fullName": "Saman Kumara",
      "sex": "MALE",
      "dateOfBirth": "1978-04-12",
      "dateOfBirthEstimated": false,
      "ageYears": 48,
      "phoneNumber": "+94771234567"
    }
  ],
  "page": 0,
  "size": 20,
  "totalItems": 1,
  "totalPages": 1
}
```


Error response example (400 VALIDATION_FAILED; all codes for this endpoint are listed under Responses)


```
{
  "type": "https://dms.example/problems/validation-failed",
  "title": "Validation failed",
  "status": 400,
  "detail": "One or more fields are invalid.",
  "instance": "/api/v1/patients",
  "code": "VALIDATION_FAILED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
```

```
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "fullName",
      "code": "NotBlank",
      "message": "must not be blank"
    }
  ]
}
```


Aspect	Details
Database entities affected	patients (R), audit_events (W)
Transaction / concurrency	Read-only, single transaction.
Audit events	`PATIENT_SEARCHED`

7.2 POST /api/v1/patients

Item	Value
Method / path	`POST /api/v1/patients`
operationId	`createPatient`
Purpose	Register a new patient
Required role(s)	DOCTOR
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Required (`Idempotency-Key` header) - see 2.6

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
Idempotency-Key	yes	Client-generated unique key (UUID v4 recommended) identifying one user intent. The frontend creates it
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters: none.

Request body - schema `CreatePatientRequest` (required)

Example: Known date of birth


```
{
  "patientNumber": "p-0042",
  "fullName": "Nadeesha Rathnayake",
  "sex": "FEMALE",
  "dateOfBirth": "1996-08-14",
  "phoneNumber": "077 123 4567",
  "address": "45 Lake Drive, Colombo 08",
  "knownAllergies": "Sulfonamides"
}
```

Example: Date of birth unknown

```
{
  "patientNumber": "P-0043",
  "fullName": "Unknown elderly male (brought by neighbour)",
  "sex": "MALE",
  "estimatedAgeYears": 75
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`patientNumber`	PatientNumber	yes	min length 3; max length 20; pattern `^[A-Za-z0-9][A-Za-z0-9-]{2,19}$`	Upper-cased by the server. Stored pattern ^[A-Z0-9][A-Z0-9-]{2,19}$.
`fullName`	string	yes	min length 1; max length 150	
`sex`	Sex	yes	one of MALE, FEMALE, OTHER, UNKNOWN	
`dateOfBirth`	string (date)	no		Exact date of birth (not in the future, >= 1900-01-01). Mutually exclusive with estimatedAgeYears.
`estimatedAgeYears`	integer	no	min 0; max 130	Use when the date of birth is unknown. Mutually exclusive with dateOfBirth.
`phoneNumber`	PhoneNumber	no	max length 25; pattern `^\+?[0-9 ()-]{7,24}$`	Optional. Spaces, hyphens and parentheses are removed by the server; the result must match ^+?[0-9]{7,15}$.
`address`	string | null	no	max length 300	
`knownAllergies`	string | null	no	max length 1000	

- body: exactly one of (dateOfBirth) or (estimatedAgeYears) must be present.

Business rules

- patientNumber and phoneNumber are normalised (trim; patientNumber upper-cased; phone spaces/hyphens/parentheses removed) before validation and storage.
- Exactly one of dateOfBirth or estimatedAgeYears must be supplied. With estimatedAgeYears N, the server stores dateOfBirth = today(APP_TIMEZONE) - N years and dateOfBirthEstimated = true.
- dateOfBirth must not be in the future (APP_TIMEZONE) and not before 1900-01-01.
- patientNumber is immutable after registration.
- Registration is a separate transaction from the consultation (Workflow B). If the following consultation submission fails, the registered patient remains valid and the doctor retries the consultation only.

Responses

Status	Meaning / error codes	Body
201	Registered.	PatientResponse
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`, `IDEMPOTENCY_KEY_REQUIRED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
409	`DUPLICATE_PATIENT_NUMBER`	Problem
422	`IDEMPOTENCY_KEY_REUSED`	Problem

Status	Meaning / error codes	Body
500	`INTERNAL_ERROR`	Problem

Successful response example (201)


```
{
  "id": "6b0f5b7e-2f43-4d1c-8e6a-6a9d7b2e1c42",
  "patientNumber": "P-0042",
  "fullName": "Nadeesha Rathnayake",
  "sex": "FEMALE",
  "dateOfBirth": "1996-08-14",
  "dateOfBirthEstimated": false,
  "ageYears": 30,
  "phoneNumber": "0771234567",
  "address": "45 Lake Drive, Colombo 08",
  "knownAllergies": "Sulfonamides",
  "createdAt": "2026-10-12T04:10:00Z",
  "updatedAt": "2026-10-12T04:10:00Z",
  "version": 0
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

409 DUPLICATE_PATIENT_NUMBER


```
{
  "type": "https://dms.example/problems/duplicate-patient-number",
  "title": "Patient number already exists",
  "status": 409,
  "detail": "Patient number P-0001 is already registered.",
  "instance": "/api/v1/patients",
  "code": "DUPLICATE_PATIENT_NUMBER",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "existingPatientId": "20000000-0000-4000-8000-000000000001",
  "errors": [
    {
      "field": "patientNumber",
      "code": "Duplicate",
      "message": "patient number already registered"
    }
  ]
}
```


Aspect	Details
Database entities affected	patients (W), idempotency_keys (W), audit_events (W)
Transaction / concurrency	uq_patients_patient_number is the final guard against concurrent registrations of the same number; the violation is mapped to 409 DUPLICATE_PATIENT_NUMBER with existingPatientId.
Audit events	`PATIENT_CREATED`

7.3 GET /api/v1/patients/by-number/{patientNumber}

Item	Value
Method / path	`GET /api/v1/patients/by-number/{patientNumber}`
operationId	`getPatientByNumber`
Purpose	Look up a patient by patient number (doctor's patient-ID search)
Required role(s)	DOCTOR
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)

Item	Value
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`patientNumber`	path	string	yes	min length 3; max length 20; pattern `^[A-Za-z0-9][A-Za-z0-9-]{2,19}$`	Patient number; trimmed and upper-cased by the server before lookup.

Request body: none.

Business rules

- 404 PATIENT_NOT_FOUND is an expected outcome; the UI then offers "Register new patient" with the number pre-filled.

Responses

Status	Meaning / error codes	Body
200	Patient found.	PatientResponse
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`PATIENT_NOT_FOUND`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "20000000-0000-4000-8000-000000000001",
  "patientNumber": "P-0001",
  "fullName": "Saman Kumara",
  "sex": "MALE",
  "dateOfBirth": "1978-04-12",
  "dateOfBirthEstimated": false,
  "ageYears": 48,
  "phoneNumber": "+94771234567",
  "address": "12 Temple Road, Kandy",
  "knownAllergies": "Penicillin (rash)",
  "createdAt": "2026-09-01T03:00:00Z",
  "updatedAt": "2026-09-01T03:00:00Z",
  "version": 0
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 PATIENT_NOT_FOUND

```
{
  "type": "https://dms.example/problems/patient-not-found",
  "title": "Patient not found",
  "status": 404,
  "detail": "No patient with patient number P-0099.",
  "instance": "/api/v1/patients/by-number/P-0099",
  "code": "PATIENT_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

Aspect	Details
Database entities affected	patients (R), audit_events (W)
Transaction / concurrency	Read-only, single transaction.
Audit events	`PATIENT_VIEWED`

7.4 GET /api/v1/patients/{patientId}

Item	Value
Method / path	`GET /api/v1/patients/{patientId}`
operationId	`getPatient`
Purpose	Get patient demographics
Required role(s)	DOCTOR
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`patientId`	path	string (uuid)	yes		Patient UUID.

Request body: none.

Responses

Status	Meaning / error codes	Body
200	Patient.	PatientResponse
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem

Status	Meaning / error codes	Body
404	`PATIENT_NOT_FOUND`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "20000000-0000-4000-8000-000000000004",
  "patientNumber": "P-0004",
  "fullName": "Piyadasa Herath",
  "sex": "MALE",
  "dateOfBirth": "1956-10-12",
  "dateOfBirthEstimated": true,
  "ageYears": 70,
  "phoneNumber": null,
  "address": "Matale",
  "knownAllergies": null,
  "createdAt": "2026-09-01T03:00:00Z",
  "updatedAt": "2026-09-01T03:00:00Z",
  "version": 0
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 PATIENT_NOT_FOUND


```
{
  "type": "https://dms.example/problems/patient-not-found",
  "title": "Patient not found",
  "status": 404,
  "detail": "No patient with patient number P-0099.",
  "instance": "/api/v1/patients/20000000-0000-4000-8000-000000000001",
  "code": "PATIENT_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


Aspect	Details
Database entities affected	patients (R), audit_events (W)
Transaction / concurrency	Read-only, single transaction.
Audit events	`PATIENT_VIEWED`

7.5 PUT /api/v1/patients/{patientId}

Item	Value
Method / path	`PUT /api/v1/patients/{patientId}`
operationId	`updatePatient`
Purpose	Update patient demographics and allergies
Required role(s)	DOCTOR
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`patientId`	path	string (uuid)	yes		Patient UUID.

Request body - schema `UpdatePatientRequest` (required)

```
{
  "fullName": "Saman Kumara",
  "sex": "MALE",
  "dateOfBirth": "1978-04-12",
  "dateOfBirthEstimated": false,
  "phoneNumber": "+94771234567",
  "address": "14 Temple Road, Kandy",
  "knownAllergies": "Penicillin (rash); Aspirin (wheeze)",
  "version": 0
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`fullName`	string	yes	min length 1; max length 150	
`sex`	Sex	yes	one of MALE, FEMALE, OTHER, UNKNOWN	
`dateOfBirth`	string (date)	no		Date of birth to store. Send together with dateOfBirthEstimated. Mutually exclusive with estimatedAgeYears.
`dateOfBirthEstimated`	boolean	no		Send the value received from GET unless the user entered an exact DOB (then false). Required with dateOfBirth.
`estimatedAgeYears`	integer	no	min 0; max 130	Re-estimates the DOB from an age (stores dateOfBirthEstimated = true). Mutually exclusive with dateOfBirth.
`phoneNumber`	PhoneNumber	no	max length 25; pattern `^\+?[0-9 ()-]{7,24}$`	Optional. Spaces, hyphens and parentheses are removed by the server; the result must match ^+?[0-9]{7,15}$.
`address`	string | null	no	max length 300	
`knownAllergies`	string | null	no	max length 1000	
`version`	integer (int64)	yes	min 0	

- body: exactly one of (dateOfBirth + dateOfBirthEstimated) or (estimatedAgeYears) must be present.

Business rules

- Full replacement of the editable fields; patientNumber cannot be changed (not part of the body).

- Send either dateOfBirth + dateOfBirthEstimated (unchanged values keep an estimated DOB estimated) or estimatedAgeYears (re-estimate); never both.
- Changes to knownAllergies are audited (field name only, not the value).

Responses

Status	Meaning / error codes	Body
200	Updated.	PatientResponse
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`PATIENT_NOT_FOUND`	Problem
409	`CONCURRENT_MODIFICATION`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "20000000-0000-4000-8000-000000000001",
  "patientNumber": "P-0001",
  "fullName": "Saman Kumara",
  "sex": "MALE",
  "dateOfBirth": "1978-04-12",
  "dateOfBirthEstimated": false,
  "ageYears": 48,
  "phoneNumber": "+94771234567",
  "address": "14 Temple Road, Kandy",
  "knownAllergies": "Penicillin (rash); Aspirin (wheeze)",
  "createdAt": "2026-09-01T03:00:00Z",
  "updatedAt": "2026-10-12T04:20:00Z",
  "version": 1
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 PATIENT_NOT_FOUND


```
{
  "type": "https://dms.example/problems/patient-not-found",
  "title": "Patient not found",
  "status": 404,
  "detail": "No patient with patient number P-0099.",
  "instance": "/api/v1/patients/20000000-0000-4000-8000-000000000001",
  "code": "PATIENT_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 CONCURRENT_MODIFICATION


```
{
  "type": "https://dms.example/problems/concurrent-modification",
  "title": "Record changed by someone else",
  "status": 409,
  "detail": "This record was modified by another user. Reload and try again.",
  "instance": "/api/v1/patients/20000000-0000-4000-8000-000000000001",
  "code": "CONCURRENT_MODIFICATION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

Aspect	Details
Database entities affected	patients (W), audit_events (W)
Transaction / concurrency	Optimistic locking with version; stale version → 409 CONCURRENT_MODIFICATION. The UI reloads and shows the newer values.
Audit events	`PATIENT_UPDATED`

8. Medical history

Read-only aggregated history of a patient's consultations, conditions and prescriptions.

8.1 GET /api/v1/patients/{patientId}/medical-history

Item	Value
Method / path	`GET /api/v1/patients/{patientId}/medical-history`
operationId	`getMedicalHistory`
Purpose	Get a patient's medical history (conditions summary + paginated consultations). Consultations are returned newest first with their diagnoses and all prescriptions (including cancelled ones), prescription items and dispensing results.
Required role(s)	DOCTOR
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`patientId`	path	string (uuid)	yes		Patient UUID.
`page`	query	integer	no	min 0; default 0	Zero-based page index.
`size`	query	integer	no	min 1; max 50; default 10	Consultations per page.

Request body: none.

Business rules

- conditions aggregates consultation_diagnoses of all consultations of the patient, grouped case-insensitively by condition text, newest first, max 50.
- Pharmacists cannot call this endpoint; they see only what GET /prescriptions/{id} returns.

Responses

Status	Meaning / error codes	Body
200	Medical history.	MedicalHistoryResponse
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`PATIENT_NOT_FOUND`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "patient": {
    "id": "20000000-0000-4000-8000-000000000001",
    "patientNumber": "P-0001",
    "fullName": "Saman Kumara",
    "sex": "MALE",
    "dateOfBirth": "1978-04-12",
    "dateOfBirthEstimated": false,
    "ageYears": 48,
    "phoneNumber": "+94771234567",
    "address": "12 Temple Road, Kandy",
    "knownAllergies": "Penicillin (rash)",
    "createdAt": "2026-09-01T03:00:00Z",
    "updatedAt": "2026-09-01T03:00:00Z",
    "version": 0
  },
  "conditions": [
    {
      "conditionText": "Essential hypertension",
      "icd10Code": "I10",
      "lastRecordedAt": "2026-10-12T03:02:31Z",
      "occurrences": 1
    },
    {
      "conditionText": "Type 2 diabetes mellitus",
      "icd10Code": "E11.9",
      "lastRecordedAt": "2026-10-12T03:02:31Z",
      "occurrences": 1
    },
    {
      "conditionText": "Acute pharyngitis",
      "icd10Code": "J02.9",
      "lastRecordedAt": "2026-09-12T16:37:31Z",
      "occurrences": 1
    }
  ],
  "consultations": {
    "items": [
      {
        "id": "40000000-0000-4000-8000-000000000001",
        "patient": {
          "id": "20000000-0000-4000-8000-000000000001",
          "patientNumber": "P-0001",
          "fullName": "Saman Kumara"
        },
        "doctor": {
          "id": "00000000-0000-4000-8000-000000000002",
          "fullName": "Dr. Nimal Perera"
        },
        "queueEntryId": null,
        "consultationAt": "2026-09-12T16:37:31Z",
        "presentingComplaint": "Fever and sore throat for 3 days",
        "clinicalNotes": "Temp 38.4 C. Tonsils inflamed, no exudate. Advised fluids.",
        "diagnoses": [
          {
            "id": "41000000-0000-4000-8000-000000000001",
            "lineNumber": 1,
            "conditionText": "Acute pharyngitis",
            "icd10Code": "J02.9"
          }
```

```
        ],
        "prescriptions": [
          {
            "id": "50000000-0000-4000-8000-000000000001",
            "prescriptionNumber": "RX-000001",
            "consultationId": "40000000-0000-4000-8000-000000000001",
            "consultationAt": "2026-09-12T16:37:31Z",
            "status": "DISPENSED",
            "notes": null,
            "submittedAt": "2026-09-12T16:37:31Z",
            "prescribedBy": {
              "id": "00000000-0000-4000-8000-000000000002",
              "fullName": "Dr. Nimal Perera"
            },
            "patient": {
              "id": "20000000-0000-4000-8000-000000000001",
              "patientNumber": "P-0001",
              "fullName": "Saman Kumara",
              "sex": "MALE",
              "ageYears": 48,
              "dateOfBirthEstimated": false,
              "knownAllergies": "Penicillin (rash)"
            },
            "items": [
              {
                "id": "51000000-0000-4000-8000-000000000001",
                "lineNumber": 1,
                "medicineId": "10000000-0000-4000-8000-000000000001",
                "medicineName": "Paracetamol",
                "strength": "500 mg",
                "dosageForm": "TABLET",
                "dose": "2 tablets",
                "frequency": "4 times daily (QDS) when needed",
                "durationDays": 3,
                "quantity": 24,
                "quantityUnit": "tablets",
                "instructions": "Maximum 8 tablets in 24 hours"
              },
              {
                "id": "51000000-0000-4000-8000-000000000002",
                "lineNumber": 2,
                "medicineId": null,
                "medicineName": "Chlorhexidine mouthwash",
                "strength": "0.2%",
                "dosageForm": "SOLUTION",
                "dose": "10 ml",
                "frequency": "Twice daily (BD)",
                "durationDays": 5,
                "quantity": 1,
                "quantityUnit": "bottle",
                "instructions": "Rinse and spit, do not swallow"
              }
            ],
            "startedAt": "2026-09-12T16:42:31Z",
            "startedBy": {
              "id": "00000000-0000-4000-8000-000000000004",
              "fullName": "Kamal Silva"
            },
            "cancelledAt": null,
            "cancelledBy": null,
            "cancelReason": null,
            "supersedesPrescriptionId": null,
            "supersededByPrescriptionId": null,
            "dispensing": {
              "id": "52000000-0000-4000-8000-000000000001",
              "pharmacist": {
                "id": "00000000-0000-4000-8000-000000000004",
                "fullName": "Kamal Silva"
              },
              "dispensedAt": "2026-09-12T16:49:31Z",
              "outcome": "PARTIAL",
              "notes": null,
              "items": [
                {
                  "prescriptionItemId": "51000000-0000-4000-8000-000000000001",
                  "quantityDispensed": 24,
                  "status": "FULL",
                  "note": null
                },
```

```
                {
                  "prescriptionItemId": "51000000-0000-4000-8000-000000000002",
                  "quantityDispensed": 0,
                  "status": "NOT_DISPENSED",
                  "note": "Out of stock - patient advised to buy outside"
                }
              ]
            },
            "version": 2
          }
        ],
        "createdAt": "2026-09-12T16:37:31Z"
      }
    ],
    "page": 0,
    "size": 10,
    "totalItems": 2,
    "totalPages": 1
  }
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 PATIENT_NOT_FOUND


```
{
  "type": "https://dms.example/problems/patient-not-found",
  "title": "Patient not found",
  "status": 404,
  "detail": "No patient with patient number P-0099.",
  "instance": "/api/v1/patients/20000000-0000-4000-8000-000000000001/medical-history",
  "code": "PATIENT_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


Aspect	Details
Database entities affected	patients (R), consultations (R), consultation_diagnoses (R), prescriptions (R), prescription_items (R), dispensings (R), dispensing_items (R), users (R), audit_events (W)
Transaction / concurrency	Read-only transaction. Loaded with a fixed number of queries (page of consultations, then batch loads by consultation id) - never one query per consultation (N+1).
Audit events	`PATIENT_HISTORY_VIEWED`

9. Consultations

Creating and reading consultations (diagnoses, notes and the initial prescription in one transaction).

9.1 POST /api/v1/consultations

Item	Value
Method / path	`POST /api/v1/consultations`
operationId	`createConsultation`
Purpose	Create a consultation with diagnoses and an optional prescription (one transaction). Persists the consultation, its diagnoses and (when `prescription` is present) the prescription with all items atomically. The prescription is created with status PENDING and therefore appears immediately in the pharmacist queue (FR-021).
Required role(s)	DOCTOR
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)

Item	Value
Idempotency	Required (`Idempotency-Key` header) - see 2.6

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
Idempotency-Key	yes	Client-generated unique key (UUID v4 recommended) identifying one user intent. The frontend creates it
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters: none.

Request body - schema `CreateConsultationRequest` (required)


```
{
  "patientId": "20000000-0000-4000-8000-000000000002",
  "queueEntryId": "31000000-0000-4000-8000-000000000002",
  "presentingComplaint": "Itchy eyes and sneezing for one week",
  "clinicalNotes": "No fever. Mild conjunctival injection bilaterally.",
  "diagnoses": [
    {
      "conditionText": "Allergic rhinitis",
      "icd10Code": "J30.4"
    }
  ],
  "prescription": {
    "notes": null,
    "items": [
      {
        "medicineId": "10000000-0000-4000-8000-000000000007",
        "dose": "1 tablet",
        "frequency": "Once daily at night (nocte)",
        "durationDays": 10,
        "quantity": 10,
        "quantityUnit": "tablets",
        "instructions": "May cause drowsiness"
      },
      {
        "medicineName": "Sodium cromoglicate eye drops",
        "strength": "2%",
        "dosageForm": "DROPS",
        "dose": "1 drop each eye",
        "frequency": "4 times daily (QDS)",
        "durationDays": 14,
        "quantity": 1,
        "quantityUnit": "bottle"
      }
    ]
  }
}
```


Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`patientId`	string (uuid)	yes		
`queueEntryId`	string | null (uuid)	no		
`presentingComplaint`	string | null	no	max length 1000	
`clinicalNotes`	string | null	no	max length 5000	

Field	Type	Required	Constraints	Notes
`diagnoses`	array of DiagnosisInput	yes	min items 1; max items 10	
`diagnoses[].conditionText`	string	yes	min length 1; max length 500	
`diagnoses[].icd10Code`	string | null	no	pattern `^[A-Z][0-9]{2}(\.[0-9A-Z]{1,4})?$`	
`prescription`	PrescriptionInput | null	no		Omit or null for an advice-only consultation.
`prescription.notes`	string | null	no	max length 1000	
`prescription.items`	array of PrescriptionItemInput	yes	min items 1; max items 30	
`prescription.items[].medicineId`	string (uuid)	no		
`prescription.items[].medicineName`	string	no	min length 1; max length 200	
`prescription.items[].strength`	string | null	no	max length 50	
`prescription.items[].dosageForm`	DosageForm | null	no	one of TABLET, CAPSULE, SYRUP, SUSPENSION, SOLUTION, INJECTION, CREAM, OINTMENT, GEL, DROPS, INHALER, SUPPOSITORY, POWDER, PATCH, OTHER	
`prescription.items[].dose`	string	yes	min length 1; max length 100	
`prescription.items[].frequency`	string	yes	min length 1; max length 100	
`prescription.items[].durationDays`	integer | null	no	min 1; max 365	
`prescription.items[].quantity`	number	yes	max 99999.99; > 0	Max 2 decimal places (BigDecimal, @Digits(integer=5, fraction=2)).
`prescription.items[].quantityUnit`	string	yes	min length 1; max length 30	
`prescription.items[].instructions`	string | null	no	max length 500	

- prescription.items[]: if `medicineId` is present then `medicineName`, `strength` and `dosageForm` must be absent; otherwise `medicineName` is required.

Business rules

- doctor = authenticated user; consultationAt = server time. Clients cannot set either.
- diagnoses 1-10 (line numbers assigned in array order). prescription.items 1-30 (line numbers in array order).
- An item with medicineId must NOT carry medicineName, strength or dosageForm; the server copies them from the catalogue (snapshot). An item without medicineId must carry medicineName (free-text medicine).
- A referenced catalogue medicine must be active, otherwise 422 MEDICINE_INACTIVE with the item path.
- queueEntryId (optional) must reference a queue entry in status SERVING or COMPLETED that is not linked to another consultation. Linking does not change the queue entry status.

- Any validation or persistence failure rolls back everything; no partial consultation, diagnosis or prescription is ever stored.
- There is no update or delete endpoint for consultations; history is append-only.

Responses

Status	Meaning / error codes	Body
201	Created (or replayed for a repeated Idempotency-Key).	ConsultationResponse
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`, `IDEMPOTENCY_KEY_REQUIRED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
409	`QUEUE_ENTRY_ALREADY_LINKED`	Problem
422	`REFERENCED_RESOURCE_NOT_FOUND`, `MEDICINE_INACTIVE`, `QUEUE_ENTRY_NOT_IN_SESSION`, `IDEMPOTENCY_KEY_REUSED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (201)


```
{
  "id": "4a7e9d10-1c2b-4f3e-8d5a-0b6c7d8e9f01",
  "patient": {
    "id": "20000000-0000-4000-8000-000000000002",
    "patientNumber": "P-0002",
    "fullName": "Dilani Wijesekara"
  },
  "doctor": {
    "id": "00000000-0000-4000-8000-000000000002",
    "fullName": "Dr. Nimal Perera"
  },
  "queueEntryId": "31000000-0000-4000-8000-000000000002",
  "consultationAt": "2026-10-12T04:30:00Z",
  "presentingComplaint": "Itchy eyes and sneezing for one week",
  "clinicalNotes": "No fever. Mild conjunctival injection bilaterally.",
  "diagnoses": [
    {
      "id": "9c2d1e0f-3a4b-4c5d-8e6f-7a8b9c0d1e2f",
      "lineNumber": 1,
      "conditionText": "Allergic rhinitis",
      "icd10Code": "J30.4"
    }
  ],
  "prescriptions": [
    {
      "id": "8b1c2d3e-4f50-4617-a8b9-c0d1e2f3a4b5",
      "prescriptionNumber": "RX-000003",
      "consultationId": "4a7e9d10-1c2b-4f3e-8d5a-0b6c7d8e9f01",
      "consultationAt": "2026-10-12T04:30:00Z",
      "status": "PENDING",
      "notes": null,
      "submittedAt": "2026-10-12T04:30:00Z",
      "prescribedBy": {
        "id": "00000000-0000-4000-8000-000000000002",
        "fullName": "Dr. Nimal Perera"
      },
      "patient": {
        "id": "20000000-0000-4000-8000-000000000002",
        "patientNumber": "P-0002",
        "fullName": "Dilani Wijesekara",
        "sex": "FEMALE",
        "ageYears": 35,
        "dateOfBirthEstimated": false,
        "knownAllergies": null
      },
      "items": [
        {
          "id": "1a2b3c4d-5e6f-4a7b-8c9d-0e1f2a3b4c5d",
          "lineNumber": 1,
```

```
          "medicineId": "10000000-0000-4000-8000-000000000007",
          "medicineName": "Cetirizine",
          "strength": "10 mg",
          "dosageForm": "TABLET",
          "dose": "1 tablet",
          "frequency": "Once daily at night (nocte)",
          "durationDays": 10,
          "quantity": 10,
          "quantityUnit": "tablets",
          "instructions": "May cause drowsiness"
        },
        {
          "id": "2b3c4d5e-6f7a-4b8c-9d0e-1f2a3b4c5d6e",
          "lineNumber": 2,
          "medicineId": null,
          "medicineName": "Sodium cromoglicate eye drops",
          "strength": "2%",
          "dosageForm": "DROPS",
          "dose": "1 drop each eye",
          "frequency": "4 times daily (QDS)",
          "durationDays": 14,
          "quantity": 1,
          "quantityUnit": "bottle",
          "instructions": null
        }
      ],
      "startedAt": null,
      "startedBy": null,
      "cancelledAt": null,
      "cancelledBy": null,
      "cancelReason": null,
      "supersedesPrescriptionId": null,
      "supersededByPrescriptionId": null,
      "dispensing": null,
      "version": 0
    }
  ],
  "createdAt": "2026-10-12T04:30:00Z"
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

422 REFERENCED_RESOURCE_NOT_FOUND


```
{
  "type": "https://dms.example/problems/referenced-resource-not-found",
  "title": "Referenced resource not found",
  "status": 422,
  "detail": "medicineId does not reference an existing medicine.",
  "instance": "/api/v1/consultations",
  "code": "REFERENCED_RESOURCE_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "prescription.items[0].medicineId",
      "code": "NotFound",
      "message": "medicine not found"
    }
  ]
}
```


422 MEDICINE_INACTIVE


```
{
  "type": "https://dms.example/problems/medicine-inactive",
  "title": "Medicine is not available",
  "status": 422,
  "detail": "Chlorphenamine 4 mg is inactive and cannot be prescribed.",
  "instance": "/api/v1/consultations",
  "code": "MEDICINE_INACTIVE",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
```

```
    {
      "field": "prescription.items[1].medicineId",
      "code": "Inactive",
      "message": "medicine is inactive"
    }
  ]
}
```

409 QUEUE_ENTRY_ALREADY_LINKED

```
{
  "type": "https://dms.example/problems/queue-entry-already-linked",
  "title": "Queue entry already linked",
  "status": 409,
  "detail": "Queue entry is already linked to another consultation.",
  "instance": "/api/v1/consultations",
  "code": "QUEUE_ENTRY_ALREADY_LINKED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

422 QUEUE_ENTRY_NOT_IN_SESSION

```
{
  "type": "https://dms.example/problems/queue-entry-not-in-session",
  "title": "Queue entry not in session",
  "status": 422,
  "detail": "Entry does not belong to this session.",
  "instance": "/api/v1/consultations",
  "code": "QUEUE_ENTRY_NOT_IN_SESSION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

Aspect	Details
Database entities affected	consultations (W), consultation_diagnoses (W), prescriptions (W), prescription_items (W), patients (R), medicines (R), queue_entries (R), idempotency_keys (W), audit_events (W)
Transaction / concurrency	Single @Transactional service method. Idempotency row inserted first in the same transaction so a double-submitted form blocks on the unique key and then replays the first result (201, same consultation).
Audit events	`CONSULTATION_CREATED`, `PRESCRIPTION_CREATED`

9.2 GET /api/v1/consultations/{consultationId}

Item	Value
Method / path	`GET /api/v1/consultations/{consultationId}`
operationId	`getConsultation`
Purpose	Get a consultation with diagnoses, prescriptions and dispensing status. Used for the submission confirmation screen and to follow dispensing status (frontend polls every NEXT_PUBLIC_PRESCRIPTION_POLL_INTERVAL_MS while any prescription is PENDING or IN_PROGRESS).
Required role(s)	DOCTOR
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`consultationId`	path	string (uuid)	yes		Consultation UUID.

Request body: none.

Responses

Status	Meaning / error codes	Body
200	Consultation.	ConsultationResponse
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`CONSULTATION_NOT_FOUND`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "40000000-0000-4000-8000-000000000002",
  "patient": {
    "id": "20000000-0000-4000-8000-000000000001",
    "patientNumber": "P-0001",
    "fullName": "Saman Kumara"
  },
  "doctor": {
    "id": "00000000-0000-4000-8000-000000000002",
    "fullName": "Dr. Nimal Perera"
  },
  "queueEntryId": "31000000-0000-4000-8000-000000000001",
  "consultationAt": "2026-10-12T03:02:31Z",
  "presentingComplaint": "Review: raised blood sugar on home testing",
  "clinicalNotes": "FBS 168 mg/dl (patient report). Lifestyle advice given.",
  "diagnoses": [
    {
      "id": "41000000-0000-4000-8000-000000000002",
      "lineNumber": 1,
      "conditionText": "Type 2 diabetes mellitus",
      "icd10Code": "E11.9"
    },
    {
      "id": "41000000-0000-4000-8000-000000000003",
      "lineNumber": 2,
      "conditionText": "Essential hypertension",
      "icd10Code": "I10"
    }
  ],
  "prescriptions": [
    {
      "id": "50000000-0000-4000-8000-000000000002",
      "prescriptionNumber": "RX-000002",
      "consultationId": "40000000-0000-4000-8000-000000000002",
      "consultationAt": "2026-10-12T03:02:31Z",
      "status": "IN_PROGRESS",
      "notes": "Patient allergic to penicillin",
```

```
      "submittedAt": "2026-10-12T03:03:31Z",
      "prescribedBy": {
        "id": "00000000-0000-4000-8000-000000000002",
        "fullName": "Dr. Nimal Perera"
      },
      "patient": {
        "id": "20000000-0000-4000-8000-000000000001",
        "patientNumber": "P-0001",
        "fullName": "Saman Kumara",
        "sex": "MALE",
        "ageYears": 48,
        "dateOfBirthEstimated": false,
        "knownAllergies": "Penicillin (rash)"
      },
      "items": [
        {
          "id": "51000000-0000-4000-8000-000000000003",
          "lineNumber": 1,
          "medicineId": "10000000-0000-4000-8000-000000000005",
          "medicineName": "Metformin",
          "strength": "500 mg",
          "dosageForm": "TABLET",
          "dose": "1 tablet",
          "frequency": "Twice daily (BD)",
          "durationDays": 30,
          "quantity": 60,
          "quantityUnit": "tablets",
          "instructions": "With meals"
        },
        {
          "id": "51000000-0000-4000-8000-000000000004",
          "lineNumber": 2,
          "medicineId": "10000000-0000-4000-8000-000000000006",
          "medicineName": "Amlodipine",
          "strength": "5 mg",
          "dosageForm": "TABLET",
          "dose": "1 tablet",
          "frequency": "Once daily (OD) in the morning",
          "durationDays": 30,
          "quantity": 30,
          "quantityUnit": "tablets",
          "instructions": null
        }
      ],
      "startedAt": "2026-10-12T03:10:00Z",
      "startedBy": {
        "id": "00000000-0000-4000-8000-000000000004",
        "fullName": "Kamal Silva"
      },
      "cancelledAt": null,
      "cancelledBy": null,
      "cancelReason": null,
      "supersedesPrescriptionId": null,
      "supersededByPrescriptionId": null,
      "dispensing": null,
      "version": 1
    }
  ],
  "createdAt": "2026-10-12T03:02:31Z"
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 CONSULTATION_NOT_FOUND


```
{
  "type": "https://dms.example/problems/consultation-not-found",
  "title": "Consultation not found",
  "status": 404,
  "detail": "Consultation not found.",
  "instance": "/api/v1/consultations/40000000-0000-4000-8000-000000000002",
  "code": "CONSULTATION_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

Aspect	Details
Database entities affected	consultations (R), consultation_diagnoses (R), prescriptions (R), prescription_items (R), dispensings (R), dispensing_items (R), audit_events (W)
Transaction / concurrency	Read-only, single transaction.
Audit events	`CONSULTATION_VIEWED`

10. Prescriptions

Listing, reading, cancelling and re-issuing prescriptions. Prescription items are created only as part of a prescription and have no endpoints of their own.

Prescription items. Items are created only inside `POST /consultations` (`prescription.items`) or `POST /consultations/{consultationId}/prescriptions` and are returned embedded in `PrescriptionResponse.items`. There are deliberately no item endpoints: a submitted prescription is immutable, so a correction is a cancel + re-issue (BR-P-07).

10.1 POST /api/v1/consultations/{consultationId}/prescriptions

Item	Value
Method / path	`POST /api/v1/consultations/{consultationId}/prescriptions`
operationId	`addPrescriptionToConsultation`
Purpose	Add (or re-issue) a prescription for an existing consultation. Used to correct a mistake: the doctor cancels the wrong prescription and issues a replacement with `supersedesPrescriptionId`, or adds a prescription to a consultation that had none.
Required role(s)	DOCTOR
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Required (`Idempotency-Key` header) - see 2.6

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
Idempotency-Key	yes	Client-generated unique key (UUID v4 recommended) identifying one user intent. The frontend creates it
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`consultationId`	path	string (uuid)	yes		Consultation UUID.

Request body - schema `AddPrescriptionRequest` (required)

```
{
  "supersedesPrescriptionId": "50000000-0000-4000-8000-000000000002",
  "notes": "Corrected Metformin quantity",
```

```
  "items": [
    {
      "medicineId": "10000000-0000-4000-8000-000000000005",
      "dose": "1 tablet",
      "frequency": "Twice daily (BD)",
      "durationDays": 30,
      "quantity": 60,
      "quantityUnit": "tablets",
      "instructions": "With meals"
    }
  ]
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`supersedesPrescriptionId`	string | null (uuid)	no		
`notes`	string | null	no	max length 1000	
`items`	array of PrescriptionItemInput	yes	min items 1; max items 30	
`items[].medicineId`	string (uuid)	no		
`items[].medicineName`	string	no	min length 1; max length 200	
`items[].strength`	string | null	no	max length 50	
`items[].dosageForm`	DosageForm | null	no	one of TABLET, CAPSULE, SYRUP, SUSPENSION, SOLUTION, INJECTION, CREAM, OINTMENT, GEL, DROPS, INHALER, SUPPOSITORY, POWDER, PATCH, OTHER	
`items[].dose`	string	yes	min length 1; max length 100	
`items[].frequency`	string	yes	min length 1; max length 100	
`items[].durationDays`	integer | null	no	min 1; max 365	
`items[].quantity`	number	yes	max 99999.99; > 0	Max 2 decimal places (BigDecimal, @Digits(integer=5, fraction=2)).
`items[].quantityUnit`	string	yes	min length 1; max length 30	
`items[].instructions`	string | null	no	max length 500	

- items[]: if `medicineId` is present then `medicineName`, `strength` and `dosageForm` must be absent; otherwise `medicineName` is required.

Business rules

- Only the consultation's doctor may add a prescription (403 NOT_RESOURCE_OWNER otherwise).
- Allowed only within 24 hours of consultationAt (422 CONSULTATION_LOCKED).
- The consultation must have no non-cancelled prescription (409 ACTIVE_PRESCRIPTION_EXISTS).
- supersedesPrescriptionId, when given, must be a CANCELLED prescription of this consultation that is not already superseded.
- Item rules are identical to createConsultation.

Responses

Status	Meaning / error codes	Body
201	Created (or replayed).	PrescriptionResponse

Status	Meaning / error codes	Body
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`, `IDEMPOTENCY_KEY_REQUIRED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`NOT_RESOURCE_OWNER`, `FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`CONSULTATION_NOT_FOUND`	Problem
409	`ACTIVE_PRESCRIPTION_EXISTS`	Problem
422	`CONSULTATION_LOCKED`, `INVALID_SUPERSEDED_PRESCRIPTION`, `REFERENCED_RESOURCE_NOT_FOUND`, `MEDICINE_INACTIVE`, `IDEMPOTENCY_KEY_REUSED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (201)


```
{
  "id": "5c6d7e8f-9a0b-4c1d-8e2f-3a4b5c6d7e8f",
  "prescriptionNumber": "RX-000004",
  "consultationId": "40000000-0000-4000-8000-000000000002",
  "consultationAt": "2026-10-12T03:02:31Z",
  "status": "PENDING",
  "notes": "Corrected Metformin quantity",
  "submittedAt": "2026-10-12T03:20:00Z",
  "prescribedBy": {
    "id": "00000000-0000-4000-8000-000000000002",
    "fullName": "Dr. Nimal Perera"
  },
  "patient": {
    "id": "20000000-0000-4000-8000-000000000001",
    "patientNumber": "P-0001",
    "fullName": "Saman Kumara",
    "sex": "MALE",
    "ageYears": 48,
    "dateOfBirthEstimated": false,
    "knownAllergies": "Penicillin (rash)"
  },
  "items": [
    {
      "id": "6d7e8f9a-0b1c-4d2e-8f3a-4b5c6d7e8f9a",
      "lineNumber": 1,
      "medicineId": "10000000-0000-4000-8000-000000000005",
      "medicineName": "Metformin",
      "strength": "500 mg",
      "dosageForm": "TABLET",
      "dose": "1 tablet",
      "frequency": "Twice daily (BD)",
      "durationDays": 30,
      "quantity": 60,
      "quantityUnit": "tablets",
      "instructions": "With meals"
    }
  ],
  "startedAt": null,
  "startedBy": null,
  "cancelledAt": null,
  "cancelledBy": null,
  "cancelReason": null,
  "supersedesPrescriptionId": "50000000-0000-4000-8000-000000000002",
  "supersededByPrescriptionId": null,
  "dispensing": null,
  "version": 0
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 CONSULTATION_NOT_FOUND

```
{
  "type": "https://dms.example/problems/consultation-not-found",
  "title": "Consultation not found",
  "status": 404,
  "detail": "Consultation not found.",
  "instance": "/api/v1/consultations/40000000-0000-4000-8000-000000000002/prescriptions",
  "code": "CONSULTATION_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


403 NOT_RESOURCE_OWNER


```
{
  "type": "https://dms.example/problems/not-resource-owner",
  "title": "Not the owner of this resource",
  "status": 403,
  "detail": "Only the prescribing doctor can cancel this prescription.",
  "instance": "/api/v1/consultations/40000000-0000-4000-8000-000000000002/prescriptions",
  "code": "NOT_RESOURCE_OWNER",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 ACTIVE_PRESCRIPTION_EXISTS


```
{
  "type": "https://dms.example/problems/active-prescription-exists",
  "title": "Active prescription exists",
  "status": 409,
  "detail": "Consultation already has an active prescription RX-000042.",
  "instance": "/api/v1/consultations/40000000-0000-4000-8000-000000000002/prescriptions",
  "code": "ACTIVE_PRESCRIPTION_EXISTS",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


422 CONSULTATION_LOCKED


```
{
  "type": "https://dms.example/problems/consultation-locked",
  "title": "Consultation locked",
  "status": 422,
  "detail": "Prescriptions can only be added within 24 hours of the consultation.",
  "instance": "/api/v1/consultations/40000000-0000-4000-8000-000000000002/prescriptions",
  "code": "CONSULTATION_LOCKED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


422 INVALID_SUPERSEDED_PRESCRIPTION


```
{
  "type": "https://dms.example/problems/invalid-superseded-prescription",
  "title": "Invalid superseded prescription",
  "status": 422,
  "detail": "Only a cancelled prescription of this consultation can be superseded.",
  "instance": "/api/v1/consultations/40000000-0000-4000-8000-000000000002/prescriptions",
  "code": "INVALID_SUPERSEDED_PRESCRIPTION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


422 REFERENCED_RESOURCE_NOT_FOUND


```
{
  "type": "https://dms.example/problems/referenced-resource-not-found",
  "title": "Referenced resource not found",
  "status": 422,
```

```
  "detail": "medicineId does not reference an existing medicine.",
  "instance": "/api/v1/consultations/40000000-0000-4000-8000-000000000002/prescriptions",
  "code": "REFERENCED_RESOURCE_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "prescription.items[0].medicineId",
      "code": "NotFound",
      "message": "medicine not found"
    }
  ]
}
```


422 MEDICINE_INACTIVE


```
{
  "type": "https://dms.example/problems/medicine-inactive",
  "title": "Medicine is not available",
  "status": 422,
  "detail": "Chlorphenamine 4 mg is inactive and cannot be prescribed.",
  "instance": "/api/v1/consultations/40000000-0000-4000-8000-000000000002/prescriptions",
  "code": "MEDICINE_INACTIVE",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "prescription.items[1].medicineId",
      "code": "Inactive",
      "message": "medicine is inactive"
    }
  ]
}
```


Aspect	Details
Database entities affected	prescriptions (W), prescription_items (W), consultations (R), medicines (R), idempotency_keys (W), audit_events (W)
Transaction / concurrency	The partial unique index uq_prescriptions_active_per_consultation guarantees at most one active prescription even under concurrent requests; a violation maps to 409 ACTIVE_PRESCRIPTION_EXISTS.
Audit events	`PRESCRIPTION_CREATED`

10.2 GET /api/v1/prescriptions

Item	Value
Method / path	`GET /api/v1/prescriptions`
operationId	`listPrescriptions`
Purpose	List prescriptions (pharmacist queue / doctor's own prescriptions). PHARMACIST: all prescriptions. DOCTOR: only prescriptions of consultations the caller recorded (the server forces prescribedBy = caller; the parameter is ignored for doctors). Users with both roles get the pharmacist view. Default filter: status=PENDING,IN_PROGRESS, oldest first.
Required role(s)	PHARMACIST, DOCTOR
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>

Header	Required	Value / notes
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`status`	query	array of PrescriptionStatus	no	unique items	Repeatable. Default PENDING and IN_PROGRESS.
`patientNumber`	query	string	no	max length 20	Exact patient number (upper-cased by the server).
`prescriptionNumber`	query	string	no	pattern `^(RX-)?[0-9]{1,12}$`	Exact prescription number, with or without RX- prefix.
`submittedFrom`	query	string (date-time)	no		Inclusive lower bound of submittedAt.
`submittedTo`	query	string (date-time)	no		Exclusive upper bound of submittedAt.
`page`	query	integer	no	min 0; default 0	Zero-based page index.
`size`	query	integer	no	min 1; max 100; default 20	Page size.
`sort`	query	string	no	default submittedAt,asc; one of submittedAt,asc, submittedAt,desc	

Request body: none.

Business rules

- Returns summaries only; no clinical notes or diagnoses. Polled by the pharmacist dashboard every NEXT_PUBLIC_PRESCRIPTION_POLL_INTERVAL_MS.
- List calls are not audited individually (to avoid one audit row per poll); opening a prescription is.

Responses

Status	Meaning / error codes	Body
200	Page of prescription summaries.	PrescriptionSummaryPage
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "items": [
    {
      "id": "50000000-0000-4000-8000-000000000002",
      "prescriptionNumber": "RX-000002",
      "consultationId": "40000000-0000-4000-8000-000000000002",
      "status": "PENDING",
      "patient": {
```

```
        "id": "20000000-0000-4000-8000-000000000001",
        "patientNumber": "P-0001",
        "fullName": "Saman Kumara"
      },
      "prescribedBy": {
        "id": "00000000-0000-4000-8000-000000000002",
        "fullName": "Dr. Nimal Perera"
      },
      "submittedAt": "2026-10-12T03:03:31Z",
      "itemCount": 2,
      "startedBy": null,
      "dispensingOutcome": null,
      "dispensedAt": null
    }
  ],
  "page": 0,
  "size": 20,
  "totalItems": 1,
  "totalPages": 1
}
```


Error response example (400 VALIDATION_FAILED; all codes for this endpoint are listed under Responses)


```
{
  "type": "https://dms.example/problems/validation-failed",
  "title": "Validation failed",
  "status": 400,
  "detail": "One or more fields are invalid.",
  "instance": "/api/v1/prescriptions",
  "code": "VALIDATION_FAILED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "fullName",
      "code": "NotBlank",
      "message": "must not be blank"
    }
  ]
}
```


Aspect	Details
Database entities affected	prescriptions (R), patients (R), consultations (R), users (R), prescription_items (R, count), dispensings (R)
Transaction / concurrency	Read-only. Single query with joins plus a grouped item count (no N+1).
Audit events	none

10.3 GET /api/v1/prescriptions/{prescriptionId}

Item	Value
Method / path	`GET /api/v1/prescriptions/{prescriptionId}`
operationId	`getPrescription`
Purpose	Get a prescription with items, minimal patient data and dispensing result. The dispensing view. Contains only the patient data needed to dispense safely (name, number, sex, age, known allergies) - no diagnoses or clinical notes.
Required role(s)	PHARMACIST, DOCTOR
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`prescriptionId`	path	string (uuid)	yes		Prescription UUID.

Request body: none.

Responses

Status	Meaning / error codes	Body
200	Prescription.	PrescriptionResponse
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`PRESCRIPTION_NOT_FOUND`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "50000000-0000-4000-8000-000000000002",
  "prescriptionNumber": "RX-000002",
  "consultationId": "40000000-0000-4000-8000-000000000002",
  "consultationAt": "2026-10-12T03:02:31Z",
  "status": "PENDING",
  "notes": "Patient allergic to penicillin",
  "submittedAt": "2026-10-12T03:03:31Z",
  "prescribedBy": {
    "id": "00000000-0000-4000-8000-000000000002",
    "fullName": "Dr. Nimal Perera"
  },
  "patient": {
    "id": "20000000-0000-4000-8000-000000000001",
    "patientNumber": "P-0001",
    "fullName": "Saman Kumara",
    "sex": "MALE",
    "ageYears": 48,
    "dateOfBirthEstimated": false,
    "knownAllergies": "Penicillin (rash)"
  },
  "items": [
    {
      "id": "51000000-0000-4000-8000-000000000003",
      "lineNumber": 1,
      "medicineId": "10000000-0000-4000-8000-000000000005",
      "medicineName": "Metformin",
      "strength": "500 mg",
      "dosageForm": "TABLET",
      "dose": "1 tablet",
      "frequency": "Twice daily (BD)",
      "durationDays": 30,
      "quantity": 60,
      "quantityUnit": "tablets",
      "instructions": "With meals"
    },
    {
```

```
      "id": "51000000-0000-4000-8000-000000000004",
      "lineNumber": 2,
      "medicineId": "10000000-0000-4000-8000-000000000006",
      "medicineName": "Amlodipine",
      "strength": "5 mg",
      "dosageForm": "TABLET",
      "dose": "1 tablet",
      "frequency": "Once daily (OD) in the morning",
      "durationDays": 30,
      "quantity": 30,
      "quantityUnit": "tablets",
      "instructions": null
    }
  ],
  "startedAt": null,
  "startedBy": null,
  "cancelledAt": null,
  "cancelledBy": null,
  "cancelReason": null,
  "supersedesPrescriptionId": null,
  "supersededByPrescriptionId": null,
  "dispensing": null,
  "version": 0
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 PRESCRIPTION_NOT_FOUND


```
{
  "type": "https://dms.example/problems/prescription-not-found",
  "title": "Prescription not found",
  "status": 404,
  "detail": "Prescription not found.",
  "instance": "/api/v1/prescriptions/50000000-0000-4000-8000-000000000002",
  "code": "PRESCRIPTION_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


Aspect	Details
Database entities affected	prescriptions (R), prescription_items (R), dispensings (R), dispensing_items (R), patients (R), consultations (R), users (R), audit_events (W)
Transaction / concurrency	Read-only, single transaction.
Audit events	`PRESCRIPTION_VIEWED`

10.4 POST /api/v1/prescriptions/{prescriptionId}/cancel

Item	Value
Method / path	`POST /api/v1/prescriptions/{prescriptionId}/cancel`
operationId	`cancelPrescription`
Purpose	Cancel a prescription that has not been dispensed
Required role(s)	DOCTOR, PHARMACIST
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`prescriptionId`	path	string (uuid)	yes		Prescription UUID.

Request body - schema `CancelPrescriptionRequest` (required)

```
{
  "reason": "Wrong strength prescribed - re-issuing"
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`reason`	string	yes	min length 3; max length 500	

Business rules

- Allowed from PENDING or IN_PROGRESS only. DISPENSED and CANCELLED are terminal → 409 INVALID_STATUS_TRANSITION.
- A DOCTOR may cancel only prescriptions of consultations they recorded (403 NOT_RESOURCE_OWNER); a PHARMACIST may cancel any (e.g. patient never collected, nothing available).
- reason is mandatory and stored in cancel_reason; it is not copied to the audit log.
- Cancelling never deletes items; the doctor can re-issue with POST /consultations/{id}/prescriptions.

Responses

Status	Meaning / error codes	Body
200	Cancelled.	PrescriptionResponse
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`NOT_RESOURCE_OWNER`, `FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`PRESCRIPTION_NOT_FOUND`	Problem
409	`INVALID_STATUS_TRANSITION`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)

```
{
  "id": "50000000-0000-4000-8000-000000000002",
  "prescriptionNumber": "RX-000002",
  "consultationId": "40000000-0000-4000-8000-000000000002",
  "consultationAt": "2026-10-12T03:02:31Z",
```

```
  "status": "CANCELLED",
  "notes": "Patient allergic to penicillin",
  "submittedAt": "2026-10-12T03:03:31Z",
  "prescribedBy": {
    "id": "00000000-0000-4000-8000-000000000002",
    "fullName": "Dr. Nimal Perera"
  },
  "patient": {
    "id": "20000000-0000-4000-8000-000000000001",
    "patientNumber": "P-0001",
    "fullName": "Saman Kumara",
    "sex": "MALE",
    "ageYears": 48,
    "dateOfBirthEstimated": false,
    "knownAllergies": "Penicillin (rash)"
  },
  "items": [
    {
      "id": "51000000-0000-4000-8000-000000000003",
      "lineNumber": 1,
      "medicineId": "10000000-0000-4000-8000-000000000005",
      "medicineName": "Metformin",
      "strength": "500 mg",
      "dosageForm": "TABLET",
      "dose": "1 tablet",
      "frequency": "Twice daily (BD)",
      "durationDays": 30,
      "quantity": 60,
      "quantityUnit": "tablets",
      "instructions": "With meals"
    }
  ],
  "startedAt": null,
  "startedBy": null,
  "cancelledAt": "2026-10-12T03:12:00Z",
  "cancelledBy": {
    "id": "00000000-0000-4000-8000-000000000002",
    "fullName": "Dr. Nimal Perera"
  },
  "cancelReason": "Wrong strength prescribed - re-issuing",
  "supersedesPrescriptionId": null,
  "supersededByPrescriptionId": null,
  "dispensing": null,
  "version": 1
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 PRESCRIPTION_NOT_FOUND


```
{
  "type": "https://dms.example/problems/prescription-not-found",
  "title": "Prescription not found",
  "status": 404,
  "detail": "Prescription not found.",
  "instance": "/api/v1/prescriptions/50000000-0000-4000-8000-000000000002/cancel",
  "code": "PRESCRIPTION_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


403 NOT_RESOURCE_OWNER


```
{
  "type": "https://dms.example/problems/not-resource-owner",
  "title": "Not the owner of this resource",
  "status": 403,
  "detail": "Only the prescribing doctor can cancel this prescription.",
  "instance": "/api/v1/prescriptions/50000000-0000-4000-8000-000000000002/cancel",
  "code": "NOT_RESOURCE_OWNER",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

409 INVALID_STATUS_TRANSITION


```
{
  "type": "https://dms.example/problems/invalid-status-transition",
  "title": "Invalid status transition",
  "status": 409,
  "detail": "Prescription is already DISPENSED.",
  "instance": "/api/v1/prescriptions/50000000-0000-4000-8000-000000000002/cancel",
  "code": "INVALID_STATUS_TRANSITION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "currentStatus": "DISPENSED"
}
```


Aspect	Details
Database entities affected	prescriptions (W), consultations (R), audit_events (W)
Transaction / concurrency	Row locked FOR UPDATE, status re-checked. Cancel and dispense racing on the same prescription serialise on the lock; the second gets 409.
Audit events	`PRESCRIPTION_CANCELLED`

11. Pharmacy and dispensing

Pharmacist workflow - start and dispense.

11.1 POST /api/v1/prescriptions/{prescriptionId}/start

Item	Value
Method / path	`POST /api/v1/prescriptions/{prescriptionId}/start`
operationId	`startPrescription`
Purpose	Mark a prescription IN_PROGRESS (pharmacist has started preparing it)
Required role(s)	PHARMACIST
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`prescriptionId`	path	string (uuid)	yes		Prescription UUID.

Request body: none.

Business rules

- PENDING → IN_PROGRESS; records started_at and started_by.

- Already IN_PROGRESS → 200 with the unchanged prescription (idempotent; any pharmacist may continue another pharmacist's work).
- DISPENSED or CANCELLED → 409 INVALID_STATUS_TRANSITION with currentStatus.

Responses

Status	Meaning / error codes	Body
200	Prescription now IN_PROGRESS.	PrescriptionResponse
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`PRESCRIPTION_NOT_FOUND`	Problem
409	`INVALID_STATUS_TRANSITION`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "50000000-0000-4000-8000-000000000002",
  "prescriptionNumber": "RX-000002",
  "consultationId": "40000000-0000-4000-8000-000000000002",
  "consultationAt": "2026-10-12T03:02:31Z",
  "status": "IN_PROGRESS",
  "notes": "Patient allergic to penicillin",
  "submittedAt": "2026-10-12T03:03:31Z",
  "prescribedBy": {
    "id": "00000000-0000-4000-8000-000000000002",
    "fullName": "Dr. Nimal Perera"
  },
  "patient": {
    "id": "20000000-0000-4000-8000-000000000001",
    "patientNumber": "P-0001",
    "fullName": "Saman Kumara",
    "sex": "MALE",
    "ageYears": 48,
    "dateOfBirthEstimated": false,
    "knownAllergies": "Penicillin (rash)"
  },
  "items": [
    {
      "id": "51000000-0000-4000-8000-000000000003",
      "lineNumber": 1,
      "medicineId": "10000000-0000-4000-8000-000000000005",
      "medicineName": "Metformin",
      "strength": "500 mg",
      "dosageForm": "TABLET",
      "dose": "1 tablet",
      "frequency": "Twice daily (BD)",
      "durationDays": 30,
      "quantity": 60,
      "quantityUnit": "tablets",
      "instructions": "With meals"
    },
    {
      "id": "51000000-0000-4000-8000-000000000004",
      "lineNumber": 2,
      "medicineId": "10000000-0000-4000-8000-000000000006",
      "medicineName": "Amlodipine",
      "strength": "5 mg",
      "dosageForm": "TABLET",
      "dose": "1 tablet",
      "frequency": "Once daily (OD) in the morning",
      "durationDays": 30,
      "quantity": 30,
      "quantityUnit": "tablets",
      "instructions": null
    }
```

```
  ],
  "startedAt": "2026-10-12T03:10:00Z",
  "startedBy": {
    "id": "00000000-0000-4000-8000-000000000004",
    "fullName": "Kamal Silva"
  },
  "cancelledAt": null,
  "cancelledBy": null,
  "cancelReason": null,
  "supersedesPrescriptionId": null,
  "supersededByPrescriptionId": null,
  "dispensing": null,
  "version": 1
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 PRESCRIPTION_NOT_FOUND


```
{
  "type": "https://dms.example/problems/prescription-not-found",
  "title": "Prescription not found",
  "status": 404,
  "detail": "Prescription not found.",
  "instance": "/api/v1/prescriptions/50000000-0000-4000-8000-000000000002/start",
  "code": "PRESCRIPTION_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 INVALID_STATUS_TRANSITION


```
{
  "type": "https://dms.example/problems/invalid-status-transition",
  "title": "Invalid status transition",
  "status": 409,
  "detail": "Prescription is already DISPENSED.",
  "instance": "/api/v1/prescriptions/50000000-0000-4000-8000-000000000002/start",
  "code": "INVALID_STATUS_TRANSITION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "currentStatus": "DISPENSED"
}
```


Aspect	Details
Database entities affected	prescriptions (W), audit_events (W)
Transaction / concurrency	The prescription row is locked with SELECT ... FOR UPDATE (PESSIMISTIC_WRITE) before the status check.
Audit events	`PRESCRIPTION_STARTED`

11.2 POST /api/v1/prescriptions/{prescriptionId}/dispense

Item	Value
Method / path	`POST /api/v1/prescriptions/{prescriptionId}/dispense`
operationId	`dispensePrescription`
Purpose	Confirm dispensing ("Mark as Dispensed") with per-item quantities. Records exactly what was handed over for every prescribed item, the pharmacist and the timestamp, and moves the prescription to DISPENSED. Allowed from PENDING or IN_PROGRESS. This is the only way a prescription becomes DISPENSED.
Required role(s)	PHARMACIST

Item	Value
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Required (`Idempotency-Key` header) - see 2.6

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
Idempotency-Key	yes	Client-generated unique key (UUID v4 recommended) identifying one user intent. The frontend creates it
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`prescriptionId`	path	string (uuid)	yes		Prescription UUID.

Request body - schema `DispenseRequest` (required)

```
{
  "notes": "Counselled on taking Metformin with meals",
  "items": [
    {
      "prescriptionItemId": "51000000-0000-4000-8000-000000000003",
      "quantityDispensed": 60
    },
    {
      "prescriptionItemId": "51000000-0000-4000-8000-000000000004",
      "quantityDispensed": 14,
      "note": "Only 14 in stock; patient to collect remainder elsewhere"
    }
  ]
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`notes`	string | null	no	max length 1000	
`items`	array of DispenseItemInput	yes	min items 1; max items 30	
`items[].prescriptionItemId`	string (uuid)	yes		
`items[].quantityDispensed`	number	yes	min 0; max 99999.99	Max 2 decimal places.
`items[].note`	string | null	no	max length 500	Required when quantityDispensed is below the prescribed quantity.

Business rules

- items must list every prescription item of this prescription exactly once (by prescriptionItemId).
- 0 <= quantityDispensed <= prescribed quantity, max 2 decimals.

- Item status is derived - equal to prescribed = FULL, between = PARTIAL, 0 = NOT_DISPENSED. A note is required for PARTIAL and NOT_DISPENSED.
- At least one item must have quantityDispensed > 0; if nothing can be supplied, cancel the prescription instead.
- outcome = FULL when all items are FULL, otherwise PARTIAL. Remaining quantities are not tracked for later dispensing in v1.
- pharmacist = authenticated user; dispensedAt = server time.
- Dispensing is irreversible in v1 (no un-dispense endpoint). Corrections are made by a note in a new consultation/prescription.

Responses

Status	Meaning / error codes	Body
200	Dispensed (or replayed).	PrescriptionResponse
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`, `IDEMPOTENCY_KEY_REQUIRED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`PRESCRIPTION_NOT_FOUND`	Problem
409	`INVALID_STATUS_TRANSITION`	Problem
422	`DISPENSE_ITEMS_MISMATCH`, `DISPENSE_QUANTITY_EXCEEDS_PRESCRIBED`, `NOTHING_DISPENSED`, `DISPENSE_NOTE_REQUIRED`, `IDEMPOTENCY_KEY_REUSED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "50000000-0000-4000-8000-000000000002",
  "prescriptionNumber": "RX-000002",
  "consultationId": "40000000-0000-4000-8000-000000000002",
  "consultationAt": "2026-10-12T03:02:31Z",
  "status": "DISPENSED",
  "notes": "Patient allergic to penicillin",
  "submittedAt": "2026-10-12T03:03:31Z",
  "prescribedBy": {
    "id": "00000000-0000-4000-8000-000000000002",
    "fullName": "Dr. Nimal Perera"
  },
  "patient": {
    "id": "20000000-0000-4000-8000-000000000001",
    "patientNumber": "P-0001",
    "fullName": "Saman Kumara",
    "sex": "MALE",
    "ageYears": 48,
    "dateOfBirthEstimated": false,
    "knownAllergies": "Penicillin (rash)"
  },
  "items": [
    {
      "id": "51000000-0000-4000-8000-000000000003",
      "lineNumber": 1,
      "medicineId": "10000000-0000-4000-8000-000000000005",
      "medicineName": "Metformin",
      "strength": "500 mg",
      "dosageForm": "TABLET",
      "dose": "1 tablet",
      "frequency": "Twice daily (BD)",
      "durationDays": 30,
      "quantity": 60,
      "quantityUnit": "tablets",
      "instructions": "With meals"
    },
    {
```

```
      "id": "51000000-0000-4000-8000-000000000004",
      "lineNumber": 2,
      "medicineId": "10000000-0000-4000-8000-000000000006",
      "medicineName": "Amlodipine",
      "strength": "5 mg",
      "dosageForm": "TABLET",
      "dose": "1 tablet",
      "frequency": "Once daily (OD) in the morning",
      "durationDays": 30,
      "quantity": 30,
      "quantityUnit": "tablets",
      "instructions": null
    }
  ],
  "startedAt": "2026-10-12T03:10:00Z",
  "startedBy": {
    "id": "00000000-0000-4000-8000-000000000004",
    "fullName": "Kamal Silva"
  },
  "cancelledAt": null,
  "cancelledBy": null,
  "cancelReason": null,
  "supersedesPrescriptionId": null,
  "supersededByPrescriptionId": null,
  "dispensing": {
    "id": "52000000-0000-4000-8000-000000000002",
    "pharmacist": {
      "id": "00000000-0000-4000-8000-000000000004",
      "fullName": "Kamal Silva"
    },
    "dispensedAt": "2026-10-12T03:15:00Z",
    "outcome": "PARTIAL",
    "notes": "Counselled on taking Metformin with meals",
    "items": [
      {
        "prescriptionItemId": "51000000-0000-4000-8000-000000000003",
        "quantityDispensed": 60,
        "status": "FULL",
        "note": null
      },
      {
        "prescriptionItemId": "51000000-0000-4000-8000-000000000004",
        "quantityDispensed": 14,
        "status": "PARTIAL",
        "note": "Only 14 in stock; patient to collect remainder elsewhere"
      }
    ]
  },
  "version": 2
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 PRESCRIPTION_NOT_FOUND


```
{
  "type": "https://dms.example/problems/prescription-not-found",
  "title": "Prescription not found",
  "status": 404,
  "detail": "Prescription not found.",
  "instance": "/api/v1/prescriptions/50000000-0000-4000-8000-000000000002/dispense",
  "code": "PRESCRIPTION_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 INVALID_STATUS_TRANSITION


```
{
  "type": "https://dms.example/problems/invalid-status-transition",
  "title": "Invalid status transition",
  "status": 409,
  "detail": "Prescription is already DISPENSED.",
  "instance": "/api/v1/prescriptions/50000000-0000-4000-8000-000000000002/dispense",
```

```
  "code": "INVALID_STATUS_TRANSITION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "currentStatus": "DISPENSED"
}
```

422 DISPENSE_ITEMS_MISMATCH

```
{
  "type": "https://dms.example/problems/dispense-items-mismatch",
  "title": "Dispense items do not match",
  "status": 422,
  "detail": "Dispense request must include each prescription item exactly once.",
  "instance": "/api/v1/prescriptions/50000000-0000-4000-8000-000000000002/dispense",
  "code": "DISPENSE_ITEMS_MISMATCH",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

422 DISPENSE_QUANTITY_EXCEEDS_PRESCRIBED

```
{
  "type": "https://dms.example/problems/dispense-quantity-exceeds-prescribed",
  "title": "Quantity exceeds prescribed",
  "status": 422,
  "detail": "Line 1: 30 tablets dispensed but 24 prescribed.",
  "instance": "/api/v1/prescriptions/50000000-0000-4000-8000-000000000002/dispense",
  "code": "DISPENSE_QUANTITY_EXCEEDS_PRESCRIBED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "items[0].quantityDispensed",
      "code": "Max",
      "message": "must not exceed prescribed quantity 24.00"
    }
  ]
}
```

422 NOTHING_DISPENSED

```
{
  "type": "https://dms.example/problems/nothing-dispensed",
  "title": "Nothing dispensed",
  "status": 422,
  "detail": "At least one item must be dispensed. Cancel the prescription if nothing can be supplied.",
  "instance": "/api/v1/prescriptions/50000000-0000-4000-8000-000000000002/dispense",
  "code": "NOTHING_DISPENSED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

422 DISPENSE_NOTE_REQUIRED

```
{
  "type": "https://dms.example/problems/dispense-note-required",
  "title": "Note required",
  "status": 422,
  "detail": "A note is required for partially or not dispensed items.",
  "instance": "/api/v1/prescriptions/50000000-0000-4000-8000-000000000002/dispense",
  "code": "DISPENSE_NOTE_REQUIRED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "items[1].note",
      "code": "Required",
      "message": "note is required when not fully dispensed"
    }
  ]
}
```

Aspect	Details
Database entities affected	prescriptions (W), prescription_items (R), dispensings (W), dispensing_items (W), idempotency_keys (W), audit_events (W)
Transaction / concurrency	Prescription row locked FOR UPDATE, status re-checked inside the transaction, then dispensing + items inserted and status set. uq_dispensings_prescription makes a second dispensing impossible even if application checks were bypassed. A double click with the same Idempotency-Key replays the first 200 response; a second pharmacist dispensing concurrently gets 409 INVALID_STATUS_TRANSITION (currentStatus DISPENSED).
Audit events	`PRESCRIPTION_DISPENSED`

12. Medicines

Lightweight medicine catalogue (no inventory).

12.1 GET /api/v1/medicines

Item	Value
Method / path	`GET /api/v1/medicines`
operationId	`listMedicines`
Purpose	Search the medicine catalogue (prescription autocomplete and catalogue management)
Required role(s)	DOCTOR, PHARMACIST, ADMIN
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`search`	query	string	no	min length 1; max length 100	Case-insensitive substring of name (1-100 chars).
`active`	query	boolean	no		true = only active (used by the prescription editor), false = only inactive, omitted = both (catalogue management).
`page`	query	integer	no	min 0; default 0	Zero-based page index.
`size`	query	integer	no	min 1; max 100; default 20	Page size.

Request body: none.

Responses

Status	Meaning / error codes	Body
200	Page of medicines sorted by name, strength.	MedicinePage
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "items": [
    {
      "id": "10000000-0000-4000-8000-000000000001",
      "name": "Paracetamol",
      "strength": "500 mg",
      "dosageForm": "TABLET",
      "active": true,
      "createdAt": "2026-09-01T03:00:00Z",
      "updatedAt": "2026-09-01T03:00:00Z",
      "version": 0
    },
    {
      "id": "10000000-0000-4000-8000-000000000002",
      "name": "Paracetamol",
      "strength": "120 mg/5 ml",
      "dosageForm": "SYRUP",
      "active": true,
      "createdAt": "2026-09-01T03:00:00Z",
      "updatedAt": "2026-09-01T03:00:00Z",
      "version": 0
    }
  ],
  "page": 0,
  "size": 20,
  "totalItems": 2,
  "totalPages": 1
}
```


Error response example (400 VALIDATION_FAILED; all codes for this endpoint are listed under Responses)


```
{
  "type": "https://dms.example/problems/validation-failed",
  "title": "Validation failed",
  "status": 400,
  "detail": "One or more fields are invalid.",
  "instance": "/api/v1/medicines",
  "code": "VALIDATION_FAILED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "fullName",
      "code": "NotBlank",
      "message": "must not be blank"
    }
  ]
}
```


Aspect	Details
Database entities affected	medicines (R)
Transaction / concurrency	Read-only, single transaction.
Audit events	none

12.2 POST /api/v1/medicines

Item	Value
Method / path	`POST /api/v1/medicines`
operationId	`createMedicine`
Purpose	Add a medicine to the catalogue
Required role(s)	PHARMACIST, ADMIN
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters: none.

Request body - schema `CreateMedicineRequest` (required)

```
{
  "name": "Losartan",
  "strength": "50 mg",
  "dosageForm": "TABLET"
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`name`	string	yes	min length 1; max length 200	
`strength`	string | null	no	max length 50	
`dosageForm`	DosageForm	yes	one of TABLET, CAPSULE, SYRUP, SUSPENSION, SOLUTION, INJECTION, CREAM, OINTMENT, GEL, DROPS, INHALER, SUPPOSITORY, POWDER, PATCH, OTHER	

Business rules

- (lower(name), lower(strength), dosageForm) must be unique.

Responses

Status	Meaning / error codes	Body
201	Created.	MedicineResponse
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
409	`DUPLICATE_MEDICINE`	Problem

Status	Meaning / error codes	Body
500	`INTERNAL_ERROR`	Problem

Successful response example (201)


```
{
  "id": "10000000-0000-4000-8000-000000000099",
  "name": "Losartan",
  "strength": "50 mg",
  "dosageForm": "TABLET",
  "active": true,
  "createdAt": "2026-10-12T05:00:00Z",
  "updatedAt": "2026-10-12T05:00:00Z",
  "version": 0
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

409 DUPLICATE_MEDICINE


```
{
  "type": "https://dms.example/problems/duplicate-medicine",
  "title": "Medicine already exists",
  "status": 409,
  "detail": "Paracetamol 500 mg TABLET already exists in the catalogue.",
  "instance": "/api/v1/medicines",
  "code": "DUPLICATE_MEDICINE",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


Aspect	Details
Database entities affected	medicines (W), audit_events (W)
Transaction / concurrency	uq_medicines_name_strength_form is the final guard; violation → 409 DUPLICATE_MEDICINE.
Audit events	`MEDICINE_CREATED`

12.3 GET /api/v1/medicines/{medicineId}

Item	Value
Method / path	`GET /api/v1/medicines/{medicineId}`
operationId	`getMedicine`
Purpose	Get a catalogue medicine
Required role(s)	DOCTOR, PHARMACIST, ADMIN
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)

Header	Required	Value / notes
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`medicineId`	path	string (uuid)	yes		Medicine UUID.

Request body: none.

Responses

Status	Meaning / error codes	Body
200	Medicine.	MedicineResponse
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`MEDICINE_NOT_FOUND`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "10000000-0000-4000-8000-000000000011",
  "name": "Chlorphenamine",
  "strength": "4 mg",
  "dosageForm": "TABLET",
  "active": false,
  "createdAt": "2026-09-01T03:00:00Z",
  "updatedAt": "2026-10-01T03:00:00Z",
  "version": 1
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 MEDICINE_NOT_FOUND


```
{
  "type": "https://dms.example/problems/medicine-not-found",
  "title": "Medicine not found",
  "status": 404,
  "detail": "Medicine not found.",
  "instance": "/api/v1/medicines/10000000-0000-4000-8000-000000000011",
  "code": "MEDICINE_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


Aspect	Details
Database entities affected	medicines (R)
Transaction / concurrency	Read-only, single transaction.
Audit events	none

12.4 PUT /api/v1/medicines/{medicineId}

Item	Value
Method / path	`PUT /api/v1/medicines/{medicineId}`
operationId	`updateMedicine`
Purpose	Update or (de)activate a catalogue medicine
Required role(s)	PHARMACIST, ADMIN
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`medicineId`	path	string (uuid)	yes		Medicine UUID.

Request body - schema `UpdateMedicineRequest` (required)

```
{
  "name": "Chlorphenamine",
  "strength": "4 mg",
  "dosageForm": "TABLET",
  "active": false,
  "version": 0
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`name`	string	yes	min length 1; max length 200	
`strength`	string | null	no	max length 50	
`dosageForm`	DosageForm	yes	one of TABLET, CAPSULE, SYRUP, SUSPENSION, SOLUTION, INJECTION, CREAM, OINTMENT, GEL, DROPS, INHALER, SUPPOSITORY, POWDER, PATCH, OTHER	
`active`	boolean	yes		
`version`	integer (int64)	yes	min 0	

Business rules

- Setting active = false marks the medicine unavailable for NEW prescriptions only; existing prescription items keep their snapshot and can still be dispensed.

- Medicines are never deleted (prescription_items reference them).

Responses

Status	Meaning / error codes	Body
200	Updated.	MedicineResponse
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`MEDICINE_NOT_FOUND`	Problem
409	`DUPLICATE_MEDICINE`, `CONCURRENT_MODIFICATION`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "10000000-0000-4000-8000-000000000011",
  "name": "Chlorphenamine",
  "strength": "4 mg",
  "dosageForm": "TABLET",
  "active": false,
  "createdAt": "2026-09-01T03:00:00Z",
  "updatedAt": "2026-10-12T05:05:00Z",
  "version": 1
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 MEDICINE_NOT_FOUND


```
{
  "type": "https://dms.example/problems/medicine-not-found",
  "title": "Medicine not found",
  "status": 404,
  "detail": "Medicine not found.",
  "instance": "/api/v1/medicines/10000000-0000-4000-8000-000000000011",
  "code": "MEDICINE_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 DUPLICATE_MEDICINE


```
{
  "type": "https://dms.example/problems/duplicate-medicine",
  "title": "Medicine already exists",
  "status": 409,
  "detail": "Paracetamol 500 mg TABLET already exists in the catalogue.",
  "instance": "/api/v1/medicines/10000000-0000-4000-8000-000000000011",
  "code": "DUPLICATE_MEDICINE",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 CONCURRENT_MODIFICATION


```
{
  "type": "https://dms.example/problems/concurrent-modification",
  "title": "Record changed by someone else",
  "status": 409,
  "detail": "This record was modified by another user. Reload and try again.",
  "instance": "/api/v1/medicines/10000000-0000-4000-8000-000000000011",
```

```
  "code": "CONCURRENT_MODIFICATION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

Aspect	Details
Database entities affected	medicines (W), audit_events (W)
Transaction / concurrency	Optimistic locking with version.
Audit events	`MEDICINE_UPDATED`

13. Bookings

Creating and cancelling bookings. A booking is a queue entry created on behalf of a patient (telephone or walk-in).

13.1 GET /api/v1/queue-sessions/{queueSessionId}/entries

Item	Value
Method / path	`GET /api/v1/queue-sessions/{queueSessionId}/entries`
operationId	`listQueueEntries`
Purpose	List bookings / queue entries of a session (staff only)
Required role(s)	DOCTOR, PHARMACIST
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`queueSessionId`	path	string (uuid)	yes		Queue session UUID.
`status`	query	array of QueueEntryStatus	no	unique items	Repeatable status filter; default all.
`page`	query	integer	no	min 0; default 0	Zero-based page index.
`size`	query	integer	no	min 1; max 100; default 20	Page size.

Request body: none.

Business rules

- Sorted by queueNumber ascending.

Responses

Status	Meaning / error codes	Body
200	Page of entries.	QueueEntryPage
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`QUEUE_SESSION_NOT_FOUND`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "items": [
    {
      "id": "31000000-0000-4000-8000-000000000003",
      "queueSessionId": "30000000-0000-4000-8000-000000000001",
      "queueNumber": 3,
      "status": "WAITING",
      "channel": "PHONE",
      "patientName": "Ishara Gunasekara (mother calling)",
      "contactPhone": "0759876543",
      "notes": null,
      "bookedAt": "2026-10-12T02:17:31Z",
      "bookedBy": {
        "id": "00000000-0000-4000-8000-000000000004",
        "fullName": "Kamal Silva"
      },
      "calledAt": null,
      "calledBy": null,
      "completedAt": null,
      "skippedAt": null,
      "skipReason": null,
      "cancelledAt": null,
      "cancelReason": null,
      "requeueCount": 0,
      "version": 0
    }
  ],
  "page": 0,
  "size": 20,
  "totalItems": 1,
  "totalPages": 1
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 QUEUE_SESSION_NOT_FOUND


```
{
  "type": "https://dms.example/problems/queue-session-not-found",
  "title": "Queue session not found",
  "status": 404,
  "detail": "Queue session not found.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001/entries",
  "code": "QUEUE_SESSION_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

Aspect	Details
Database entities affected	queue_entries (R), users (R)
Transaction / concurrency	Read-only, single transaction.
Audit events	none

13.2 POST /api/v1/queue-sessions/{queueSessionId}/entries

Item	Value
Method / path	`POST /api/v1/queue-sessions/{queueSessionId}/entries`
operationId	`createBooking`
Purpose	Register a telephone or walk-in booking and assign the next queue number
Required role(s)	DOCTOR, PHARMACIST
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Required (`Idempotency-Key` header) - see 2.6

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
Idempotency-Key	yes	Client-generated unique key (UUID v4 recommended) identifying one user intent. The frontend creates it
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`queueSessionId`	path	string (uuid)	yes		Queue session UUID.

Request body - schema `CreateBookingRequest` (required)

```
{
  "patientName": "Mrs. Wickramasinghe",
  "contactPhone": "071-234 5678",
  "channel": "PHONE",
  "notes": "Needs wheelchair access",
  "allowDuplicate": false
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`patientName`	string	yes	min length 1; max length 150	

Field	Type	Required	Constraints	Notes
`contactPhone`	PhoneNumber	no	max length 25; pattern `^\+?[0-9 ()-]{7,24}$`	Optional. Spaces, hyphens and parentheses are removed by the server; the result must match ^+?[0-9]{7,15}$.
`channel`	BookingChannel	yes	one of PHONE, WALK_IN	
`notes`	string | null	no	max length 300	
`allowDuplicate`	boolean	no	default False	

Business rules

- Allowed while the session is SCHEDULED or OPEN.
- queueNumber = session.lastIssuedNumber + 1 (1..999). Numbers are never reused, even after cancellation.
- contactPhone is normalised (spaces, hyphens, parentheses removed). If another WAITING/SERVING entry of the session has the same phone and allowDuplicate is not true → 409 DUPLICATE_BOOKING with existingQueueNumber (staff can confirm, e.g. two family members, and resend with allowDuplicate = true and a NEW Idempotency-Key).
- patientName is what the caller says; it is not linked to a patient record and is never shown publicly. notes must not contain clinical information (UI hint).

Responses

Status	Meaning / error codes	Body
201	Booking created; tell the caller the queueNumber.	QueueEntryResponse
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`, `IDEMPOTENCY_KEY_REQUIRED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`QUEUE_SESSION_NOT_FOUND`	Problem
409	`QUEUE_SESSION_CLOSED`, `QUEUE_SESSION_FULL`, `DUPLICATE_BOOKING`	Problem
422	`IDEMPOTENCY_KEY_REUSED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (201)


```
{
  "id": "31000000-0000-4000-8000-000000000006",
  "queueSessionId": "30000000-0000-4000-8000-000000000001",
  "queueNumber": 6,
  "status": "WAITING",
  "channel": "PHONE",
  "patientName": "Mrs. Wickramasinghe",
  "contactPhone": "0712345678",
  "notes": "Needs wheelchair access",
  "bookedAt": "2026-10-12T04:40:00Z",
  "bookedBy": {
    "id": "00000000-0000-4000-8000-000000000004",
    "fullName": "Kamal Silva"
  },
  "calledAt": null,
  "calledBy": null,
  "completedAt": null,
  "skippedAt": null,
  "skipReason": null,
  "cancelledAt": null,
  "cancelReason": null,
  "requeueCount": 0,
```

```
  "version": 0
}
```

Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 QUEUE_SESSION_NOT_FOUND

```
{
  "type": "https://dms.example/problems/queue-session-not-found",
  "title": "Queue session not found",
  "status": 404,
  "detail": "Queue session not found.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001/entries",
  "code": "QUEUE_SESSION_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

409 QUEUE_SESSION_CLOSED

```
{
  "type": "https://dms.example/problems/queue-session-closed",
  "title": "Queue session closed",
  "status": 409,
  "detail": "Queue session is closed.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001/entries",
  "code": "QUEUE_SESSION_CLOSED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

409 QUEUE_SESSION_FULL

```
{
  "type": "https://dms.example/problems/queue-session-full",
  "title": "Queue session full",
  "status": 409,
  "detail": "This session has reached the maximum of 999 bookings.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001/entries",
  "code": "QUEUE_SESSION_FULL",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

409 DUPLICATE_BOOKING

```
{
  "type": "https://dms.example/problems/duplicate-booking",
  "title": "Possible duplicate booking",
  "status": 409,
  "detail": "Phone 0771234567 already holds queue number 7 in this session.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001/entries",
  "code": "DUPLICATE_BOOKING",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "existingQueueNumber": 7
}
```

Aspect	Details
Database entities affected	queue_sessions (W), queue_entries (W), idempotency_keys (W), audit_events (W)

Aspect	Details
Transaction / concurrency	One transaction: SELECT session FOR UPDATE → check status/limit/duplicate → last_issued_number += 1 → INSERT entry. All bookings of a session serialise on the session row, so numbers are unique and gap-free; a rollback undoes the increment. uq_queue_entries_session_number is the database backstop. Retries with the same Idempotency-Key return the original entry (no second number).
Audit events	`BOOKING_CREATED`

13.3 GET /api/v1/queue-entries/{queueEntryId}

Item	Value
Method / path	`GET /api/v1/queue-entries/{queueEntryId}`
operationId	`getQueueEntry`
Purpose	Get one booking / queue entry
Required role(s)	DOCTOR, PHARMACIST
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`queueEntryId`	path	string (uuid)	yes		Queue entry (booking) UUID.

Request body: none.

Responses

Status	Meaning / error codes	Body
200	Entry.	QueueEntryResponse
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`QUEUE_ENTRY_NOT_FOUND`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "31000000-0000-4000-8000-000000000004",
  "queueSessionId": "30000000-0000-4000-8000-000000000001",
  "queueNumber": 4,
```

```
  "status": "WAITING",
  "channel": "PHONE",
  "patientName": "Mr. Bandara",
  "contactPhone": "0723456789",
  "notes": null,
  "bookedAt": "2026-10-12T02:37:31Z",
  "bookedBy": {
    "id": "00000000-0000-4000-8000-000000000004",
    "fullName": "Kamal Silva"
  },
  "calledAt": null,
  "calledBy": null,
  "completedAt": null,
  "skippedAt": null,
  "skipReason": null,
  "cancelledAt": null,
  "cancelReason": null,
  "requeueCount": 0,
  "version": 0
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 QUEUE_ENTRY_NOT_FOUND


```
{
  "type": "https://dms.example/problems/queue-entry-not-found",
  "title": "Queue entry not found",
  "status": 404,
  "detail": "Queue entry not found.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003",
  "code": "QUEUE_ENTRY_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


Aspect	Details
Database entities affected	queue_entries (R)
Transaction / concurrency	Read-only, single transaction.
Audit events	none

13.4 POST /api/v1/queue-entries/{queueEntryId}/cancel

Item	Value
Method / path	`POST /api/v1/queue-entries/{queueEntryId}/cancel`
operationId	`cancelBooking`
Purpose	Cancel a waiting booking (patient called to cancel)
Required role(s)	DOCTOR, PHARMACIST
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>

Header	Required	Value / notes
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`queueEntryId`	path	string (uuid)	yes		Queue entry (booking) UUID.

Request body - schema `CancelBookingRequest` (required)

```
{
  "reason": "Patient phoned to cancel"
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`reason`	string	yes	min length 3; max length 500	

Business rules

- WAITING → CANCELLED with mandatory reason. Already CANCELLED → 200 unchanged. SERVING/COMPLETED/SKIPPED → 409.
- The queue number is not reused.

Responses

Status	Meaning / error codes	Body
200	Cancelled.	QueueEntryResponse
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`QUEUE_ENTRY_NOT_FOUND`	Problem
409	`QUEUE_SESSION_CLOSED`, `INVALID_STATUS_TRANSITION`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "31000000-0000-4000-8000-000000000004",
  "queueSessionId": "30000000-0000-4000-8000-000000000001",
  "queueNumber": 4,
  "status": "CANCELLED",
  "channel": "PHONE",
  "patientName": "Mr. Bandara",
  "contactPhone": "0723456789",
  "notes": null,
  "bookedAt": "2026-10-12T02:37:31Z",
  "bookedBy": {
    "id": "00000000-0000-4000-8000-000000000004",
    "fullName": "Kamal Silva"
```

```
  },
  "calledAt": null,
  "calledBy": null,
  "completedAt": null,
  "skippedAt": null,
  "skipReason": null,
  "cancelledAt": "2026-10-12T04:00:00Z",
  "cancelReason": "Patient phoned to cancel",
  "requeueCount": 0,
  "version": 1
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 QUEUE_ENTRY_NOT_FOUND


```
{
  "type": "https://dms.example/problems/queue-entry-not-found",
  "title": "Queue entry not found",
  "status": 404,
  "detail": "Queue entry not found.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/cancel",
  "code": "QUEUE_ENTRY_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 QUEUE_SESSION_CLOSED


```
{
  "type": "https://dms.example/problems/queue-session-closed",
  "title": "Queue session closed",
  "status": 409,
  "detail": "Queue session is closed.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/cancel",
  "code": "QUEUE_SESSION_CLOSED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 INVALID_STATUS_TRANSITION


```
{
  "type": "https://dms.example/problems/invalid-status-transition",
  "title": "Invalid status transition",
  "status": 409,
  "detail": "Prescription is already DISPENSED.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/cancel",
  "code": "INVALID_STATUS_TRANSITION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "currentStatus": "DISPENSED"
}
```


Aspect	Details
Database entities affected	queue_sessions (W), queue_entries (W), audit_events (W)
Transaction / concurrency	Session row locked FOR UPDATE.
Audit events	`BOOKING_CANCELLED`

14. Queue management

Queue sessions and queue-entry transitions (call, complete, skip, requeue).

14.1 GET /api/v1/queue-sessions

Item	Value
Method / path	`GET /api/v1/queue-sessions`
operationId	`listQueueSessions`
Purpose	List queue sessions for a date
Required role(s)	DOCTOR, PHARMACIST
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`date`	query	string (date)	no		Session date (YYYY-MM-DD). Default today in APP_TIMEZONE.
`status`	query	array of QueueSessionStatus	no	unique items	
`page`	query	integer	no	min 0; default 0	Zero-based page index.
`size`	query	integer	no	min 1; max 100; default 20	Page size.

Request body: none.

Responses

Status	Meaning / error codes	Body
200	Sessions sorted by name.	QueueSessionPage
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "items": [
    {
      "id": "30000000-0000-4000-8000-000000000001",
      "sessionDate": "2026-10-12",
      "name": "Morning OPD - Room 1",
      "status": "OPEN",
      "assignedDoctor": {
        "id": "00000000-0000-4000-8000-000000000002",
```

```
        "fullName": "Dr. Nimal Perera"
      },
      "lastIssuedNumber": 5,
      "currentServingNumber": 2,
      "nextWaitingNumber": 3,
      "counts": {
        "waiting": 3,
        "serving": 1,
        "completed": 1,
        "skipped": 0,
        "cancelled": 0
      },
      "servingEntries": [
        {
          "id": "31000000-0000-4000-8000-000000000002",
          "queueNumber": 2,
          "patientName": "Dilani Wijesekara",
          "calledAt": "2026-10-12T04:27:31Z",
          "calledBy": {
            "id": "00000000-0000-4000-8000-000000000002",
            "fullName": "Dr. Nimal Perera"
          }
        }
      ],
      "openedAt": "2026-10-12T02:37:31Z",
      "closedAt": null,
      "createdAt": "2026-10-12T01:00:00Z",
      "updatedAt": "2026-10-12T04:27:31Z",
      "version": 9
    }
  ],
  "page": 0,
  "size": 20,
  "totalItems": 1,
  "totalPages": 1
}
```


Error response example (400 VALIDATION_FAILED; all codes for this endpoint are listed under Responses)


```
{
  "type": "https://dms.example/problems/validation-failed",
  "title": "Validation failed",
  "status": 400,
  "detail": "One or more fields are invalid.",
  "instance": "/api/v1/queue-sessions",
  "code": "VALIDATION_FAILED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "fullName",
      "code": "NotBlank",
      "message": "must not be blank"
    }
  ]
}
```


Aspect	Details
Database entities affected	queue_sessions (R), queue_entries (R, counts), users (R)
Transaction / concurrency	Read-only, single transaction.
Audit events	none

14.2 POST /api/v1/queue-sessions

Item	Value
Method / path	`POST /api/v1/queue-sessions`

Item	Value
operationId	`createQueueSession`
Purpose	Create a queue session (e.g. today's clinic for a room/doctor)
Required role(s)	DOCTOR, PHARMACIST
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters: none.

Request body - schema `CreateQueueSessionRequest` (required)

```
{
  "sessionDate": "2026-10-13",
  "name": "Evening OPD - Room 2",
  "assignedDoctorId": "00000000-0000-4000-8000-000000000003"
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`sessionDate`	string (date)	yes		
`name`	string	yes	min length 1; max length 100	
`assignedDoctorId`	string | null (uuid)	no		

Business rules

- sessionDate must be between today and today + 30 days (APP_TIMEZONE).
- name is trimmed; unique per date case-insensitively.
- Created in status SCHEDULED with lastIssuedNumber 0. Bookings are accepted immediately; calling requires opening.

Responses

Status	Meaning / error codes	Body
201	Created.	QueueSessionResponse
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
409	`DUPLICATE_QUEUE_SESSION`	Problem

Status	Meaning / error codes	Body
422	`INVALID_SESSION_DATE`, `ASSIGNED_USER_NOT_DOCTOR`, `REFERENCED_RESOURCE_NOT_FOUND`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (201)


```
{
  "id": "30000000-0000-4000-8000-000000000003",
  "sessionDate": "2026-10-13",
  "name": "Evening OPD - Room 2",
  "status": "SCHEDULED",
  "assignedDoctor": {
    "id": "00000000-0000-4000-8000-000000000003",
    "fullName": "Dr. Anusha Fernando"
  },
  "lastIssuedNumber": 0,
  "currentServingNumber": null,
  "nextWaitingNumber": null,
  "counts": {
    "waiting": 0,
    "serving": 0,
    "completed": 0,
    "skipped": 0,
    "cancelled": 0
  },
  "servingEntries": [],
  "openedAt": null,
  "closedAt": null,
  "createdAt": "2026-10-12T05:00:00Z",
  "updatedAt": "2026-10-12T05:00:00Z",
  "version": 0
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

409 DUPLICATE_QUEUE_SESSION


```
{
  "type": "https://dms.example/problems/duplicate-queue-session",
  "title": "Queue session already exists",
  "status": 409,
  "detail": "A session named 'Morning OPD - Room 1' already exists on 2026-10-12.",
  "instance": "/api/v1/queue-sessions",
  "code": "DUPLICATE_QUEUE_SESSION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


422 INVALID_SESSION_DATE


```
{
  "type": "https://dms.example/problems/invalid-session-date",
  "title": "Invalid session date",
  "status": 422,
  "detail": "Session date must be between today and 30 days ahead.",
  "instance": "/api/v1/queue-sessions",
  "code": "INVALID_SESSION_DATE",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


422 ASSIGNED_USER_NOT_DOCTOR


```
{
  "type": "https://dms.example/problems/assigned-user-not-doctor",
  "title": "Assigned user is not a doctor",
```

```
  "status": 422,
  "detail": "Assigned user must be an enabled doctor.",
  "instance": "/api/v1/queue-sessions",
  "code": "ASSIGNED_USER_NOT_DOCTOR",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


422 REFERENCED_RESOURCE_NOT_FOUND


```
{
  "type": "https://dms.example/problems/referenced-resource-not-found",
  "title": "Referenced resource not found",
  "status": 422,
  "detail": "medicineId does not reference an existing medicine.",
  "instance": "/api/v1/queue-sessions",
  "code": "REFERENCED_RESOURCE_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "prescription.items[0].medicineId",
      "code": "NotFound",
      "message": "medicine not found"
    }
  ]
}
```


Aspect	Details
Database entities affected	queue_sessions (W), users (R), audit_events (W)
Transaction / concurrency	uq_queue_sessions_date_name guards concurrent creation; violation → 409 DUPLICATE_QUEUE_SESSION.
Audit events	`QUEUE_SESSION_CREATED`

14.3 GET /api/v1/queue-sessions/{queueSessionId}

Item	Value
Method / path	`GET /api/v1/queue-sessions/{queueSessionId}`
operationId	`getQueueSession`
Purpose	Get queue session status for staff (counts, now serving, next number). Polled by the staff queue dashboard every NEXT_PUBLIC_QUEUE_POLL_INTERVAL_MS. Supports If-None-Match / ETag like the public endpoint.
Required role(s)	DOCTOR, PHARMACIST
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
If-None-Match	no	
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`queueSessionId`	path	string (uuid)	yes		Queue session UUID.

Request body: none.

Responses

Status	Meaning / error codes	Body
200	Session.	QueueSessionResponse
304	Not modified (ETag matched).	(no body)
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`QUEUE_SESSION_NOT_FOUND`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "30000000-0000-4000-8000-000000000001",
  "sessionDate": "2026-10-12",
  "name": "Morning OPD - Room 1",
  "status": "OPEN",
  "assignedDoctor": {
    "id": "00000000-0000-4000-8000-000000000002",
    "fullName": "Dr. Nimal Perera"
  },
  "lastIssuedNumber": 5,
  "currentServingNumber": 2,
  "nextWaitingNumber": 3,
  "counts": {
    "waiting": 3,
    "serving": 1,
    "completed": 1,
    "skipped": 0,
    "cancelled": 0
  },
  "servingEntries": [
    {
      "id": "31000000-0000-4000-8000-000000000002",
      "queueNumber": 2,
      "patientName": "Dilani Wijesekara",
      "calledAt": "2026-10-12T04:27:31Z",
      "calledBy": {
        "id": "00000000-0000-4000-8000-000000000002",
        "fullName": "Dr. Nimal Perera"
      }
    }
  ],
  "openedAt": "2026-10-12T02:37:31Z",
  "closedAt": null,
  "createdAt": "2026-10-12T01:00:00Z",
  "updatedAt": "2026-10-12T04:27:31Z",
  "version": 9
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 QUEUE_SESSION_NOT_FOUND

```
{
  "type": "https://dms.example/problems/queue-session-not-found",
  "title": "Queue session not found",
  "status": 404,
  "detail": "Queue session not found.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001",
  "code": "QUEUE_SESSION_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


Aspect	Details
Database entities affected	queue_sessions (R), queue_entries (R), users (R)
Transaction / concurrency	Read-only, single transaction.
Audit events	none

14.4 POST /api/v1/queue-sessions/{queueSessionId}/open

Item	Value
Method / path	`POST /api/v1/queue-sessions/{queueSessionId}/open`
operationId	`openQueueSession`
Purpose	Open a scheduled session so patients can be called
Required role(s)	DOCTOR, PHARMACIST
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`queueSessionId`	path	string (uuid)	yes		Queue session UUID.

Request body: none.

Business rules

- SCHEDULED → OPEN; only when sessionDate = today (APP_TIMEZONE), otherwise 422 INVALID_SESSION_DATE.
- Already OPEN → 200 unchanged (idempotent). CLOSED → 409 QUEUE_SESSION_CLOSED.

Responses

Status	Meaning / error codes	Body
200	Session open.	QueueSessionResponse
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`QUEUE_SESSION_NOT_FOUND`	Problem
409	`QUEUE_SESSION_CLOSED`	Problem
422	`INVALID_SESSION_DATE`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "30000000-0000-4000-8000-000000000001",
  "sessionDate": "2026-10-12",
  "name": "Morning OPD - Room 1",
  "status": "OPEN",
  "assignedDoctor": {
    "id": "00000000-0000-4000-8000-000000000002",
    "fullName": "Dr. Nimal Perera"
  },
  "lastIssuedNumber": 5,
  "currentServingNumber": null,
  "nextWaitingNumber": 1,
  "counts": {
    "waiting": 5,
    "serving": 0,
    "completed": 0,
    "skipped": 0,
    "cancelled": 0
  },
  "servingEntries": [],
  "openedAt": "2026-10-12T02:37:31Z",
  "closedAt": null,
  "createdAt": "2026-10-12T01:00:00Z",
  "updatedAt": "2026-10-12T02:37:31Z",
  "version": 6
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 QUEUE_SESSION_NOT_FOUND


```
{
  "type": "https://dms.example/problems/queue-session-not-found",
  "title": "Queue session not found",
  "status": 404,
  "detail": "Queue session not found.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001/open",
  "code": "QUEUE_SESSION_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 QUEUE_SESSION_CLOSED


```
{
  "type": "https://dms.example/problems/queue-session-closed",
  "title": "Queue session closed",
  "status": 409,
  "detail": "Queue session is closed.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001/open",
  "code": "QUEUE_SESSION_CLOSED",
```

```
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


422 INVALID_SESSION_DATE


```
{
  "type": "https://dms.example/problems/invalid-session-date",
  "title": "Invalid session date",
  "status": 422,
  "detail": "Session date must be between today and 30 days ahead.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001/open",
  "code": "INVALID_SESSION_DATE",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


Aspect	Details
Database entities affected	queue_sessions (W), audit_events (W)
Transaction / concurrency	Session row locked FOR UPDATE.
Audit events	`QUEUE_SESSION_OPENED`

14.5 POST /api/v1/queue-sessions/{queueSessionId}/close

Item	Value
Method / path	`POST /api/v1/queue-sessions/{queueSessionId}/close`
operationId	`closeQueueSession`
Purpose	Close a session (end of clinic)
Required role(s)	DOCTOR, PHARMACIST
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`queueSessionId`	path	string (uuid)	yes		Queue session UUID.

Request body - schema `CloseQueueSessionRequest` (optional)

```
{
  "force": true
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`force`	boolean	no	default False	

Business rules

- SCHEDULED or OPEN → CLOSED. Already CLOSED → 200 unchanged.
- Without force, 409 QUEUE_SESSION_HAS_ACTIVE_ENTRIES if any entry is WAITING or SERVING.
- With force = true, WAITING entries become CANCELLED (cancel_reason "Session closed") and SERVING entries become COMPLETED, in the same transaction.
- A nightly job (00:05 APP_TIMEZONE) force-closes sessions whose sessionDate is before today.
- CLOSED is terminal; no reopening in v1 (create a new session instead).

Responses

Status	Meaning / error codes	Body
200	Session closed.	QueueSessionResponse
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`QUEUE_SESSION_NOT_FOUND`	Problem
409	`QUEUE_SESSION_HAS_ACTIVE_ENTRIES`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "30000000-0000-4000-8000-000000000001",
  "sessionDate": "2026-10-12",
  "name": "Morning OPD - Room 1",
  "status": "CLOSED",
  "assignedDoctor": {
    "id": "00000000-0000-4000-8000-000000000002",
    "fullName": "Dr. Nimal Perera"
  },
  "lastIssuedNumber": 5,
  "currentServingNumber": 2,
  "nextWaitingNumber": null,
  "counts": {
    "waiting": 0,
    "serving": 0,
    "completed": 2,
    "skipped": 0,
    "cancelled": 3
  },
  "servingEntries": [],
  "openedAt": "2026-10-12T02:37:31Z",
  "closedAt": "2026-10-12T07:30:00Z",
  "createdAt": "2026-10-12T01:00:00Z",
  "updatedAt": "2026-10-12T07:30:00Z",
  "version": 12
}
```

Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 QUEUE_SESSION_NOT_FOUND


```
{
  "type": "https://dms.example/problems/queue-session-not-found",
  "title": "Queue session not found",
  "status": 404,
  "detail": "Queue session not found.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001/close",
  "code": "QUEUE_SESSION_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 QUEUE_SESSION_HAS_ACTIVE_ENTRIES


```
{
  "type": "https://dms.example/problems/queue-session-has-active-entries",
  "title": "Queue session has active entries",
  "status": 409,
  "detail": "3 patients are still waiting. Close with force=true to cancel them.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001/close",
  "code": "QUEUE_SESSION_HAS_ACTIVE_ENTRIES",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "waitingCount": 3,
  "servingCount": 0
}
```


Aspect	Details
Database entities affected	queue_sessions (W), queue_entries (W), audit_events (W)
Transaction / concurrency	Session row locked FOR UPDATE; entry updates are bulk UPDATEs inside the same transaction.
Audit events	`QUEUE_SESSION_CLOSED`

14.6 POST /api/v1/queue-sessions/{queueSessionId}/call-next

Item	Value
Method / path	`POST /api/v1/queue-sessions/{queueSessionId}/call-next`
operationId	`callNextPatient`
Purpose	Call the next waiting patient (the doctor's "Next patient" button). Atomically (optionally) completes the entry currently being served and moves the lowest-numbered WAITING entry to SERVING. This is the only meaning of the doctor's "increase queue number" button: it advances the serving number. It never creates a booking.
Required role(s)	DOCTOR
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>

Header	Required	Value / notes
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`queueSessionId`	path	string (uuid)	yes		Queue session UUID.

Request body - schema `CallNextRequest` (required)

```
{
  "expectedLastCalledNumber": 2,
  "completeCurrentEntryId": "31000000-0000-4000-8000-000000000002"
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`expectedLastCalledNumber`	integer | null	yes	min 1; max 999	currentServingNumber the client last saw (null if none).
`completeCurrentEntryId`	string | null (uuid)	no		

Business rules

- expectedLastCalledNumber must equal the session's currentServingNumber as last seen by the client (null if nobody has been called yet). Mismatch → 409 QUEUE_STATE_CHANGED with currentServingNumber; nothing changes. This makes double clicks and two doctors pressing at once safe - a patient can never be skipped silently.
- completeCurrentEntryId (optional) must be an entry of this session; if SERVING it becomes COMPLETED, if already COMPLETED it is ignored, any other status → 409 INVALID_STATUS_TRANSITION.
- The next entry is the WAITING entry with the lowest queueNumber (FIFO; requeued entries keep their original number and are therefore called next).
- If no entry is WAITING the response has calledEntry = null (200); the completion, if requested, is still applied.
- Session must be OPEN.

Responses

Status	Meaning / error codes	Body
200	Queue advanced (or nothing waiting).	CallNextResponse
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`QUEUE_SESSION_NOT_FOUND`	Problem
409	`QUEUE_SESSION_NOT_OPEN`, `QUEUE_SESSION_CLOSED`, `QUEUE_STATE_CHANGED`, `INVALID_STATUS_TRANSITION`	Problem

Status	Meaning / error codes	Body
422	`QUEUE_ENTRY_NOT_IN_SESSION`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "completedEntry": {
    "id": "31000000-0000-4000-8000-000000000002",
    "queueNumber": 2,
    "status": "COMPLETED"
  },
  "calledEntry": {
    "id": "31000000-0000-4000-8000-000000000003",
    "queueSessionId": "30000000-0000-4000-8000-000000000001",
    "queueNumber": 3,
    "status": "SERVING",
    "channel": "PHONE",
    "patientName": "Ishara Gunasekara (mother calling)",
    "contactPhone": "0759876543",
    "notes": null,
    "bookedAt": "2026-10-12T02:17:31Z",
    "bookedBy": {
      "id": "00000000-0000-4000-8000-000000000004",
      "fullName": "Kamal Silva"
    },
    "calledAt": "2026-10-12T04:45:00Z",
    "calledBy": {
      "id": "00000000-0000-4000-8000-000000000002",
      "fullName": "Dr. Nimal Perera"
    },
    "completedAt": null,
    "skippedAt": null,
    "skipReason": null,
    "cancelledAt": null,
    "cancelReason": null,
    "requeueCount": 0,
    "version": 1
  },
  "session": {
    "id": "30000000-0000-4000-8000-000000000001",
    "sessionDate": "2026-10-12",
    "name": "Morning OPD - Room 1",
    "status": "OPEN",
    "assignedDoctor": {
      "id": "00000000-0000-4000-8000-000000000002",
      "fullName": "Dr. Nimal Perera"
    },
    "lastIssuedNumber": 5,
    "currentServingNumber": 3,
    "nextWaitingNumber": 4,
    "counts": {
      "waiting": 2,
      "serving": 1,
      "completed": 2,
      "skipped": 0,
      "cancelled": 0
    },
    "servingEntries": [
      {
        "id": "31000000-0000-4000-8000-000000000003",
        "queueNumber": 3,
        "patientName": "Ishara Gunasekara (mother calling)",
        "calledAt": "2026-10-12T04:45:00Z",
        "calledBy": {
          "id": "00000000-0000-4000-8000-000000000002",
          "fullName": "Dr. Nimal Perera"
        }
      }
    ],
    "openedAt": "2026-10-12T02:37:31Z",
    "closedAt": null,
    "createdAt": "2026-10-12T01:00:00Z",
    "updatedAt": "2026-10-12T04:45:00Z",
```

```
    "version": 10
  }
}
```

Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 QUEUE_SESSION_NOT_FOUND

```
{
  "type": "https://dms.example/problems/queue-session-not-found",
  "title": "Queue session not found",
  "status": 404,
  "detail": "Queue session not found.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001/call-next",
  "code": "QUEUE_SESSION_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

409 QUEUE_SESSION_NOT_OPEN

```
{
  "type": "https://dms.example/problems/queue-session-not-open",
  "title": "Queue session not open",
  "status": 409,
  "detail": "Queue session is SCHEDULED; open it first.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001/call-next",
  "code": "QUEUE_SESSION_NOT_OPEN",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

409 QUEUE_SESSION_CLOSED

```
{
  "type": "https://dms.example/problems/queue-session-closed",
  "title": "Queue session closed",
  "status": 409,
  "detail": "Queue session is closed.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001/call-next",
  "code": "QUEUE_SESSION_CLOSED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

409 QUEUE_STATE_CHANGED

```
{
  "type": "https://dms.example/problems/queue-state-changed",
  "title": "Queue changed",
  "status": 409,
  "detail": "The queue has moved on: number 7 is now being served.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001/call-next",
  "code": "QUEUE_STATE_CHANGED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "currentServingNumber": 7
}
```

422 QUEUE_ENTRY_NOT_IN_SESSION

```
{
  "type": "https://dms.example/problems/queue-entry-not-in-session",
  "title": "Queue entry not in session",
  "status": 422,
  "detail": "Entry does not belong to this session.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001/call-next",
  "code": "QUEUE_ENTRY_NOT_IN_SESSION",
```

```
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 INVALID_STATUS_TRANSITION


```
{
  "type": "https://dms.example/problems/invalid-status-transition",
  "title": "Invalid status transition",
  "status": 409,
  "detail": "Prescription is already DISPENSED.",
  "instance": "/api/v1/queue-sessions/30000000-0000-4000-8000-000000000001/call-next",
  "code": "INVALID_STATUS_TRANSITION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "currentStatus": "DISPENSED"
}
```


Aspect	Details
Database entities affected	queue_sessions (W), queue_entries (W), audit_events (W)
Transaction / concurrency	One transaction: SELECT session FOR UPDATE → compare expectedLastCalledNumber → optional complete → SELECT lowest WAITING → UPDATE entry SERVING → session.last_called_number = number, version += 1. Concurrent calls serialise on the session row; the second sees the new last_called_number and fails the compare.
Audit events	`QUEUE_ENTRY_COMPLETED`, `QUEUE_ENTRY_CALLED`

14.7 POST /api/v1/queue-entries/{queueEntryId}/call

Item	Value
Method / path	`POST /api/v1/queue-entries/{queueEntryId}/call`
operationId	`callQueueEntry`
Purpose	Call a specific waiting entry out of order (e.g. emergency or priority patient)
Required role(s)	DOCTOR
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`queueEntryId`	path	string (uuid)	yes		Queue entry (booking) UUID.

Request body: none.

Business rules

- WAITING → SERVING; sets session.last_called_number to this number. Already SERVING → 200 unchanged.
- Other statuses → 409 INVALID_STATUS_TRANSITION.

Responses

Status	Meaning / error codes	Body
200	Entry is being served.	QueueEntryResponse
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`QUEUE_ENTRY_NOT_FOUND`	Problem
409	`QUEUE_SESSION_NOT_OPEN`, `QUEUE_SESSION_CLOSED`, `INVALID_STATUS_TRANSITION`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "31000000-0000-4000-8000-000000000005",
  "queueSessionId": "30000000-0000-4000-8000-000000000001",
  "queueNumber": 5,
  "status": "SERVING",
  "channel": "WALK_IN",
  "patientName": "Ms. Rizvi",
  "contactPhone": null,
  "notes": null,
  "bookedAt": "2026-10-12T04:07:31Z",
  "bookedBy": {
    "id": "00000000-0000-4000-8000-000000000004",
    "fullName": "Kamal Silva"
  },
  "calledAt": "2026-10-12T04:50:00Z",
  "calledBy": {
    "id": "00000000-0000-4000-8000-000000000002",
    "fullName": "Dr. Nimal Perera"
  },
  "completedAt": null,
  "skippedAt": null,
  "skipReason": null,
  "cancelledAt": null,
  "cancelReason": null,
  "requeueCount": 0,
  "version": 1
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 QUEUE_ENTRY_NOT_FOUND


```
{
  "type": "https://dms.example/problems/queue-entry-not-found",
  "title": "Queue entry not found",
  "status": 404,
  "detail": "Queue entry not found.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/call",
  "code": "QUEUE_ENTRY_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```

409 QUEUE_SESSION_NOT_OPEN


```
{
  "type": "https://dms.example/problems/queue-session-not-open",
  "title": "Queue session not open",
  "status": 409,
  "detail": "Queue session is SCHEDULED; open it first.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/call",
  "code": "QUEUE_SESSION_NOT_OPEN",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 QUEUE_SESSION_CLOSED


```
{
  "type": "https://dms.example/problems/queue-session-closed",
  "title": "Queue session closed",
  "status": 409,
  "detail": "Queue session is closed.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/call",
  "code": "QUEUE_SESSION_CLOSED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 INVALID_STATUS_TRANSITION


```
{
  "type": "https://dms.example/problems/invalid-status-transition",
  "title": "Invalid status transition",
  "status": 409,
  "detail": "Prescription is already DISPENSED.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/call",
  "code": "INVALID_STATUS_TRANSITION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "currentStatus": "DISPENSED"
}
```


Aspect	Details
Database entities affected	queue_sessions (W), queue_entries (W), audit_events (W)
Transaction / concurrency	Session row locked FOR UPDATE (all queue mutations take the session lock first, then the entry - fixed lock order, no deadlocks).
Audit events	`QUEUE_ENTRY_CALLED`

14.8 POST /api/v1/queue-entries/{queueEntryId}/complete

Item	Value
Method / path	`POST /api/v1/queue-entries/{queueEntryId}/complete`
operationId	`completeQueueEntry`
Purpose	Mark the served entry as completed without calling the next one
Required role(s)	DOCTOR
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`queueEntryId`	path	string (uuid)	yes		Queue entry (booking) UUID.

Request body: none.

Business rules

- SERVING → COMPLETED. Already COMPLETED → 200 unchanged. Other statuses → 409.

Responses

Status	Meaning / error codes	Body
200	Completed.	QueueEntryResponse
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`QUEUE_ENTRY_NOT_FOUND`	Problem
409	`QUEUE_SESSION_NOT_OPEN`, `QUEUE_SESSION_CLOSED`, `INVALID_STATUS_TRANSITION`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "31000000-0000-4000-8000-000000000002",
  "queueSessionId": "30000000-0000-4000-8000-000000000001",
  "queueNumber": 2,
  "status": "COMPLETED",
  "channel": "WALK_IN",
  "patientName": "Dilani Wijesekara",
  "contactPhone": null,
  "notes": null,
  "bookedAt": "2026-10-12T02:07:31Z",
  "bookedBy": {
    "id": "00000000-0000-4000-8000-000000000004",
    "fullName": "Kamal Silva"
  },
  "calledAt": "2026-10-12T04:27:31Z",
  "calledBy": {
    "id": "00000000-0000-4000-8000-000000000002",
    "fullName": "Dr. Nimal Perera"
  },
  "completedAt": "2026-10-12T04:44:00Z",
  "skippedAt": null,
  "skipReason": null,
  "cancelledAt": null,
  "cancelReason": null,
  "requeueCount": 0,
  "version": 2
}
```

Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 QUEUE_ENTRY_NOT_FOUND


```
{
  "type": "https://dms.example/problems/queue-entry-not-found",
  "title": "Queue entry not found",
  "status": 404,
  "detail": "Queue entry not found.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/complete",
  "code": "QUEUE_ENTRY_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 QUEUE_SESSION_NOT_OPEN


```
{
  "type": "https://dms.example/problems/queue-session-not-open",
  "title": "Queue session not open",
  "status": 409,
  "detail": "Queue session is SCHEDULED; open it first.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/complete",
  "code": "QUEUE_SESSION_NOT_OPEN",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 QUEUE_SESSION_CLOSED


```
{
  "type": "https://dms.example/problems/queue-session-closed",
  "title": "Queue session closed",
  "status": 409,
  "detail": "Queue session is closed.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/complete",
  "code": "QUEUE_SESSION_CLOSED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 INVALID_STATUS_TRANSITION


```
{
  "type": "https://dms.example/problems/invalid-status-transition",
  "title": "Invalid status transition",
  "status": 409,
  "detail": "Prescription is already DISPENSED.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/complete",
  "code": "INVALID_STATUS_TRANSITION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "currentStatus": "DISPENSED"
}
```


Aspect	Details
Database entities affected	queue_sessions (W), queue_entries (W), audit_events (W)
Transaction / concurrency	Session row locked FOR UPDATE.
Audit events	`QUEUE_ENTRY_COMPLETED`

14.9 POST /api/v1/queue-entries/{queueEntryId}/skip

Item	Value
Method / path	`POST /api/v1/queue-entries/{queueEntryId}/skip`
operationId	`skipQueueEntry`
Purpose	Mark a called patient as not present (no-show)
Required role(s)	DOCTOR
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Content-Type	yes	application/json
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`queueEntryId`	path	string (uuid)	yes		Queue entry (booking) UUID.

Request body - schema `SkipQueueEntryRequest` (optional)

```
{
  "reason": "Called twice, not present"
}
```

Validation rules (Bean Validation on the request DTO; violations → 400 `VALIDATION_FAILED` with `errors[]`)

Field	Type	Required	Constraints	Notes
`reason`	string | null	no	max length 300	

Business rules

- SERVING → SKIPPED. Already SKIPPED → 200 unchanged. A WAITING entry cannot be skipped (cancel it instead).

Responses

Status	Meaning / error codes	Body
200	Skipped.	QueueEntryResponse
400	`VALIDATION_FAILED`, `MALFORMED_REQUEST`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`QUEUE_ENTRY_NOT_FOUND`	Problem

Status	Meaning / error codes	Body
409	`QUEUE_SESSION_NOT_OPEN`, `QUEUE_SESSION_CLOSED`, `INVALID_STATUS_TRANSITION`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "31000000-0000-4000-8000-000000000003",
  "queueSessionId": "30000000-0000-4000-8000-000000000001",
  "queueNumber": 3,
  "status": "SKIPPED",
  "channel": "PHONE",
  "patientName": "Ishara Gunasekara (mother calling)",
  "contactPhone": "0759876543",
  "notes": null,
  "bookedAt": "2026-10-12T02:17:31Z",
  "bookedBy": {
    "id": "00000000-0000-4000-8000-000000000004",
    "fullName": "Kamal Silva"
  },
  "calledAt": "2026-10-12T04:45:00Z",
  "calledBy": {
    "id": "00000000-0000-4000-8000-000000000002",
    "fullName": "Dr. Nimal Perera"
  },
  "completedAt": null,
  "skippedAt": "2026-10-12T04:48:00Z",
  "skipReason": "Called twice, not present",
  "cancelledAt": null,
  "cancelReason": null,
  "requeueCount": 0,
  "version": 2
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 QUEUE_ENTRY_NOT_FOUND


```
{
  "type": "https://dms.example/problems/queue-entry-not-found",
  "title": "Queue entry not found",
  "status": 404,
  "detail": "Queue entry not found.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/skip",
  "code": "QUEUE_ENTRY_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 QUEUE_SESSION_NOT_OPEN


```
{
  "type": "https://dms.example/problems/queue-session-not-open",
  "title": "Queue session not open",
  "status": 409,
  "detail": "Queue session is SCHEDULED; open it first.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/skip",
  "code": "QUEUE_SESSION_NOT_OPEN",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 QUEUE_SESSION_CLOSED


```
{
  "type": "https://dms.example/problems/queue-session-closed",
  "title": "Queue session closed",
```

```
  "status": 409,
  "detail": "Queue session is closed.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/skip",
  "code": "QUEUE_SESSION_CLOSED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 INVALID_STATUS_TRANSITION


```
{
  "type": "https://dms.example/problems/invalid-status-transition",
  "title": "Invalid status transition",
  "status": 409,
  "detail": "Prescription is already DISPENSED.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/skip",
  "code": "INVALID_STATUS_TRANSITION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "currentStatus": "DISPENSED"
}
```


Aspect	Details
Database entities affected	queue_sessions (W), queue_entries (W), audit_events (W)
Transaction / concurrency	Session row locked FOR UPDATE.
Audit events	`QUEUE_ENTRY_SKIPPED`

14.10 POST /api/v1/queue-entries/{queueEntryId}/requeue

Item	Value
Method / path	`POST /api/v1/queue-entries/{queueEntryId}/requeue`
operationId	`requeueQueueEntry`
Purpose	Return a skipped patient to the waiting list (arrived late)
Required role(s)	DOCTOR, PHARMACIST
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`queueEntryId`	path	string (uuid)	yes		Queue entry (booking) UUID.

Request body: none.

Business rules

- SKIPPED → WAITING; keeps the original queueNumber, so the patient is next in line; requeueCount += 1.
- Only from SKIPPED. A WAITING entry → 200 unchanged.

Responses

Status	Meaning / error codes	Body
200	Waiting again.	QueueEntryResponse
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
404	`QUEUE_ENTRY_NOT_FOUND`	Problem
409	`QUEUE_SESSION_NOT_OPEN`, `QUEUE_SESSION_CLOSED`, `INVALID_STATUS_TRANSITION`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "31000000-0000-4000-8000-000000000003",
  "queueSessionId": "30000000-0000-4000-8000-000000000001",
  "queueNumber": 3,
  "status": "WAITING",
  "channel": "PHONE",
  "patientName": "Ishara Gunasekara (mother calling)",
  "contactPhone": "0759876543",
  "notes": null,
  "bookedAt": "2026-10-12T02:17:31Z",
  "bookedBy": {
    "id": "00000000-0000-4000-8000-000000000004",
    "fullName": "Kamal Silva"
  },
  "calledAt": "2026-10-12T04:45:00Z",
  "calledBy": {
    "id": "00000000-0000-4000-8000-000000000002",
    "fullName": "Dr. Nimal Perera"
  },
  "completedAt": null,
  "skippedAt": "2026-10-12T04:48:00Z",
  "skipReason": "Called twice, not present",
  "cancelledAt": null,
  "cancelReason": null,
  "requeueCount": 1,
  "version": 3
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 QUEUE_ENTRY_NOT_FOUND


```
{
  "type": "https://dms.example/problems/queue-entry-not-found",
  "title": "Queue entry not found",
  "status": 404,
  "detail": "Queue entry not found.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/requeue",
  "code": "QUEUE_ENTRY_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 QUEUE_SESSION_NOT_OPEN

```
{
  "type": "https://dms.example/problems/queue-session-not-open",
  "title": "Queue session not open",
  "status": 409,
  "detail": "Queue session is SCHEDULED; open it first.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/requeue",
  "code": "QUEUE_SESSION_NOT_OPEN",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 QUEUE_SESSION_CLOSED


```
{
  "type": "https://dms.example/problems/queue-session-closed",
  "title": "Queue session closed",
  "status": 409,
  "detail": "Queue session is closed.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/requeue",
  "code": "QUEUE_SESSION_CLOSED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


409 INVALID_STATUS_TRANSITION


```
{
  "type": "https://dms.example/problems/invalid-status-transition",
  "title": "Invalid status transition",
  "status": 409,
  "detail": "Prescription is already DISPENSED.",
  "instance": "/api/v1/queue-entries/31000000-0000-4000-8000-000000000003/requeue",
  "code": "INVALID_STATUS_TRANSITION",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "currentStatus": "DISPENSED"
}
```


Aspect	Details
Database entities affected	queue_sessions (W), queue_entries (W), audit_events (W)
Transaction / concurrency	Session row locked FOR UPDATE.
Audit events	`QUEUE_ENTRY_REQUEUED`

15. Public queue display

Unauthenticated, privacy-safe queue status for patients and waiting-room screens.

15.1 GET /api/v1/public/queue-sessions

Item	Value
Method / path	`GET /api/v1/public/queue-sessions`
operationId	`listPublicQueueSessions`
Purpose	Today's queue sessions (public, no authentication)
Required role(s)	Public - no role
Authentication	None (public)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters: none.

Request body: none.

Business rules

- Returns sessions whose sessionDate = today (APP_TIMEZONE), any status, sorted by name.
- Contains no personal data and no staff names.

Responses

Status	Meaning / error codes	Body
200	Today's sessions.	PublicQueueSessionList
429	`RATE_LIMITED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "date": "2026-10-12",
  "items": [
    {
      "id": "30000000-0000-4000-8000-000000000001",
      "name": "Morning OPD - Room 1",
      "sessionDate": "2026-10-12",
      "status": "OPEN",
      "currentServingNumber": 2,
      "waitingCount": 3
    }
  ]
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

429 RATE_LIMITED


```
{
  "type": "https://dms.example/problems/rate-limited",
  "title": "Too many requests",
  "status": 429,
  "detail": "Too many requests. Retry after 30 seconds.",
  "instance": "/api/v1/public/queue-sessions",
  "code": "RATE_LIMITED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "retryAfterSeconds": 30
}
```


Aspect	Details
Database entities affected	queue_sessions (R), queue_entries (R, counts)
Transaction / concurrency	Read-only, single transaction.

Aspect	Details
Audit events	none

15.2 GET /api/v1/public/queue-sessions/{queueSessionId}

Item	Value
Method / path	`GET /api/v1/public/queue-sessions/{queueSessionId}`
operationId	`getPublicQueueStatus`
Purpose	Privacy-safe live status of one queue (waiting-room display and patients' phones). Polled every NEXT_PUBLIC_QUEUE_POLL_INTERVAL_MS (default 5 s). Send If-None-Match with the last ETag; an unchanged queue returns 304 with no body.
Required role(s)	Public - no role
Authentication	None (public)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Accept	no	application/json (errors: application/problem+json)
If-None-Match	no	ETag from the previous response.
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`queueSessionId`	path	string (uuid)	yes		Queue session UUID.

Request body: none.

Business rules

- Only sessions with sessionDate >= today (APP_TIMEZONE) are visible; past sessions → 404 QUEUE_SESSION_NOT_FOUND.
- Response is built from a dedicated DTO containing only numbers, counts, status and the session label. It never contains patient names, phone numbers, patient numbers, notes, diagnoses, prescriptions or staff names.
- waitingNumbers lists at most the first 50 waiting numbers in order.
- ETag = W/"-"; Cache-Control - no-cache.

Responses

Status	Meaning / error codes	Body
200	Current queue status.	PublicQueueStatus
304	Not modified.	(no body)
400	`VALIDATION_FAILED`	Problem
404	`QUEUE_SESSION_NOT_FOUND`	Problem

Status	Meaning / error codes	Body
429	`RATE_LIMITED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "id": "30000000-0000-4000-8000-000000000001",
  "name": "Morning OPD - Room 1",
  "sessionDate": "2026-10-12",
  "status": "OPEN",
  "currentServingNumber": 2,
  "servingNumbers": [
    2
  ],
  "waitingCount": 3,
  "waitingNumbers": [
    3,
    4,
    5
  ],
  "lastIssuedNumber": 5,
  "updatedAt": "2026-10-12T04:27:31Z"
}
```


Error response examples (operation-specific; the standard 400/401/403/500 bodies are shown in section 3)

404 QUEUE_SESSION_NOT_FOUND


```
{
  "type": "https://dms.example/problems/queue-session-not-found",
  "title": "Queue session not found",
  "status": 404,
  "detail": "Queue session not found.",
  "instance": "/api/v1/public/queue-sessions/30000000-0000-4000-8000-000000000001",
  "code": "QUEUE_SESSION_NOT_FOUND",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z"
}
```


429 RATE_LIMITED


```
{
  "type": "https://dms.example/problems/rate-limited",
  "title": "Too many requests",
  "status": 429,
  "detail": "Too many requests. Retry after 30 seconds.",
  "instance": "/api/v1/public/queue-sessions/30000000-0000-4000-8000-000000000001",
  "code": "RATE_LIMITED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "retryAfterSeconds": 30
}
```


Aspect	Details
Database entities affected	queue_sessions (R), queue_entries (R, numbers only)
Transaction / concurrency	Read-only; consistent snapshot in one read-only transaction.
Audit events	none

16. Audit

Read access to the audit trail (ADMIN).

16.1 GET /api/v1/audit-events

Item	Value
Method / path	`GET /api/v1/audit-events`
operationId	`listAuditEvents`
Purpose	Search the audit trail
Required role(s)	ADMIN
Authentication	Bearer JWT (`Authorization: Bearer <accessToken>`)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Authorization	yes	Bearer <accessToken>
Accept	no	application/json (errors: application/problem+json)
X-Request-Id	no	Optional client correlation id (8-64 chars of [A-Za-z0-9-]). Generated by the server when absent and echoed in the response and in Problem.requestId.

Path and query parameters

Name	In	Type	Required	Constraints	Description
`patientId`	query	string (uuid)	no		
`actorUserId`	query	string (uuid)	no		
`action`	query	AuditAction	no		
`entityType`	query	string	no	one of USER, SESSION, PATIENT, CONSULTATION, PRESCRIPTION, MEDICINE, QUEUE_SESSION, QUEUE_ENTRY, AUDIT	
`entityId`	query	string (uuid)	no		
`from`	query	string (date-time)	no		Inclusive lower bound of occurredAt.
`to`	query	string (date-time)	no		Exclusive upper bound of occurredAt.
`page`	query	integer	no	min 0; default 0	Zero-based page index.
`size`	query	integer	no	min 1; max 100; default 20	Page size.

Request body: none.

Business rules

- Sorted by occurredAt descending. The range from..to may span at most 366 days (VALIDATION_FAILED otherwise).
- Viewing the audit log is itself audited.

Responses

Status	Meaning / error codes	Body
200	Page of audit events.	AuditEventPage
400	`VALIDATION_FAILED`	Problem
401	`UNAUTHENTICATED`, `TOKEN_EXPIRED`	Problem
403	`FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`	Problem
500	`INTERNAL_ERROR`	Problem

Successful response example (200)


```
{
  "items": [
    {
      "id": "7c8d9e0f-1a2b-4c3d-8e4f-5a6b7c8d9e0f",
      "occurredAt": "2026-10-12T04:12:00Z",
      "actor": {
        "id": "00000000-0000-4000-8000-000000000002",
        "username": "dr.perera"
      },
      "action": "PATIENT_HISTORY_VIEWED",
      "entityType": "PATIENT",
      "entityId": "20000000-0000-4000-8000-000000000001",
      "patientId": "20000000-0000-4000-8000-000000000001",
      "outcome": "SUCCESS",
      "details": {
        "page": 0,
        "size": 10
      },
      "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
      "ipAddress": "192.168.1.23"
    }
  ],
  "page": 0,
  "size": 20,
  "totalItems": 1,
  "totalPages": 1
}
```


Error response example (400 VALIDATION_FAILED; all codes for this endpoint are listed under Responses)


```
{
  "type": "https://dms.example/problems/validation-failed",
  "title": "Validation failed",
  "status": 400,
  "detail": "One or more fields are invalid.",
  "instance": "/api/v1/audit-events",
  "code": "VALIDATION_FAILED",
  "requestId": "0b5e2d4c-6f1a-4e3b-9c7d-2a1f0e9d8c7b",
  "timestamp": "2026-10-12T04:30:00Z",
  "errors": [
    {
      "field": "fullName",
      "code": "NotBlank",
      "message": "must not be blank"
    }
  ]
}
```

Aspect	Details
Database entities affected	audit_events (R/W), users (R)
Transaction / concurrency	Read-only, single transaction.
Audit events	`AUDIT_LOG_VIEWED`

17. Health checks

Spring Boot Actuator liveness and readiness.

17.1 GET /actuator/health

Item	Value
Method / path	`GET /actuator/health`
operationId	`health`
Purpose	Aggregate health (UP/DOWN); no component details are exposed
Required role(s)	Public - no role
Authentication	None (public)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Accept	no	application/json (errors: application/problem+json)

Path and query parameters: none.

Request body: none.

Responses

Status	Meaning / error codes	Body
200	Application and database are up.	HealthStatus
503	Down (Actuator format, not Problem Details).	HealthStatus

Successful response example (200)

```
{
  "status": "UP"
}
```

Aspect	Details
Database entities affected	none
Transaction / concurrency	Read-only, single transaction.
Audit events	none

17.2 GET /actuator/health/liveness

Item	Value
Method / path	`GET /actuator/health/liveness`
operationId	`healthLiveness`
Purpose	Liveness probe (process is running; does not check the database)
Required role(s)	Public - no role
Authentication	None (public)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Accept	no	application/json (errors: application/problem+json)

Path and query parameters: none.

Request body: none.

Responses

Status	Meaning / error codes	Body
200	Live.	HealthStatus

Successful response example (200)

```
{
  "status": "UP"
}
```

Aspect	Details
Database entities affected	none
Transaction / concurrency	Read-only, single transaction.
Audit events	none

17.3 GET /actuator/health/readiness

Item	Value
Method / path	`GET /actuator/health/readiness`
operationId	`healthReadiness`
Purpose	Readiness probe (includes database connectivity)
Required role(s)	Public - no role
Authentication	None (public)
Idempotency	Not applicable

Request headers

Header	Required	Value / notes
Accept	no	application/json (errors: application/problem+json)

Path and query parameters: none.

Request body: none.

Responses

Status	Meaning / error codes	Body
200	Ready to receive traffic.	HealthStatus
503	Not ready.	HealthStatus

Successful response example (200)

```
{
  "status": "UP"
}
```

Aspect	Details
Database entities affected	none
Transaction / concurrency	Read-only, single transaction.
Audit events	none

18. Schemas

All component schemas of `openapi.yaml`. Request DTOs reject unknown properties.

18.1 AddPrescriptionRequest

Property	Type	Required	Constraints	Description
`supersedesPrescriptionId`	string | null (uuid)	no		
`notes`	string | null	no	max length 1000	
`items`	array of PrescriptionItemInput	yes	min items 1; max items 30	
`items[].medicineId`	string (uuid)	no		
`items[].medicineName`	string	no	min length 1; max length 200	
`items[].strength`	string | null	no	max length 50	
`items[].dosageForm`	DosageForm | null	no	one of TABLET, CAPSULE, SYRUP, SUSPENSION, SOLUTION, INJECTION, CREAM, OINTMENT, GEL, DROPS, INHALER, SUPPOSITORY, POWDER, PATCH, OTHER	
`items[].dose`	string	yes	min length 1; max length 100	
`items[].frequency`	string	yes	min length 1; max length 100	
`items[].durationDays`	integer | null	no	min 1; max 365	
`items[].quantity`	number	yes	max 99999.99; > 0	Max 2 decimal places (BigDecimal, @Digits(integer=5, fraction=2)).
`items[].quantityUnit`	string	yes	min length 1; max length 30	
`items[].instructions`	string | null	no	max length 500	

- items[]: if `medicineId` is present then `medicineName`, `strength` and `dosageForm` must be absent; otherwise `medicineName` is required.

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.2 AuditAction

Audit action recorded in audit_events.action.

Values: `AUTH_LOGIN_SUCCEEDED`, `AUTH_LOGIN_FAILED`, `AUTH_ACCOUNT_LOCKED`, `AUTH_LOGOUT`, `AUTH_REFRESH_REUSE_DETECTED`, `AUTH_PASSWORD_CHANGED`, `ACCESS_DENIED`, `USER_CREATED`, `USER_UPDATED`, `USER_PASSWORD_RESET`, `PATIENT_CREATED`, `PATIENT_UPDATED`, `PATIENT_VIEWED`, `PATIENT_SEARCHED`, `PATIENT_HISTORY_VIEWED`, `CONSULTATION_CREATED`, `CONSULTATION_VIEWED`, `PRESCRIPTION_CREATED`, `PRESCRIPTION_VIEWED`, `PRESCRIPTION_STARTED`, `PRESCRIPTION_DISPENSED`, `PRESCRIPTION_CANCELLED`, `MEDICINE_CREATED`, `MEDICINE_UPDATED`, `QUEUE_SESSION_CREATED`, `QUEUE_SESSION_OPENED`, `QUEUE_SESSION_CLOSED`, `BOOKING_CREATED`, `BOOKING_CANCELLED`, `QUEUE_ENTRY_CALLED`, `QUEUE_ENTRY_COMPLETED`, `QUEUE_ENTRY_SKIPPED`, `QUEUE_ENTRY_REQUEUED`, `AUDIT_LOG_VIEWED`, `RETENTION_JOB_EXECUTED`

18.3 AuditActor

Property	Type	Required	Constraints	Description
id	string (uuid)	yes		
username	string | null	yes		

18.4 AuditEventPage

Page of AuditEventResponse.

Property	Type	Required	Constraints	Description
items	array of AuditEventResponse	yes		
items[].id	string (uuid)	yes		
items[].occurredAt	string (date-time)	yes		
items[].actor	AuditActor | null	no		
items[].action	AuditAction	yes		Audit action recorded in audit_events.action.
items[].entityType	string	yes		
items[].entityId	string | null (uuid)	no		
items[].patientId	string | null (uuid)	no		
items[].outcome	AuditOutcome	yes	one of SUCCESS, FAILURE, DENIED	
items[].details	object	yes		
items[].requestId	string | null	no		
items[].ipAddress	string | null	no		
page	integer	yes	min 0	
size	integer	yes	min 1; max 100	
totalItems	integer (int64)	yes	min 0	
totalPages	integer	yes	min 0	

18.5 AuditEventResponse

Property	Type	Required	Constraints	Description
id	string (uuid)	yes		
occurredAt	string (date-time)	yes		
actor	AuditActor | null	no		
actor.id	string (uuid)	yes		
actor.username	string | null	yes		
action	AuditAction	yes		Audit action recorded in audit_events.action.
entityType	string	yes		
entityId	string | null (uuid)	no		

Property	Type	Required	Constraints	Description
`patientId`	string | null (uuid)	no		
`outcome`	AuditOutcome	yes	one of SUCCESS, FAILURE, DENIED	
`details`	object	yes		
`requestId`	string | null	no		
`ipAddress`	string | null	no		

18.6 AuditOutcome

Values: `SUCCESS`, `FAILURE`, `DENIED`

18.7 BookingChannel

Values: `PHONE`, `WALK_IN`

18.8 CallNextRequest

Property	Type	Required	Constraints	Description
`expectedLastCalledNumber`	integer | null	yes	min 1; max 999	currentServingNumber the client last saw (null if none).
`completeCurrentEntryId`	string | null (uuid)	no		

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.9 CallNextResponse

Property	Type	Required	Constraints	Description
`calledEntry`	QueueEntryResponse | null	yes		null when nobody is waiting.
`calledEntry.id`	string (uuid)	yes		
`calledEntry.queueSessionId`	string (uuid)	yes		
`calledEntry.queueNumber`	integer	yes	min 1; max 999	
`calledEntry.status`	QueueEntryStatus	yes	one of WAITING, SERVING, COMPLETED, SKIPPED, CANCELLED	WAITING → SERVING → COMPLETED | SKIPPED; WAITING → CANCELLED; SKIPPED → WAITING (requeue).
`calledEntry.channel`	BookingChannel	yes	one of PHONE, WALK_IN	
`calledEntry.patientName`	string | null	no		null after retention anonymisation.
`calledEntry.contactPhone`	string | null	no		
`calledEntry.notes`	string | null	no		
`calledEntry.bookedAt`	string (date-time)	yes		
`calledEntry.bookedBy`	StaffRef	yes		Minimal reference to a staff user.
`calledEntry.calledAt`	string | null (date-time)	no		
`calledEntry.calledBy`	StaffRef | null	no		

Property	Type	Required	Constraints	Description
`calledEntry.completedAt`	string | null (date-time)	no		
`calledEntry.skippedAt`	string | null (date-time)	no		
`calledEntry.skipReason`	string | null	no		
`calledEntry.cancelledAt`	string | null (date-time)	no		
`calledEntry.cancelReason`	string | null	no		
`calledEntry.requeueCount`	integer	yes	min 0	
`calledEntry.version`	integer (int64)	yes		
`completedEntry`	QueueEntryRef | null	yes		
`completedEntry.id`	string (uuid)	yes		
`completedEntry.queueNumber`	integer	yes		
`completedEntry.status`	QueueEntryStatus	yes	one of WAITING, SERVING, COMPLETED, SKIPPED, CANCELLED	WAITING → SERVING → COMPLETED | SKIPPED; WAITING → CANCELLED; SKIPPED → WAITING (requeue).
`session`	QueueSessionResponse	yes		
`session.id`	string (uuid)	yes		
`session.sessionDate`	string (date)	yes		
`session.name`	string	yes		
`session.status`	QueueSessionStatus	yes	one of SCHEDULED, OPEN, CLOSED	SCHEDULED → OPEN → CLOSED; SCHEDULED → CLOSED.
`session.assignedDoctor`	StaffRef | null	no		
`session.lastIssuedNumber`	integer	yes	min 0; max 999	
`session.currentServingNumber`	integer | null	no	min 1; max 999	Most recently called number (session.last_called_number). Send it back as expectedLastCalledNumber.
`session.nextWaitingNumber`	integer | null	no		Lowest WAITING number - who call-next would call.
`session.counts`	QueueCounts	yes		
`session.servingEntries`	array of ServingEntry	yes		
`session.openedAt`	string | null (date-time)	no		
`session.closedAt`	string | null (date-time)	no		
`session.createdAt`	string (date-time)	yes		
`session.updatedAt`	string (date-time)	yes		
`session.version`	integer (int64)	yes		

18.10 CancelBookingRequest

Property	Type	Required	Constraints	Description
`reason`	string	yes	min length 3; max length 500	

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.11 CancelPrescriptionRequest

Property	Type	Required	Constraints	Description
`reason`	string	yes	min length 3; max length 500	

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.12 ChangePasswordRequest

Property	Type	Required	Constraints	Description
`currentPassword`	string (password)	yes	min length 1; max length 128	
`newPassword`	string (password)	yes	min length 12; max length 128	

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.13 CloseQueueSessionRequest

Property	Type	Required	Constraints	Description
`force`	boolean	no	default False	

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.14 ConditionSummary

Property	Type	Required	Constraints	Description
`conditionText`	string	yes		
`icd10Code`	string | null	no		
`lastRecordedAt`	string (date-time)	yes		
`occurrences`	integer	yes	min 1	

18.15 ConsultationPage

Page of ConsultationResponse.

Property	Type	Required	Constraints	Description
`items`	array of ConsultationResponse	yes		
`items[].id`	string (uuid)	yes		
`items[].patient`	PatientRef	yes		
`items[].doctor`	StaffRef	yes		Minimal reference to a staff user.
`items[].queueEntryId`	string | null (uuid)	no		
`items[].consultationAt`	string (date-time)	yes		
`items[].presentingComplaint`	string | null	no		
`items[].clinicalNotes`	string | null	no		
`items[].diagnoses`	array of Diagnosis	yes		
`items[].prescriptions`	array of PrescriptionResponse	yes		All prescriptions of the consultation including cancelled ones, newest first. At most one is not CANCELLED.
`items[].createdAt`	string (date-time)	yes		

Property	Type	Required	Constraints	Description
page	integer	yes	min 0	
size	integer	yes	min 1; max 100	
totalItems	integer (int64)	yes	min 0	
totalPages	integer	yes	min 0	

18.16 ConsultationResponse

Property	Type	Required	Constraints	Description
id	string (uuid)	yes		
patient	PatientRef	yes		
patient.id	string (uuid)	yes		
patient.patientNumber	string	yes		
patient.fullName	string	yes		
doctor	StaffRef	yes		Minimal reference to a staff user.
doctor.id	string (uuid)	yes		
doctor.fullName	string	yes		
queueEntryId	string | null (uuid)	no		
consultationAt	string (date-time)	yes		
presentingComplaint	string | null	no		
clinicalNotes	string | null	no		
diagnoses	array of Diagnosis	yes		
diagnoses[].id	string (uuid)	yes		
diagnoses[].lineNumber	integer	yes	min 1; max 10	
diagnoses[].conditionText	string	yes		
diagnoses[].icd10Code	string | null	no		
prescriptions	array of PrescriptionResponse	yes		All prescriptions of the consultation including cancelled ones, newest first. At most one is not CANCELLED.
prescriptions[].id	string (uuid)	yes		
prescriptions[].prescriptionNumber	string	yes	pattern ^RX-[0-9]{6,}$	Formatted from prescriptions.prescription_number as RX-%06d.
prescriptions[].consultationId	string (uuid)	yes		
prescriptions[].consultationAt	string (date-time)	yes		
prescriptions[].status	PrescriptionStatus	yes	one of PENDING, IN_PROGRESS, DISPENSED, CANCELLED	PENDING → IN_PROGRESS → DISPENSED; PENDING| IN_PROGRESS → CANCELLED; PENDING → DISPENSED allowed. DISPENSED and CANCELLED are terminal.
prescriptions[].notes	string | null	no		
prescriptions[].submittedAt	string (date-time)	yes		

Property	Type	Required	Constraints	Description
`prescriptions[].prescribedBy`	StaffRef	yes		Minimal reference to a staff user.
`prescriptions[].patient`	PrescriptionPatient	yes		Minimum patient data needed to dispense safely.
`prescriptions[].items`	array of PrescriptionItem	yes	min items 1	
`prescriptions[].startedAt`	string | null (date-time)	no		
`prescriptions[].startedBy`	StaffRef | null	no		
`prescriptions[].cancelledAt`	string | null (date-time)	no		
`prescriptions[].cancelledBy`	StaffRef | null	no		
`prescriptions[].cancelReason`	string | null	no		
`prescriptions[].supersedesPrescriptionId`	string | null (uuid)	no		
`prescriptions[].supersededByPrescriptionId`	string | null (uuid)	no		
`prescriptions[].dispensing`	Dispensing | null	no		Present only when status = DISPENSED.
`prescriptions[].version`	integer (int64)	yes		
`createdAt`	string (date-time)	yes		

18.17 CreateBookingRequest

Property	Type	Required	Constraints	Description
`patientName`	string	yes	min length 1; max length 150	
`contactPhone`	PhoneNumber	no	max length 25; pattern `^\+?[0-9 ()-]{7,24}$`	Optional. Spaces, hyphens and parentheses are removed by the server; the result must match ^+?[0-9]{7,15}$.
`channel`	BookingChannel	yes	one of PHONE, WALK_IN	
`notes`	string | null	no	max length 300	
`allowDuplicate`	boolean	no	default False	

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.18 CreateConsultationRequest

Property	Type	Required	Constraints	Description
`patientId`	string (uuid)	yes		
`queueEntryId`	string | null (uuid)	no		
`presentingComplaint`	string | null	no	max length 1000	
`clinicalNotes`	string | null	no	max length 5000	
`diagnoses`	array of DiagnosisInput	yes	min items 1; max items 10	
`diagnoses[].conditionText`	string	yes	min length 1; max length 500	
`diagnoses[].icd10Code`	string | null	no	pattern `^[A-Z][0-9]{2}(\.[0-9A-Z]{1,4})?$`	

Property	Type	Required	Constraints	Description
`prescription`	PrescriptionInput | null	no		Omit or null for an advice-only consultation.
`prescription.notes`	string | null	no	max length 1000	
`prescription.items`	array of PrescriptionItemInput	yes	min items 1; max items 30	

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.19 CreateMedicineRequest

Property	Type	Required	Constraints	Description
`name`	string	yes	min length 1; max length 200	
`strength`	string | null	no	max length 50	
`dosageForm`	DosageForm	yes	one of TABLET, CAPSULE, SYRUP, SUSPENSION, SOLUTION, INJECTION, CREAM, OINTMENT, GEL, DROPS, INHALER, SUPPOSITORY, POWDER, PATCH, OTHER	

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.20 CreatePatientRequest

Property	Type	Required	Constraints	Description
`patientNumber`	PatientNumber	yes	min length 3; max length 20; pattern `^[A-Za-z0-9][A-Za-z0-9-]{2,19}$`	Upper-cased by the server. Stored pattern ^[A-Z0-9][A-Z0-9-]{2,19}$.
`fullName`	string	yes	min length 1; max length 150	
`sex`	Sex	yes	one of MALE, FEMALE, OTHER, UNKNOWN	
`dateOfBirth`	string (date)	no		Exact date of birth (not in the future, >= 1900-01-01). Mutually exclusive with estimatedAgeYears.
`estimatedAgeYears`	integer	no	min 0; max 130	Use when the date of birth is unknown. Mutually exclusive with dateOfBirth.
`phoneNumber`	PhoneNumber	no	max length 25; pattern `^\+?[0-9 ()-]{7,24}$`	Optional. Spaces, hyphens and parentheses are removed by the server; the result must match ^+?[0-9]{7,15}$.
`address`	string | null	no	max length 300	
`knownAllergies`	string | null	no	max length 1000	

- body: exactly one of (dateOfBirth) or (estimatedAgeYears) must be present.

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.21 CreateQueueSessionRequest

Property	Type	Required	Constraints	Description
`sessionDate`	string (date)	yes		
`name`	string	yes	min length 1; max length 100	
`assignedDoctorId`	string | null (uuid)	no		

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.22 CreateUserRequest

Property	Type	Required	Constraints	Description
`username`	string	yes	pattern `^[A-Za-z0-9._-]{3,50}$`	Lower-cased by the server.
`fullName`	string	yes	min length 1; max length 150	
`roles`	array of Role	yes	min items 1; max items 3; unique items	
`professionalRegistrationNumber`	string | null	no	max length 50	
`temporaryPassword`	string (password)	yes	min length 12; max length 128	

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.23 CurrentUser

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`username`	string	yes		
`fullName`	string	yes		
`roles`	array of Role	yes	min items 1; unique items	
`mustChangePassword`	boolean	yes		
`professionalRegistrationNumber`	string | null	no		

18.24 Diagnosis

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`lineNumber`	integer	yes	min 1; max 10	
`conditionText`	string	yes		
`icd10Code`	string | null	no		

18.25 DiagnosisInput

Property	Type	Required	Constraints	Description
`conditionText`	string	yes	min length 1; max length 500	
`icd10Code`	string | null	no	pattern `^[A-Z][0-9]{2}(\.[0-9A-Z]{1,4})?$`	

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.26 DispenseItemInput

Property	Type	Required	Constraints	Description
`prescriptionItemId`	string (uuid)	yes		

Property	Type	Required	Constraints	Description
`quantityDispensed`	number	yes	min 0; max 99999.99	Max 2 decimal places.
`note`	string | null	no	max length 500	Required when quantityDispensed is below the prescribed quantity.

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.27 DispenseRequest

Property	Type	Required	Constraints	Description
`notes`	string | null	no	max length 1000	
`items`	array of DispenseItemInput	yes	min items 1; max items 30	
`items[].prescriptionItemId`	string (uuid)	yes		
`items[].quantityDispensed`	number	yes	min 0; max 99999.99	Max 2 decimal places.
`items[].note`	string | null	no	max length 500	Required when quantityDispensed is below the prescribed quantity.

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.28 DispensedItemStatus

Derived by the server from quantityDispensed vs prescribed quantity.

Values: `FULL`, `PARTIAL`, `NOT_DISPENSED`

18.29 Dispensing

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`pharmacist`	StaffRef	yes		Minimal reference to a staff user.
`pharmacist.id`	string (uuid)	yes		
`pharmacist.fullName`	string	yes		
`dispensedAt`	string (date-time)	yes		
`outcome`	DispensingOutcome	yes	one of FULL, PARTIAL	FULL = every item fully dispensed; PARTIAL = at least one item partial or not dispensed.
`notes`	string | null	no		
`items`	array of DispensingItem	yes		
`items[].prescriptionItemId`	string (uuid)	yes		
`items[].quantityDispensed`	number	yes	min 0	
`items[].status`	DispensedItemStatus	yes	one of FULL, PARTIAL, NOT_DISPENSED	Derived by the server from quantityDispensed vs prescribed quantity.
`items[].note`	string | null	no		

18.30 DispensingItem

Property	Type	Required	Constraints	Description
`prescriptionItemId`	string (uuid)	yes		
`quantityDispensed`	number	yes	min 0	
`status`	DispensedItemStatus	yes	one of FULL, PARTIAL, NOT_DISPENSED	Derived by the server from quantityDispensed vs prescribed quantity.
`note`	string | null	no		

18.31 DispensingOutcome

FULL = every item fully dispensed; PARTIAL = at least one item partial or not dispensed.

Values: `FULL`, `PARTIAL`

18.32 DosageForm

Values: `TABLET`, `CAPSULE`, `SYRUP`, `SUSPENSION`, `SOLUTION`, `INJECTION`, `CREAM`, `OINTMENT`, `GEL`, `DROPS`, `INHALER`, `SUPPOSITORY`, `POWDER`, `PATCH`, `OTHER`

18.33 ErrorCode

Stable machine-readable error code. Catalogue with HTTP status and meaning: API documentation section 3.

Values: `MALFORMED_REQUEST`, `VALIDATION_FAILED`, `IDEMPOTENCY_KEY_REQUIRED`, `UNAUTHENTICATED`, `TOKEN_EXPIRED`, `INVALID_CREDENTIALS`, `ACCOUNT_LOCKED`, `ACCOUNT_DISABLED`, `REFRESH_TOKEN_INVALID`, `REFRESH_TOKEN_ROTATED`, `SESSION_REVOKED`, `FORBIDDEN`, `PASSWORD_CHANGE_REQUIRED`, `NOT_RESOURCE_OWNER`, `RESOURCE_NOT_FOUND`, `PATIENT_NOT_FOUND`, `CONSULTATION_NOT_FOUND`, `PRESCRIPTION_NOT_FOUND`, `QUEUE_SESSION_NOT_FOUND`, `QUEUE_ENTRY_NOT_FOUND`, `USER_NOT_FOUND`, `MEDICINE_NOT_FOUND`, `DUPLICATE_PATIENT_NUMBER`, `DUPLICATE_USERNAME`, `DUPLICATE_MEDICINE`, `DUPLICATE_QUEUE_SESSION`, `DUPLICATE_BOOKING`, `CONCURRENT_MODIFICATION`, `INVALID_STATUS_TRANSITION`, `QUEUE_STATE_CHANGED`, `QUEUE_SESSION_NOT_OPEN`, `QUEUE_SESSION_CLOSED`, `QUEUE_SESSION_FULL`, `QUEUE_SESSION_HAS_ACTIVE_ENTRIES`, `QUEUE_ENTRY_ALREADY_LINKED`, `ACTIVE_PRESCRIPTION_EXISTS`, `REFERENCED_RESOURCE_NOT_FOUND`, `MEDICINE_INACTIVE`, `CONSULTATION_LOCKED`, `INVALID_SUPERSEDED_PRESCRIPTION`, `QUEUE_ENTRY_NOT_IN_SESSION`, `DISPENSE_ITEMS_MISMATCH`, `DISPENSE_QUANTITY_EXCEEDS_PRESCRIBED`, `NOTHING_DISPENSED`, `DISPENSE_NOTE_REQUIRED`, `INVALID_SESSION_DATE`, `ASSIGNED_USER_NOT_DOCTOR`, `SELF_MODIFICATION_NOT_ALLOWED`, `LAST_ADMIN_REQUIRED`, `CURRENT_PASSWORD_INCORRECT`, `PASSWORD_POLICY_VIOLATION`, `IDEMPOTENCY_KEY_REUSED`, `RATE_LIMITED`, `INTERNAL_ERROR`, `SERVICE_UNAVAILABLE`

18.34 FieldError

Property	Type	Required	Constraints	Description
`field`	string	yes		JSON path of the field, e.g. prescription.items[0].quantity, or the parameter name.
`code`	string	yes		Constraint name, e.g. NotBlank, Size, Pattern, Max, NotFound, Inactive.
`message`	string	yes		
`rejectedValue`	object	no		Rejected value; omitted for passwords and free-text clinical fields.

18.35 HealthStatus

Property	Type	Required	Constraints	Description
`status`	string	yes	one of UP, DOWN, OUT_OF_SERVICE, UNKNOWN	

18.36 LoginRequest

Property	Type	Required	Constraints	Description
`username`	string	yes	min length 1; max length 50	
`password`	string (password)	yes	min length 1; max length 128	

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.37 MedicalHistoryResponse

Property	Type	Required	Constraints	Description
`patient`	PatientResponse	yes		
`patient.id`	string (uuid)	yes		
`patient.patientNumber`	string	yes		
`patient.fullName`	string	yes		
`patient.sex`	Sex	yes	one of MALE, FEMALE, OTHER, UNKNOWN	
`patient.dateOfBirth`	string (date)	yes		
`patient.dateOfBirthEstimated`	boolean	yes		
`patient.ageYears`	integer	yes	min 0	Completed years at today's date in APP_TIMEZONE (computed).
`patient.phoneNumber`	string | null	no		
`patient.address`	string | null	no		
`patient.knownAllergies`	string | null	no		
`patient.createdAt`	string (date-time)	yes		
`patient.updatedAt`	string (date-time)	yes		
`patient.version`	integer (int64)	yes		
`conditions`	array of ConditionSummary	yes	max items 50	
`conditions[].conditionText`	string	yes		
`conditions[].icd10Code`	string | null	no		
`conditions[].lastRecordedAt`	string (date-time)	yes		
`conditions[].occurrences`	integer	yes	min 1	
`consultations`	ConsultationPage	yes		Page of ConsultationResponse.
`consultations.items`	array of ConsultationResponse	yes		
`consultations.page`	integer	yes	min 0	
`consultations.size`	integer	yes	min 1; max 100	
`consultations.totalItems`	integer (int64)	yes	min 0	

Property	Type	Required	Constraints	Description
`consultations.totalPages`	integer	yes	min 0	

18.38 MedicinePage

Page of MedicineResponse.

Property	Type	Required	Constraints	Description
`items`	array of MedicineResponse	yes		
`items[].id`	string (uuid)	yes		
`items[].name`	string	yes		
`items[].strength`	string | null	no		
`items[].dosageForm`	DosageForm	yes	one of TABLET, CAPSULE, SYRUP, SUSPENSION, SOLUTION, INJECTION, CREAM, OINTMENT, GEL, DROPS, INHALER, SUPPOSITORY, POWDER, PATCH, OTHER	
`items[].active`	boolean	yes		
`items[].createdAt`	string (date-time)	yes		
`items[].updatedAt`	string (date-time)	yes		
`items[].version`	integer (int64)	yes		
`page`	integer	yes	min 0	
`size`	integer	yes	min 1; max 100	
`totalItems`	integer (int64)	yes	min 0	
`totalPages`	integer	yes	min 0	

18.39 MedicineResponse

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`name`	string	yes		
`strength`	string | null	no		
`dosageForm`	DosageForm	yes	one of TABLET, CAPSULE, SYRUP, SUSPENSION, SOLUTION, INJECTION, CREAM, OINTMENT, GEL, DROPS, INHALER, SUPPOSITORY, POWDER, PATCH, OTHER	
`active`	boolean	yes		
`createdAt`	string (date-time)	yes		
`updatedAt`	string (date-time)	yes		
`version`	integer (int64)	yes		

18.40 PatientNumber

Upper-cased by the server. Stored pattern ^[A-Z0-9][A-Z0-9-]{2,19}$.

18.41 PatientRef

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`patientNumber`	string	yes		
`fullName`	string	yes		

18.42 PatientResponse

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`patientNumber`	string	yes		
`fullName`	string	yes		
`sex`	Sex	yes	one of MALE, FEMALE, OTHER, UNKNOWN	
`dateOfBirth`	string (date)	yes		
`dateOfBirthEstimated`	boolean	yes		
`ageYears`	integer	yes	min 0	Completed years at today's date in APP_TIMEZONE (computed).
`phoneNumber`	string | null	no		
`address`	string | null	no		
`knownAllergies`	string | null	no		
`createdAt`	string (date-time)	yes		
`updatedAt`	string (date-time)	yes		
`version`	integer (int64)	yes		

18.43 PatientSummary

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`patientNumber`	string	yes		
`fullName`	string	yes		
`sex`	Sex	yes	one of MALE, FEMALE, OTHER, UNKNOWN	
`dateOfBirth`	string (date)	yes		
`dateOfBirthEstimated`	boolean	yes		
`ageYears`	integer	yes		
`phoneNumber`	string | null	no		

18.44 PatientSummaryPage

Page of PatientSummary.

Property	Type	Required	Constraints	Description
`items`	array of PatientSummary	yes		
`items[].id`	string (uuid)	yes		
`items[].patientNumber`	string	yes		
`items[].fullName`	string	yes		
`items[].sex`	Sex	yes	one of MALE, FEMALE, OTHER, UNKNOWN	
`items[].dateOfBirth`	string (date)	yes		
`items[].dateOfBirthEstimated`	boolean	yes		
`items[].ageYears`	integer	yes		
`items[].phoneNumber`	string | null	no		
`page`	integer	yes	min 0	
`size`	integer	yes	min 1; max 100	
`totalItems`	integer (int64)	yes	min 0	
`totalPages`	integer	yes	min 0	

18.45 PhoneNumber

Optional. Spaces, hyphens and parentheses are removed by the server; the result must match ^+?[0-9]{7,15}$.

18.46 PrescriptionInput

Property	Type	Required	Constraints	Description
`notes`	string | null	no	max length 1000	
`items`	array of PrescriptionItemInput	yes	min items 1; max items 30	
`items[].medicineId`	string (uuid)	no		
`items[].medicineName`	string	no	min length 1; max length 200	
`items[].strength`	string | null	no	max length 50	
`items[].dosageForm`	DosageForm | null	no	one of TABLET, CAPSULE, SYRUP, SUSPENSION, SOLUTION, INJECTION, CREAM, OINTMENT, GEL, DROPS, INHALER, SUPPOSITORY, POWDER, PATCH, OTHER	
`items[].dose`	string	yes	min length 1; max length 100	
`items[].frequency`	string	yes	min length 1; max length 100	
`items[].durationDays`	integer | null	no	min 1; max 365	
`items[].quantity`	number	yes	max 99999.99; > 0	Max 2 decimal places (BigDecimal, @Digits(integer=5, fraction=2)).
`items[].quantityUnit`	string	yes	min length 1; max length 30	

Property	Type	Required	Constraints	Description
`items[].instructions`	string | null	no	max length 500	

- items[]: if `medicineId` is present then `medicineName`, `strength` and `dosageForm` must be absent; otherwise `medicineName` is required.

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.47 PrescriptionItem

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`lineNumber`	integer	yes	min 1; max 30	
`medicineId`	string | null (uuid)	no		
`medicineName`	string	yes		
`strength`	string | null	no		
`dosageForm`	DosageForm | null	no	one of TABLET, CAPSULE, SYRUP, SUSPENSION, SOLUTION, INJECTION, CREAM, OINTMENT, GEL, DROPS, INHALER, SUPPOSITORY, POWDER, PATCH, OTHER	
`dose`	string	yes		
`frequency`	string	yes		
`durationDays`	integer | null	no		
`quantity`	number	yes		
`quantityUnit`	string	yes		
`instructions`	string | null	no		

18.48 PrescriptionItemInput

Either medicineId (catalogue) or medicineName (free text), never both.

Property	Type	Required	Constraints	Description
`medicineId`	string (uuid)	no		
`medicineName`	string	no	min length 1; max length 200	
`strength`	string | null	no	max length 50	
`dosageForm`	DosageForm | null	no	one of TABLET, CAPSULE, SYRUP, SUSPENSION, SOLUTION, INJECTION, CREAM, OINTMENT, GEL, DROPS, INHALER, SUPPOSITORY, POWDER, PATCH, OTHER	
`dose`	string	yes	min length 1; max length 100	
`frequency`	string	yes	min length 1; max length 100	
`durationDays`	integer | null	no	min 1; max 365	
`quantity`	number	yes	max 99999.99; > 0	Max 2 decimal places (BigDecimal, @Digits(integer=5, fraction=2)).
`quantityUnit`	string	yes	min length 1; max length 30	

Property	Type	Required	Constraints	Description
`instructions`	string | null	no	max length 500	

- body: if `medicineId` is present then `medicineName`, `strength` and `dosageForm` must be absent; otherwise `medicineName` is required.

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.49 PrescriptionPatient

Minimum patient data needed to dispense safely.

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`patientNumber`	string	yes		
`fullName`	string	yes		
`sex`	Sex	yes	one of MALE, FEMALE, OTHER, UNKNOWN	
`ageYears`	integer	yes		
`dateOfBirthEstimated`	boolean	yes		
`knownAllergies`	string | null	no		

18.50 PrescriptionResponse

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`prescriptionNumber`	string	yes	pattern `^RX-[0-9]{6,}$`	Formatted from prescriptions.prescription_number as RX-%06d.
`consultationId`	string (uuid)	yes		
`consultationAt`	string (date-time)	yes		
`status`	PrescriptionStatus	yes	one of PENDING, IN_PROGRESS, DISPENSED, CANCELLED	PENDING → IN_PROGRESS → DISPENSED; PENDING|IN_PROGRESS → CANCELLED; PENDING → DISPENSED allowed. DISPENSED and CANCELLED are terminal.
`notes`	string | null	no		
`submittedAt`	string (date-time)	yes		
`prescribedBy`	StaffRef	yes		Minimal reference to a staff user.
`prescribedBy.id`	string (uuid)	yes		
`prescribedBy.fullName`	string	yes		
`patient`	PrescriptionPatient	yes		Minimum patient data needed to dispense safely.
`patient.id`	string (uuid)	yes		
`patient.patientNumber`	string	yes		
`patient.fullName`	string	yes		
`patient.sex`	Sex	yes	one of MALE, FEMALE, OTHER, UNKNOWN	

Property	Type	Required	Constraints	Description
`patient.ageYears`	integer	yes		
`patient.dateOfBirthEstimated`	boolean	yes		
`patient.knownAllergies`	string | null	no		
`items`	array of PrescriptionItem	yes	min items 1	
`items[].id`	string (uuid)	yes		
`items[].lineNumber`	integer	yes	min 1; max 30	
`items[].medicineId`	string | null (uuid)	no		
`items[].medicineName`	string	yes		
`items[].strength`	string | null	no		
`items[].dosageForm`	DosageForm | null	no	one of TABLET, CAPSULE, SYRUP, SUSPENSION, SOLUTION, INJECTION, CREAM, OINTMENT, GEL, DROPS, INHALER, SUPPOSITORY, POWDER, PATCH, OTHER	
`items[].dose`	string	yes		
`items[].frequency`	string	yes		
`items[].durationDays`	integer | null	no		
`items[].quantity`	number	yes		
`items[].quantityUnit`	string	yes		
`items[].instructions`	string | null	no		
`startedAt`	string | null (date-time)	no		
`startedBy`	StaffRef | null	no		
`startedBy.id`	string (uuid)	yes		
`startedBy.fullName`	string	yes		
`cancelledAt`	string | null (date-time)	no		
`cancelledBy`	StaffRef | null	no		
`cancelledBy.id`	string (uuid)	yes		
`cancelledBy.fullName`	string	yes		
`cancelReason`	string | null	no		
`supersedesPrescriptionId`	string | null (uuid)	no		
`supersededByPrescriptionId`	string | null (uuid)	no		
`dispensing`	Dispensing | null	no		Present only when status = DISPENSED.
`dispensing.id`	string (uuid)	yes		
`dispensing.pharmacist`	StaffRef	yes		Minimal reference to a staff user.

Property	Type	Required	Constraints	Description
`dispensing.dispensedAt`	string (date-time)	yes		
`dispensing.outcome`	DispensingOutcome	yes	one of FULL, PARTIAL	FULL = every item fully dispensed; PARTIAL = at least one item partial or not dispensed.
`dispensing.notes`	string | null	no		
`dispensing.items`	array of DispensingItem	yes		
`version`	integer (int64)	yes		

18.51 PrescriptionStatus

PENDING → IN_PROGRESS → DISPENSED; PENDING\|IN_PROGRESS → CANCELLED; PENDING → DISPENSED allowed. DISPENSED and CANCELLED are terminal.

Values: `PENDING`, `IN_PROGRESS`, `DISPENSED`, `CANCELLED`

18.52 PrescriptionSummary

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`prescriptionNumber`	string	yes		
`consultationId`	string (uuid)	yes		
`status`	PrescriptionStatus	yes	one of PENDING, IN_PROGRESS, DISPENSED, CANCELLED	PENDING → IN_PROGRESS → DISPENSED; PENDING|IN_PROGRESS → CANCELLED; PENDING → DISPENSED allowed. DISPENSED and CANCELLED are terminal.
`patient`	PatientRef	yes		
`patient.id`	string (uuid)	yes		
`patient.patientNumber`	string	yes		
`patient.fullName`	string	yes		
`prescribedBy`	StaffRef	yes		Minimal reference to a staff user.
`prescribedBy.id`	string (uuid)	yes		
`prescribedBy.fullName`	string	yes		
`submittedAt`	string (date-time)	yes		
`itemCount`	integer	yes	min 1	
`startedBy`	StaffRef | null	no		
`startedBy.id`	string (uuid)	yes		
`startedBy.fullName`	string	yes		
`dispensingOutcome`	DispensingOutcome | null	no	one of FULL, PARTIAL	
`dispensedAt`	string | null (date-time)	no		

18.53 PrescriptionSummaryPage

Page of PrescriptionSummary.

Property	Type	Required	Constraints	Description
`items`	array of PrescriptionSummary	yes		
`items[].id`	string (uuid)	yes		
`items[].prescriptionNumber`	string	yes		
`items[].consultationId`	string (uuid)	yes		
`items[].status`	PrescriptionStatus	yes	one of PENDING, IN_PROGRESS, DISPENSED, CANCELLED	PENDING → IN_PROGRESS → DISPENSED; PENDING| IN_PROGRESS → CANCELLED; PENDING → DISPENSED allowed. DISPENSED and CANCELLED are terminal.
`items[].patient`	PatientRef	yes		
`items[].prescribedBy`	StaffRef	yes		Minimal reference to a staff user.
`items[].submittedAt`	string (date-time)	yes		
`items[].itemCount`	integer	yes	min 1	
`items[].startedBy`	StaffRef | null	no		
`items[].dispensingOutcome`	DispensingOutcome | null	no	one of FULL, PARTIAL	
`items[].dispensedAt`	string | null (date-time)	no		
`page`	integer	yes	min 0	
`size`	integer	yes	min 1; max 100	
`totalItems`	integer (int64)	yes	min 0	
`totalPages`	integer	yes	min 0	

18.54 Problem

RFC 9457 Problem Details used for every error response (media type application/problem+json). `code` is the stable machine-readable value clients switch on; `title` is stable per code; `detail` is human-readable and may vary. Context fields (existingPatientId, currentStatus, ...) are present only for the codes that define them.

Property	Type	Required	Constraints	Description
`type`	string (uri)	yes		https://dms.example/problems/<code-in-kebab-case>
`title`	string	yes		
`status`	integer	yes	min 400; max 599	
`detail`	string	no		
`instance`	string	no		Request path.
`code`	ErrorCode	yes		Stable machine-readable error code. Catalogue with HTTP status and meaning: API documentation section 3.
`requestId`	string	yes		
`timestamp`	string (date-time)	yes		
`errors`	array of FieldError	no		Field-level problems (VALIDATION_FAILED and some 409/422 codes).

Property	Type	Required	Constraints	Description
errors[].field	string	yes		JSON path of the field, e.g. prescription.items[0].quantity, or the parameter name.
errors[].code	string	yes		Constraint name, e.g. NotBlank, Size, Pattern, Max, NotFound, Inactive.
errors[].message	string	yes		
errors[].rejectedValue	object	no		Rejected value; omitted for passwords and free-text clinical fields.
existingPatientId	string (uuid)	no		DUPLICATE_PATIENT_NUMBER only.
existingQueueNumber	integer	no		DUPLICATE_BOOKING only.
currentStatus	string	no		INVALID_STATUS_TRANSITION only - status of the resource now.
currentServingNumber	integer | null	no		QUEUE_STATE_CHANGED only.
waitingCount	integer	no		QUEUE_SESSION_HAS_ACTIVE_ENTRIES only.
servingCount	integer	no		QUEUE_SESSION_HAS_ACTIVE_ENTRIES only.
lockedUntil	string (date-time)	no		ACCOUNT_LOCKED only.
retryAfterSeconds	integer	no		RATE_LIMITED only.

18.55 PublicQueueSessionList

Property	Type	Required	Constraints	Description
date	string (date)	yes		
items	array of PublicQueueSessionSummary	yes		
items[].id	string (uuid)	yes		
items[].name	string	yes		
items[].sessionDate	string (date)	yes		
items[].status	QueueSessionStatus	yes	one of SCHEDULED, OPEN, CLOSED	SCHEDULED → OPEN → CLOSED; SCHEDULED → CLOSED.
items[].currentServingNumber	integer | null	yes		
items[].waitingCount	integer	yes	min 0	

18.56 PublicQueueSessionSummary

Property	Type	Required	Constraints	Description
id	string (uuid)	yes		
name	string	yes		
sessionDate	string (date)	yes		
status	QueueSessionStatus	yes	one of SCHEDULED, OPEN, CLOSED	SCHEDULED → OPEN → CLOSED; SCHEDULED → CLOSED.
currentServingNumber	integer | null	yes		
waitingCount	integer	yes	min 0	

18.57 PublicQueueStatus

The ONLY queue representation exposed without authentication. Adding a field requires a privacy review (BR-Q-10).

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`name`	string	yes		
`sessionDate`	string (date)	yes		
`status`	QueueSessionStatus	yes	one of SCHEDULED, OPEN, CLOSED	SCHEDULED → OPEN → CLOSED; SCHEDULED → CLOSED.
`currentServingNumber`	integer | null	yes		
`servingNumbers`	array of integer	yes		
`waitingCount`	integer	yes	min 0	
`waitingNumbers`	array of integer	yes	max items 50	
`lastIssuedNumber`	integer	yes	min 0	
`updatedAt`	string (date-time)	yes		

18.58 QueueCounts

Property	Type	Required	Constraints	Description
`waiting`	integer	yes	min 0	
`serving`	integer	yes	min 0	
`completed`	integer	yes	min 0	
`skipped`	integer	yes	min 0	
`cancelled`	integer	yes	min 0	

18.59 QueueEntryPage

Page of QueueEntryResponse.

Property	Type	Required	Constraints	Description
`items`	array of QueueEntryResponse	yes		
`items[].id`	string (uuid)	yes		
`items[].queueSessionId`	string (uuid)	yes		
`items[].queueNumber`	integer	yes	min 1; max 999	
`items[].status`	QueueEntryStatus	yes	one of WAITING, SERVING, COMPLETED, SKIPPED, CANCELLED	WAITING → SERVING → COMPLETED | SKIPPED; WAITING -> CANCELLED; SKIPPED → WAITING (requeue).
`items[].channel`	BookingChannel	yes	one of PHONE, WALK_IN	
`items[].patientName`	string | null	no		null after retention anonymisation.
`items[].contactPhone`	string | null	no		
`items[].notes`	string | null	no		
`items[].bookedAt`	string (date-time)	yes		
`items[].bookedBy`	StaffRef	yes		Minimal reference to a staff user.

Property	Type	Required	Constraints	Description
`items[].calledAt`	string | null (date-time)	no		
`items[].calledBy`	StaffRef | null	no		
`items[].completedAt`	string | null (date-time)	no		
`items[].skippedAt`	string | null (date-time)	no		
`items[].skipReason`	string | null	no		
`items[].cancelledAt`	string | null (date-time)	no		
`items[].cancelReason`	string | null	no		
`items[].requeueCount`	integer	yes	min 0	
`items[].version`	integer (int64)	yes		
`page`	integer	yes	min 0	
`size`	integer	yes	min 1; max 100	
`totalItems`	integer (int64)	yes	min 0	
`totalPages`	integer	yes	min 0	

18.60 QueueEntryRef

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`queueNumber`	integer	yes		
`status`	QueueEntryStatus	yes	one of WAITING, SERVING, COMPLETED, SKIPPED, CANCELLED	WAITING → SERVING → COMPLETED | SKIPPED; WAITING → CANCELLED; SKIPPED → WAITING (requeue).

18.61 QueueEntryResponse

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`queueSessionId`	string (uuid)	yes		
`queueNumber`	integer	yes	min 1; max 999	
`status`	QueueEntryStatus	yes	one of WAITING, SERVING, COMPLETED, SKIPPED, CANCELLED	WAITING → SERVING → COMPLETED | SKIPPED; WAITING → CANCELLED; SKIPPED → WAITING (requeue).
`channel`	BookingChannel	yes	one of PHONE, WALK_IN	
`patientName`	string | null	no		null after retention anonymisation.
`contactPhone`	string | null	no		
`notes`	string | null	no		
`bookedAt`	string (date-time)	yes		
`bookedBy`	StaffRef	yes		Minimal reference to a staff user.
`bookedBy.id`	string (uuid)	yes		
`bookedBy.fullName`	string	yes		
`calledAt`	string | null (date-time)	no		
`calledBy`	StaffRef | null	no		

Property	Type	Required	Constraints	Description
`calledBy.id`	string (uuid)	yes		
`calledBy.fullName`	string	yes		
`completedAt`	string | null (date-time)	no		
`skippedAt`	string | null (date-time)	no		
`skipReason`	string | null	no		
`cancelledAt`	string | null (date-time)	no		
`cancelReason`	string | null	no		
`requeueCount`	integer	yes	min 0	
`version`	integer (int64)	yes		

18.62 QueueEntryStatus

WAITING → SERVING → COMPLETED | SKIPPED; WAITING → CANCELLED; SKIPPED → WAITING (requeue).

Values: `WAITING`, `SERVING`, `COMPLETED`, `SKIPPED`, `CANCELLED`

18.63 QueueSessionPage

Page of QueueSessionResponse.

Property	Type	Required	Constraints	Description
`items`	array of QueueSessionResponse	yes		
`items[].id`	string (uuid)	yes		
`items[].sessionDate`	string (date)	yes		
`items[].name`	string	yes		
`items[].status`	QueueSessionStatus	yes	one of SCHEDULED, OPEN, CLOSED	SCHEDULED → OPEN → CLOSED; SCHEDULED → CLOSED.
`items[].assignedDoctor`	StaffRef | null	no		
`items[].lastIssuedNumber`	integer	yes	min 0; max 999	
`items[].currentServingNumber`	integer | null	no	min 1; max 999	Most recently called number (session.last_called_number). Send it back as expectedLastCalledNumber.
`items[].nextWaitingNumber`	integer | null	no		Lowest WAITING number - who call-next would call.
`items[].counts`	QueueCounts	yes		
`items[].servingEntries`	array of ServingEntry	yes		
`items[].openedAt`	string | null (date-time)	no		
`items[].closedAt`	string | null (date-time)	no		
`items[].createdAt`	string (date-time)	yes		
`items[].updatedAt`	string (date-time)	yes		
`items[].version`	integer (int64)	yes		

Property	Type	Required	Constraints	Description
page	integer	yes	min 0	
size	integer	yes	min 1; max 100	
totalItems	integer (int64)	yes	min 0	
totalPages	integer	yes	min 0	

18.64 QueueSessionResponse

Property	Type	Required	Constraints	Description
id	string (uuid)	yes		
sessionDate	string (date)	yes		
name	string	yes		
status	QueueSessionStatus	yes	one of SCHEDULED, OPEN, CLOSED	SCHEDULED → OPEN → CLOSED; SCHEDULED → CLOSED.
assignedDoctor	StaffRef | null	no		
assignedDoctor.id	string (uuid)	yes		
assignedDoctor.fullName	string	yes		
lastIssuedNumber	integer	yes	min 0; max 999	
currentServingNumber	integer | null	no	min 1; max 999	Most recently called number (session.last_called_number). Send it back as expectedLastCalledNumber.
nextWaitingNumber	integer | null	no		Lowest WAITING number - who call-next would call.
counts	QueueCounts	yes		
counts.waiting	integer	yes	min 0	
counts.serving	integer	yes	min 0	
counts.completed	integer	yes	min 0	
counts.skipped	integer	yes	min 0	
counts.cancelled	integer	yes	min 0	
servingEntries	array of ServingEntry	yes		
servingEntries[].id	string (uuid)	yes		
servingEntries[].queueNumber	integer	yes		
servingEntries[].patientName	string | null	no		
servingEntries[].calledAt	string (date-time)	yes		
servingEntries[].calledBy	StaffRef | null	no		
openedAt	string | null (date-time)	no		
closedAt	string | null (date-time)	no		
createdAt	string (date-time)	yes		
updatedAt	string (date-time)	yes		
version	integer (int64)	yes		

18.65 QueueSessionStatus

SCHEDULED → OPEN → CLOSED; SCHEDULED → CLOSED.

Values: `SCHEDULED`, `OPEN`, `CLOSED`

18.66 ResetPasswordRequest

Property	Type	Required	Constraints	Description
`temporaryPassword`	string (password)	yes	min length 12; max length 128	

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.67 Role

Values: `ADMIN`, `DOCTOR`, `PHARMACIST`

18.68 ServingEntry

Staff-only view of an entry currently being served.

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`queueNumber`	integer	yes		
`patientName`	string | null	no		
`calledAt`	string (date-time)	yes		
`calledBy`	StaffRef | null	no		
`calledBy.id`	string (uuid)	yes		
`calledBy.fullName`	string	yes		

18.69 Sex

Values: `MALE`, `FEMALE`, `OTHER`, `UNKNOWN`

18.70 SkipQueueEntryRequest

Property	Type	Required	Constraints	Description
`reason`	string | null	no	max length 300	

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.71 StaffList

Property	Type	Required	Constraints	Description
`items`	array of StaffMember	yes	max items 200	
`items[].id`	string (uuid)	yes		
`items[].fullName`	string	yes		
`items[].roles`	array of Role	yes		

18.72 StaffMember

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`fullName`	string	yes		
`roles`	array of Role	yes		

18.73 StaffRef

Minimal reference to a staff user.

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`fullName`	string	yes		

18.74 TokenResponse

Property	Type	Required	Constraints	Description
`accessToken`	string	yes		
`tokenType`	string	yes	= Bearer	
`expiresIn`	integer	yes		Access-token lifetime in seconds.
`user`	CurrentUser	yes		
`user.id`	string (uuid)	yes		
`user.username`	string	yes		
`user.fullName`	string	yes		
`user.roles`	array of Role	yes	min items 1; unique items	
`user.mustChangePassword`	boolean	yes		
`user.professionalRegistrationNumber`	string | null	no		

18.75 UpdateMedicineRequest

Property	Type	Required	Constraints	Description
`name`	string	yes	min length 1; max length 200	
`strength`	string | null	no	max length 50	
`dosageForm`	DosageForm	yes	one of TABLET, CAPSULE, SYRUP, SUSPENSION, SOLUTION, INJECTION, CREAM, OINTMENT, GEL, DROPS, INHALER, SUPPOSITORY, POWDER, PATCH, OTHER	
`active`	boolean	yes		
`version`	integer (int64)	yes	min 0	

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.76 UpdatePatientRequest

Property	Type	Required	Constraints	Description
`fullName`	string	yes	min length 1; max length 150	
`sex`	Sex	yes	one of MALE, FEMALE, OTHER, UNKNOWN	
`dateOfBirth`	string (date)	no		Date of birth to store. Send together with dateOfBirthEstimated. Mutually exclusive with estimatedAgeYears.
`dateOfBirthEstimated`	boolean	no		Send the value received from GET unless the user entered an exact DOB (then false). Required with dateOfBirth.
`estimatedAgeYears`	integer	no	min 0; max 130	Re-estimates the DOB from an age (stores dateOfBirthEstimated = true). Mutually exclusive with dateOfBirth.
`phoneNumber`	PhoneNumber	no	max length 25; pattern `^\+?[0-9 ()-]{7,24}$`	Optional. Spaces, hyphens and parentheses are removed by the server; the result must match ^+?[0-9]{7,15}$.
`address`	string | null	no	max length 300	
`knownAllergies`	string | null	no	max length 1000	
`version`	integer (int64)	yes	min 0	

- body: exactly one of (dateOfBirth + dateOfBirthEstimated) or (estimatedAgeYears) must be present.

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.77 UpdateUserRequest

Property	Type	Required	Constraints	Description
`fullName`	string	yes	min length 1; max length 150	
`roles`	array of Role	yes	min items 1; max items 3; unique items	
`enabled`	boolean	yes		
`professionalRegistrationNumber`	string | null	no	max length 50	
`version`	integer (int64)	yes	min 0	

Unknown properties are rejected (`FAIL_ON_UNKNOWN_PROPERTIES` for request DTOs).

18.78 UserPage

Page of UserResponse.

Property	Type	Required	Constraints	Description
`items`	array of UserResponse	yes		
`items[].id`	string (uuid)	yes		
`items[].username`	string	yes		
`items[].fullName`	string	yes		
`items[].roles`	array of Role	yes	min items 1; unique items	
`items[].enabled`	boolean	yes		
`items[].mustChangePassword`	boolean	yes		

Property	Type	Required	Constraints	Description
`items[].professionalRegistrationNumber`	string | null	no		
`items[].lastLoginAt`	string | null (date-time)	no		
`items[].lockedUntil`	string | null (date-time)	no		
`items[].createdAt`	string (date-time)	yes		
`items[].updatedAt`	string (date-time)	yes		
`items[].version`	integer (int64)	yes		
`page`	integer	yes	min 0	
`size`	integer	yes	min 1; max 100	
`totalItems`	integer (int64)	yes	min 0	
`totalPages`	integer	yes	min 0	

18.79 UserResponse

Property	Type	Required	Constraints	Description
`id`	string (uuid)	yes		
`username`	string	yes		
`fullName`	string	yes		
`roles`	array of Role	yes	min items 1; unique items	
`enabled`	boolean	yes		
`mustChangePassword`	boolean	yes		
`professionalRegistrationNumber`	string | null	no		
`lastLoginAt`	string | null (date-time)	no		
`lockedUntil`	string | null (date-time)	no		
`createdAt`	string (date-time)	yes		
`updatedAt`	string (date-time)	yes		
`version`	integer (int64)	yes		